

Jack Mackerel Technical Annex 2026

Cumulative assessment update Version 2

Jack Mackerel Working Group

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1 Introduction

This cumulative technical annex carries forward the assessment background and methods documented for SC13, records the May 2026 benchmark (SCW16) and subsequent model development, and presents the annual assessment and analyses available during SC14 through **11 September 2026**. The working assessment uses Models h1_1.06 and h2_1.06, with data through 2026. Results, input tables and diagnostic figures below are read from those saved assessment products.

The SC13 annex provides the preceding assessment record. SCW16 reviewed the data and model specification; subsequent development revised the Peruvian indices, selectivity and productivity assumptions. During SC14, analysts incorporated the annual data update, examined conflicting abundance indices, and checked projections, reference points and uncertainty. The 2026 results retain working status pending the agreed Committee assessment and advice record. The numerical projection results are the saved 11 September runs identified in their manifest.

Table 1: Assessment work covered by this cumulative update.

Stage	Date and place	Contribution to this annex
SC13 assessment	2025	Assessment background, model methods and the preceding annual assessment record.
SCW16 benchmark	18–22 May 2026, Lima	Data review, CPUE and acoustic treatments, biological inputs and model development.
Post-benchmark development	May–June 2026	Separate Peruvian CPUE series, selectivity and productivity revisions.
SC14 working analyses	7–11 September 2026, Tórshavn	Annual data bridge, Model 1.06, LOI/LOO diagnostics, uncertainty analysis and projection checks.

The assessment history and development are documented in the SC13 technical annex, benchmark report, model-development paper and SC14 working assessment ([Jack Mackerel Assessment SC14, 2026](#); [Model Developments, 2026](#); [SPRFMO Jack Mackerel Working Group, 2026a](#); [SPRFMO Scientific Committee, 2025](#)).

1.1 Scientific Name and General Distribution

The Chilean jack mackerel (*Trachurus murphyi*, Nichols 1920) is widespread throughout the South Pacific. It is found along the shelf and oceanic waters adjacent to Ecuador, Peru, and Chile, and across the South Pacific along the Subtropical Convergence Zone in what has been described as the “jack mackerel belt” that goes from the coast of Chile to New Zealand within a 35° to 50° S variable band across the South Pacific.

1.2 Main Management Units

At least five management units of *T. murphyi* associated to distinct fisheries are identified in the SE Pacific: the Ecuadorian fishery, which is managed as part of a more general pelagic fishery within the Ecuadorian EEZ; the Peruvian fishery, which is managed as part of a jack mackerel, mackerel and sardine fishery directed exclusively for direct human consumption taking place almost entirely within the Peruvian EEZ; the northern and the central-southern Chilean fisheries which are managed as separate management units, with the northern fishery being mostly within the Chilean EEZ and the central-southern Chilean fishery which straddles the Chilean EEZ and the adjacent high sea; and, the purely high sea fishery which is a multinational fishery being managed entirely within the context of the SPRFMO. At present there is no directed fishery for *T. murphyi* in the central and western South Pacific and around New Zealand, where incidental catches are very small.

1.3 Stock Structure

There are a number of competing stock structure hypotheses, and up to five and more separate stocks have been suggested: i) a Peruvian stock (northern stock) which is a straddling stock with respect to the high seas; ii) a Chilean stock (southern stock) which is also a straddling stock with respect to the high seas; iii) a central Pacific stock which exists solely in the high seas; iv) a southwest Pacific stock which exists solely in the high seas; v) and, a New Zealand-Australian stock which straddles the high seas and both the New Zealand and Australian EEZs. Regarding specifically the eastern and central South Pacific, the SPRFMO has identified the following four alternative stock structure working hypotheses: 1) jack mackerel caught off the coasts of Peru and Chile each constitute separate stocks which straddle the high seas; 2) jack mackerel caught off the coasts of Peru and Chile constitute a single shared stock which straddles the high seas; 3) jack mackerel caught off the Chilean area constitute a single straddling stock extending from the coast out to about 120°W; and, 4) jack mackerel caught off the Chilean area constitute separate straddling and high seas stocks.

Accordingly, the Jack Mackerel Sub-group (JMSG) of the Science Working Group (SWG) of the SPRFMO at its 11th Session (SWG-11) carried out parallel assessments of the jack mackerel stock(s) in the Eastern South Pacific under the two main working hypotheses already identified. That is: jack mackerel caught off the coasts of Peru and Chile constitute a single shared stock which straddles the high seas (hypothesis 1); or that jack mackerel caught off the coasts of Peru and Chile each constitute separate stocks (the Peruvian or northern and the Chilean or southern stock) which straddle the high seas (hypothesis 2). In following up on the SWG-11 recommendations, the SPRFMO Commission at its 1st Commission Meeting requested the newly established Scientific Committee (SC) to continue the work on evaluating alternative hypotheses on jack mackerel stock population. Pending more conclusive findings on the stock population structure of jack mackerel, the 2nd Commission meeting requested the SC to continue and expand the stock assessment work under both stock hypotheses considered in the 11th SWG Meeting, and this continues to be one of the main tasks retained in the 2026 assessment.

1.4 Fishery

The fishery for jack mackerel in the south-eastern Pacific is conducted by fleets from the coastal states (Chile, Peru and Ecuador), and by distant water fleets from various countries, operating beyond the EEZ of the coastal states.

The fishery by the coastal states is conducted by purse seiners. The largest fishery is in Chile; fishmeal production has been an important part of its history. In Peru, the fishery is variable from year to year. Here the fish are taken by purse seiners that also fish for other pelagic species (e.g., anchovy, mackerel, sardines). According to government regulations, the jack mackerel in Peru may only be used for human consumption. Ecuador constitutes the northern fringe of the distribution of jack mackerel. Here the fish only occur in certain years, when the local purse seiners may take substantial quantities (70,000 tons in 2011). Part of the catch is processed into fish meal but recently jack mackerel has been promoted to be used for human consumption.

The distant water fleets operating for jack mackerel outside the EEZs have been from a number of parties including Belize, China, Cook Islands, Cuba, European Union (Netherlands, Germany, Poland and Lithuania), Faroe Islands, Korea, Japan, Russian Federation, Ukraine and Vanuatu. These fleets consist exclusively of pelagic trawlers that freeze the catch for human consumption. In the 1980s a large fleet from Russia and other Eastern European countries operated as far west as 130° W. After the economic reforms in the communist countries around 1990, the fishery by these countries in the eastern Pacific was halted. Foreign trawlers returned to the waters outside the EEZs of the coastal states in 2003.

The fishery in Chilean and offshore waters targets jack mackerel. In the offshore fishery, the catch consists of 90 – 98% jack mackerel, with minor bycatch of chub mackerel (*Scomber japonicus*) and Pacific bream (*Brama australis*). The available time series of jack mackerel catches in the south-eastern Pacific by Member are documented in the annual catch compilation; the fleet totals used here are shown in Figure 4 and Table 10.

1.5 Management

Jack mackerel were managed by coastal states beginning in the mid-1990s. National catch quotas for jack mackerel were introduced by Peru in 1995 and by Chile in 1999. Peru introduced a ban on the use of jack mackerel for fish meal in 2002. For the international waters, the first voluntary agreement to limit the number of fishing vessels was introduced in 2010. Catch limits for jack mackerel were established for the south-eastern Pacific starting from 2011.

1.6 Information on the environment in relation to the fisheries

Important environmental events such as the El Niño effect of 2016 affect oceanographic dynamics. During such events, the depth of the 15 °C isotherm and oxycline change significantly affecting the spatial distribution of jack mackerel and their availability in different regions (see for example the work of the Habitat Monitoring Working Group of the Scientific Committee as reported in previous meetings of the Scientific Committee). The extent that such changes affect the overall population productivity is unclear.

1.7 Reproductive Biology

The main spawning season happens from October to December; however, spawning has been described from July to March. Gonadosomatic index and egg surveys have been used to determine the time of spawning.

2 Data used in the assessment

2.1 Current data and provisional values

The assessment retains four fisheries: North Chile purse seine, South-Central Chile purse seine, FarNorth, and Offshore trawl. The data bridge updates catches through 2025, adds provisional 2026 catches, and incorporates revised fishery compositions, weights at age and abundance indices. The 2026 observations describe the part of the year available at assessment time; 2026 catches are model inputs based on the current annual estimates. They should be distinguished from completed annual landings.

The active Model 1.06 contains **six abundance indices**: Chile Acoustic Central-South, Chile Acoustic North, Chile CPUE, Peruvian artisanal CPUE, Peruvian industrial CPUE, and Offshore CPUE. The current data coverage and inputs are shown in Table 27, Table 17 and Figure 5. The DEPM and Peruvian acoustic series discussed in earlier assessments and at the benchmark are absent from this annual model. Their workshop history remains relevant to the development record, while the tables below describe the actual 2026 inputs.

The current model fleet catch totals are in Table 10. Model 1.05 adds **22,799 t** of Chilean high-seas catch to the North Chile fleet in 2025. Model 1.06 adds **10,000 t** to the FarNorth fleet in 2026. The latter is an addition to the assessment catch total. The current input files and update script establish that allocation.

2.2 Benchmark review of fishery indices

SCW16 selected the revised South-Central Chile CPUE index beginning in 1998, using the selected INLA standardization and annual coverage-adjusted uncertainty. The workshop agreed a 1% compounded efficiency adjustment through 2004 and a fixed multiplicative correction of 0.669 from 2005, with the EU objection recorded in the report. A diagnostic alternative with an estimated 2005 catchability change was also requested. Coverage was used to increase uncertainty in poorly sampled years, by dividing the base CV by the square root of annual coverage relative to maximum coverage. These are the benchmark decisions; the alternative trip-database index examined at SC14 is identified separately below ([SPRFMO Jack Mackerel Working Group, 2026a](#)).

Peruvian CPUE treatment changed after the benchmark. SCW16 preferred the shorter standardized combined series for 2015–2025. Post-benchmark Models 0.15/0.16 instead used separate nominal industrial and artisanal series for 2002–2025, removed the Peruvian CPUE catchability random walks, and applied coverage-based uncertainty. Industrial CPUE received a 1% annual efficiency adjustment; artisanal CPUE remained unadjusted. The documented uncertainty used a mean CV of 30%, annual coverage adjustment, and a 20% floor. A June implementation correction changed the industrial efficiency operation from multiplication to division, lowering the terminal industrial index; the artisanal series was unchanged. The 2026 bridge updates these separate series ([Model Developments, 2026](#); [SPRFMO Jack Mackerel Working Group, 2026b](#)).

The offshore CPUE combines observations across the participating offshore fleets. Its 2026 assessment update extends through 2025. Changes in this index, the Chilean index and the North Chile acoustic survey motivate the index-influence analyses described under Results. CPUE reflects both abundance and availability to the sampled fleet; changes in fishing location, targeting, catchability and sampling coverage require consideration alongside modelled stock-wide biomass.

2.3 Survey indices and composition data

SCW16 retained the design-based Chile Acoustic North biomass series. Spatio-temporal acoustic analyses remained diagnostic and research options. Subsequent 2026 Model 1.01 removes the pre-2006 Acoustic North observations; this treatment is inherited by Model 1.06. The 2026 North Chile acoustic estimate and associated age compositions and weights at age enter the annual data bridge. The decline in that survey index is a regional observation and should be interpreted separately from the estimated trend in total stock spawning biomass (*Jack Mackerel Assessment SC14, 2026*; *SPRFMO Jack Mackerel Working Group, 2026a*).

For Peru, SCW16 agreed to replace the older acoustic series with seasonal **availability indices** and associated survey length compositions. Survey sample-size vectors were based on tows containing jack mackerel, capped at 30 and then rescaled to mean 15. These workshop choices describe a development stage; the present annual Model 1.06 omits the Peruvian acoustic indices. This distinction also separates the Peruvian CPUE update from acoustic information.

SCW16 retained fork-length data and matching growth units for the FarNorth fishery, using 10–50 cm model bins with a 50 cm plus group. Its fishery length-composition sample-size vector was rescaled to mean 30. This fishery weighting differs from the survey weighting described above. The input compositions and weights used by the current annual model are provided in the tables and figures below (*SPRFMO Jack Mackerel Working Group, 2026a*).

2.4 Biological Parameters

The maturity-at-age for jack mackerel in Chile was estimated by Leal et al. (2012) and has been updated by applying the new ageing criteria (Cerna, 2022) to the otoliths and histological maturity data collected between September 2011 and January 2012. Overall, the changes caused by the new aging criteria led to the understanding of a faster-growing species that is earlier to mature. Maturity-at-length was consistently observed with L_{50} at about 22–23 cm fork length (FL). The maturity-at-age values, for the single/Southern stock and those for the far north stock, are shown in Table 20.

To fit the length composition data from the far north fleet, a growth curve was used to convert age compositions predicted by the model to predicted lengths, with the conversion occurring within the model. The values for the von Bertalanffy growth parameters are given in Table 21. It was noted in SC10 that the growth parameters reflected fish Total Length, whereas the data were in Fork Length. The parameters were since corrected. Ageing imprecision was previously acknowledged using an age-error matrix, as shown in Table 22. However, because this matrix is based on expert judgement instead of empirical data, the discussions during SC4 led to selecting the assessment model with this ageing error option turned off. The option also remains off in the current 2026 configurations.

Mean weight-at-age is required for all fishing fleets and biomass indices in order to relate biomass quantities to the underlying model estimates of jack mackerel abundance (in numbers). The four weight-at-age matrices for the fishing fleets correspond to: Fleet 1 (northern Chile), Fleet 2 (south-central Chile), Fleet 3 (the far north fleet) and Fleet 4 (the offshore trawl fleet). These values are shown in Table 23, Table 24, Table 25, and Table 26. Historically, missing weight-at-age data were replaced with data from the previous year. In SCW14, it was recommended that those missing data be replaced with appropriate fleet-specific mean values from the previous five years instead. This procedure was adopted in SC11.

For the Chilean fleets, the mean weight-at-age was calculated by year by taking the mean length-at-age in the catch and a length-weight relationship derived for the year. Before SC3, the same weight-at-age matrix was used for the Northern Chilean Fleet (Fleet 1) and the Southern Chilean Fleet (Fleet 2). Beginning in SC3, a weight-at-age matrix specific for Northern Chile was applied. The method used two information sources: the length-age keys and the parameters of the weight-at-length relationship from IFOP's monitoring programme of the Chilean fisheries. The information was separated into two zones which correspond to fishing areas (and acoustic surveys) that occur in Chile. Annual weight-at-length relationship was fitted to the data by each fleet independently, and these relationships were applied to mean length-at-age within each zone, resulting in the weights-at-ages seen in Table 23 and Table 24. The information covers the period 1974–2026; for earlier years the weight-at-age from 1974 was used.

For the far north fleet, biological sampling data were collected by length and converted post-hoc to weights-at-age. Weight-length parameters for the von Bertalanffy weight-length curve (Bertalanffy, 1957) were estimated annually from these biological sampling data. The calculated weight-length parameters were then converted to weight-at-age by applying a mean length-at-age vector, calculated using a fixed (non-annually-varying) von Bertalanffy growth curve ($L_{\infty} = 80.77$, $t_0 = -0.356$, $k = 0.155$, $L_0 = 15.31$). These growth parameters are different from the ones used in the assessment Table 21, as they are based on Total Length, whereas the assessment uses parameters based on Fork Length. A procedure was developed (and adopted in SC11) to convert the mean length-at-age vector to a mean weight-at-age vector for the given year Table 25. This procedure was designed to better reflect the dynamics of the far north fishery, and to match the assessment model used for the Peruvian Area of National Jurisdiction (ANJ). Prior to SC11, the far north fleet weights-at-age for 2015–2022 were fixed at historical mean values.

The weights-at-age for the offshore fleet were derived from EU and Russian age-length keys as well as age-length keys from the south-central Chilean fleet. The EU and Russia reported both age, length, and weight data, allowing for weight-at-age to be reported for their catches based on observer programme data compiled in 2019. For China, Vanuatu and Korea, length-weight information is transformed using quarter-specific age-length keys. For most countries in this fleet, weight-at-length information is available. In some years, however, including 2018, weight-at-length data from the Chinese fleet were missing, which resulted in the use of the length-weight relationship from the south-central Chilean fleet. As of SCW14, due to the update in the Chilean ageing criteria, these weight-at-age data were updated for the time series beginning in 2015 (Table 26).

In SCW14, the [Natural Mortality Tool](#) was used to derive values of M range from roughly 0.1 to 0.35 with a mode at 0.28. The L_{∞} was assumed to be 80.4cm, k was assumed at 0.16 and t_0 at -0.356 . The value of 0.28 was used for the assessment beginning in SC10. In the current model, M is fixed at 0.28 yr^{-1} for the single stock and for the southern component under the two-stock hypothesis, and at 0.33 yr^{-1} for the Far North component. Each value is constant over ages and years within its component (Table 21).

For the 2026 working model, the 2025 and 2026 Offshore fleet weights at age are replaced by the **2020–2024 mean**. This is the five-year period implemented in the update script and reproduced in the current input files. The associated CPUE weights are updated consistently. The resulting inputs are shown in Table 26 and Figure 12.

3 The assessment model

3.1 Population dynamics and estimation

The Joint Jack Mackerel Model (JJM) is an age-structured statistical catch-at-age model implemented in AD Model Builder. It combines catches, age and length compositions, biological inputs and abundance indices to estimate recruitment, stock size and fishing mortality. The annual model covers 1970–2026 and ages 1–12+, with a plus group at age 12. Recruitment occurs at age 1. The current model evaluates spawning biomass 9.5 months into the year (spawning-month input 10.5), approximately mid-October; this model timing differs from the seasonal biological description above. The population dynamics use natural mortality and fleet-specific fishing mortality, with a Beverton–Holt stock–recruitment relationship and productivity regimes. Model equations and definitions are summarized in Table 28 and Table 29; the full technical background is provided by the [JJM user guide](#) (Beverton & Holt, 1957; Fournier et al., 2012).

Catch and abundance-index observations enter through lognormal components. Age and length compositions enter through multinomial components scaled by their effective sample sizes. Priors and smoothing penalties constrain biological, recruitment and selectivity parameters. The current index uncertainties and composition sample sizes are derived from the input data, with annual coverage adjustments where specified; a constant CV description would therefore be inappropriate for all series. Parameter estimation proceeds in phases. The saved mode, retrospective analyses and separate MCMC exploration provide different types of diagnostic evidence (Francis, 2011).

The single-stock hypothesis (h1) models one shared stock. The two-stock hypothesis (h2) separates southern and Far North components. Both remain in the assessment programme. The FarNorth fishery and its indices inform the northern component in h2; the North Chile, South-Central Chile and Offshore fisheries inform the southern component. Comparing the two hypotheses helps evaluate stock-structure uncertainty.

3.2 Development from the benchmark to SC14

The workshop model labels and annual assessment labels refer to different stages. SC13 Model 1.14 was the workshop starting model (0.00). Benchmark development led to Model 0.14; post-benchmark Model 0.15 split Peruvian nominal CPUE into two series, and Model 0.16 combined the Far North productivity regimes under h2. The h1 biological and data specifications were unchanged between 0.15 and 0.16. The development files show DEPM removed by workshop Model 0.12 and the Peruvian acoustic availability indices removed by 0.14. The annual SC14 Model 0.00 starts from the post-benchmark Model 0.16 configuration ([Model Developments, 2026](#)).

The June model review considered reduced FarNorth selectivity parameterization and a differentiable penalty discouraging implausibly high mean fishing mortality at ages 2–5. The annual 2026 assessment subsequently incorporated new data and model sensitivities. Improvements in total objective function include changes in penalties, so they require examination alongside the direct data likelihoods ([SPRFMO Jack Mackerel Working Group, 2026b](#)).

The SC14 data bridge (Models 0.01–0.10) adds updated catches, fishery compositions, fishery weights, Offshore and Chilean CPUE, Peruvian CPUE, and the 2026 North Chile acoustic data. Model 1.00 updates the control settings for the new terminal year. The subsequent comparisons are:

- **1.01:** remove pre-2006 North Chile acoustic observations.
- **1.02 and 1.03:** evaluate the Chilean fishing-trip CPUE series (SC14-JM15), including a version excluding its 2026 value. These are sensitivity alternatives.
- **1.04:** continue from 1.01 and replace 2025–2026 Offshore weights at age with the 2020–2024 mean.
- **1.05:** add 22,799 t of 2025 Chilean high-seas catch to North Chile.
- **1.06:** add 10,000 t to FarNorth catch in 2026; this is the current working base under both stock hypotheses.

The complete bridge is shown in Table 35. Where older run descriptions differ from the implemented changes, this annex follows the current script and input files, particularly for the Offshore weight averaging period and the FarNorth catch addition (*Jack Mackerel Assessment SC14, 2026*).

3.3 Reference points and projection conventions

Three reference-point constructions need to remain explicit. The model estimates annual equilibrium MSY quantities from biological, selectivity and recruitment assumptions. The assessment convention for the fixed biomass target is the mean of annual model-estimated B_{MSY} over the most recent ten years, **2017–2026** in this update. The status table and assessment summary figures below use that fixed mean; the annual F/F_{MSY} values use their corresponding annual model estimates.

A separate dynamic biomass reference can describe stock status relative to the unfished biomass implied by the estimated recruitment history:

$$B_{MSY,y}^{dynamic} = \left(\frac{B_{MSY}}{B_0} \right) B_{0,y}^{dynamic}.$$

Here $B_{0,y}^{dynamic}$ is unfished spawning biomass in year y under the corresponding recruitment history. The chosen B_{MSY}/B_0 ratio and its reference period are part of the definition. For MCMC-based calculations, the ratio and dynamic unfished series are paired **within the same draw**. This construction is separate from the fixed ten-year biomass target used in the status and risk tables.

The canonical annual-report source contains a call to `fixed_bmsy(dyn = TRUE)`, whereas the installed `jjmR` function used for the saved annex supports the fixed ten-year convention. A separately repaired report snapshot calculates the dynamic diagnostic directly. Reconciliation of the canonical source remains a follow-up item; the numerical status table here uses the explicitly labelled fixed reference. The exploratory MCMC Kobe display has its own reference definition and validation status.

The saved calculations retain separate annual, 15-year and three-completed-year selectivity paths. The annual MSY series used in the fixed 2017–2026 biomass target uses each year’s selectivity. The `jjm.tpl` version used for the 11 September projections reports global MSY with the 15-year mean (2011–2025), while projection scenario 4 both optimizes and applies a separate F_{MSY} using `sel_fut` (2023–2025 when `fut_sel = 3`). The ten-year risk tables use those saved projections and their recalculated ADMB standard errors. These paths and their numerical values are distinguished in Section 3.3.1.

The saved low-productivity projection variants (`h1_1.06.ls` and `h2_1.06.ls`) use steepness **0.65**. The recruitment period is **2001–2015** for the single stock and southern component, and **1970–2016** for the Far North component. Their projection figures are conditional scenarios. Catch advice requires reconciliation of the scenario definitions, fleet allocation, recruitment and selectivity assumptions, and final agreed reference points.

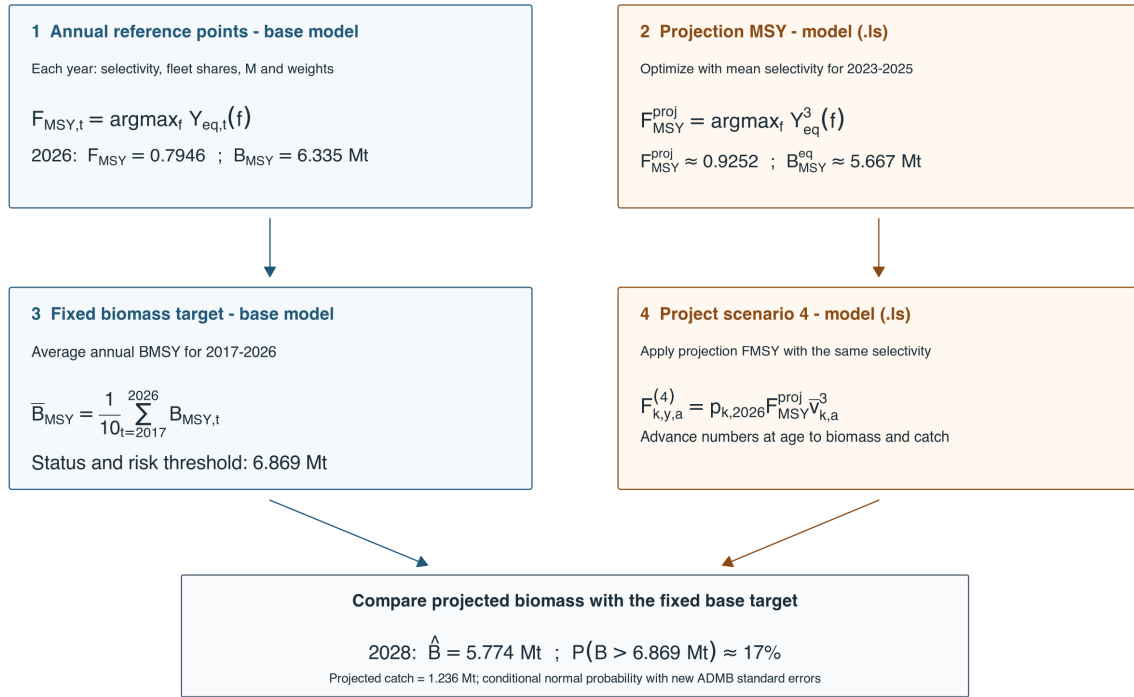
3.3.1 Reference-point and projection diagram

Figure 1 separates the reference values used to describe stock status from the scalar fishing rate used in projection scenario 4. The **single-stock example** uses the base Model 1.06 for status and its `.ls` variant for the saved low-productivity projections. The same calculations for all three stock components are tabulated below. The supplied 2022 Annex K guidance adds the catch-decision step described in Section 3.3.6.

Saved source and projections, 11 September 2026: the `jjm.tpl` version identified in the [run manifest](#) reports global MSY using the default `sel_msy` average for 2011–2025, and separately optimizes `fmsy_proj` using `sel_fut` for 2023–2025. Scenario 4 applies that same future selectivity when calculating mortality, spawning biomass and catch. These ten-year projections and their ADMB standard errors were recalculated at the saved assessment parameter values. The fixed base-model biomass targets retain their original annual-average definition. Later template revisions and executable changes require their own run validation before their results can replace this saved set.

Reference points and projections in jjm.tpl

2026 single-stock example | Biomass = spawning biomass in million tonnes (Mt)



Separate global reference, using 2011-2025 selectivity; base BMSY = 6.546 Mt; precautionary (.ls) BMSY = 5.750 Mt.
Corrected jjm.tpl, 11 September 2026. Fitted parameters retained; projections and standard errors extend through 2036.

Figure 1: Reference points and projection scenario 4 in the saved 11 September calculations. The fixed base biomass target comes from annual reference points. Scenario 4 calculates and applies FMSY with the same future selectivity; the global fifteen-year reference is reported separately.

3.3.2 Reference values

Table 2: Annual base references from `msy_mt` columns 5 and 10, and the fixed ten-year mean of column 10.

Base model / component	Annual 2026 FMSY	Annual 2026 BMSY (Mt)	Fixed 2017–2026 BMSY (Mt)
h1_1.06 / Single stock	0.7946	6.335	6.869
h2_1.06 / South	0.8854	6.074	6.524
h2_1.06 / Far North	0.1109	1.309	1.305

The scalar values below are named outputs from the saved 11 September `jjm.tpl` runs. Independent equilibrium calculations reproduce the global and projection optima for all six model/component combinations. The associated validation checks found that changing `fut_sel` between 0, 3 and 15 leaves the global reference unchanged; setting both windows to 15 gives matching global and projection optima for these constant-M configurations. [Reference values and replacement diagnostics](#) retain the native `kt` values.

Table 3: Global and projection equilibrium reference points from the saved 11 September runs. The `.ls` projection FMSY values feed scenario 4. All scalar BMSY values remain separate from the fixed **base-model annual-average** risk thresholds.

Run / component	Global FMSY (15-year)	Global BMSY (Mt; 15-year)	Projection FMSY (3-year)	Projection equilibrium BMSY (Mt; 3-year)
h1_1.06 / Single stock	0.3916	6.546	0.9128	6.434
h2_1.06 / South	0.3855	6.040	0.9502	6.205

Run / component	Global FMSY (15-year)	Global BMSY (Mt; 15-year)	Projection FMSY (3-year)	Projection equilibrium BMSY (Mt; 3-year)
h2_1.06 / Far North	0.0955	1.314	0.1120	1.293
h1_1.06.ls / Single stock	0.3932	5.750	0.9252	5.667
h2_1.06.ls / South	0.3874	5.096	0.9751	5.254
h2_1.06.ls / Far North	0.0955	1.314	0.1120	1.293

All configurations use Beverton–Holt steepness 0.65. The base single and southern stocks fit recruitment over 2000–2023; their .ls variants use 2001–2015. The Far North stock uses 1970–2016 in both. This changes productivity as well as preserving the distinction between reference-point selectivity windows.

3.3.3 Equations behind the diagram

Let $T = 2026$, k index fleets belonging to stock s , $a = 1, \dots, A$ index ages, and $A = 12+$ denote the plus group. The model's B is **spawning biomass**. Biomass and catch in the native report are in thousand tonnes (kt); the tables here divide biomass by 1,000 to show million tonnes (Mt). F and M are instantaneous annual rates. Selectivity v retains the model's age scaling, so the reported F is the sum of fleet means over model ages, rather than a fully selected maximum.

The fleet allocation and trial fishing mortality are

$$p_{k,t} = \frac{\sum_a F_{k,t,a}}{\sum_{k \in s} \sum_a F_{k,t,a}}, \quad F_{k,a}(f) = p_{k,t} f v_{k,a}, \quad Z_a(f) = M_a + \sum_{k \in s} F_{k,a}(f). \quad (1)$$

For one recruit, the equilibrium age distribution, spawning biomass per recruit and yield per recruit are

$$\begin{aligned} \ell_1(f) &= 1, \\ \ell_a(f) &= \exp \left[- \sum_{j=1}^{a-1} Z_j(f) \right] \quad (a < A), \\ \ell_A(f) &= \frac{\exp[-\sum_{j=1}^{A-1} Z_j(f)]}{1 - \exp[-Z_A(f)]}, \\ \phi(f) &= \sum_a \ell_a(f) e^{-\tau Z_a(f)} w_a^{\text{mat}}, \\ y(f) &= \sum_{k \in s} \sum_a w_{k,a}^{\text{fsh}} \ell_a(f) \frac{F_{k,a}(f)}{Z_a(f)} (1 - e^{-Z_a(f)}). \end{aligned} \quad (2)$$

Here w_a^{mat} is population weight multiplied by maturity, and $\tau = (10.5 - 1)/12 = 9.5/12$ is the spawning fraction used by these runs. The active Beverton–Holt form (`SrType = 2`) is

$$\begin{aligned} R(B) &= \frac{B}{\alpha + \beta B}, \\ \alpha &= \frac{B_0(1-h)}{4hR_0}, \\ \beta &= \frac{5h-1}{4hR_0}, \\ R_{\text{eq}}(f) &= \frac{\phi(f) - \alpha}{\beta \phi(f)}, \\ Y_{\text{eq}}(f) &= R_{\text{eq}}(f) y(f), \\ B_{\text{eq}}(f) &= R_{\text{eq}}(f) \phi(f). \end{aligned} \quad (3)$$

Thus F_{MSY} seeks the maximum of $Y_{\text{eq}}(f)$ and $B_{\text{MSY}} = B_{\text{eq}}(F_{\text{MSY}})$. In the saved template version, the annual report searches 501 logarithmically spaced trial rates from 0.0001 to 5 and retains feasible positive equilibria; the global scalar calculation uses eight Newton updates. The projection calculation locates a feasible interior maximum on a 101-point logarithmic grid and refines it with eight differentiable Newton updates, using the same equilibrium function for optimization and final evaluation. `Requil()` evaluates equilibrium recruitment without a $\exp(\sigma_R^2/2)$ multiplier.

3.3.4 Which quantities are used

Calculation	Selectivity and biological inputs	Use
Annual MSY series	Each year's selectivity, fleet shares, natural mortality, fishery weights and stock-recruit regime	<code>msy_mt</code> : annual $F_{\text{MSY},t}$ and $B_{\text{MSY},t}$
Fixed biomass target	Arithmetic mean of annual base-model $B_{\text{MSY},t}$ for 2017–2026	Stock-status summary and the restored projection-risk tables
Global scalar MSY in <code>jjm.tpl</code>	<code>sel_msy</code> : mean selectivity for 2011–2025 when <code>msy_sel</code> = 15; terminal fleet shares and fishery weights, initial-year M , terminal stock-recruit regime	Reports <code>Fmsy_global</code> , <code>Bmsy_global</code> and <code>MSY_global</code> independently of projection selectivity
Projection MSY and scenario 4	<code>sel_fut</code> : mean selectivity for 2023–2025 when <code>fut_sel</code> = 3; terminal fleet shares, weights and M	Optimizes <code>Fmsy_proj</code> and applies it to future numbers at age with the same <code>sel_fut</code>

The fixed target and annual fishing-mortality ratio are

$$\bar{B}_{\text{MSY}}^{\text{base}} = \frac{1}{10} \sum_{t=2017}^{2026} B_{\text{MSY},t}^{\text{base}}, \quad \frac{F_T}{F_{\text{MSY},T}} = \frac{\sum_{k \in s} \text{mean}_a F_{k,T,a}}{F_{\text{MSY},T}^{\text{annual}}}. \quad (4)$$

The future-selectivity averages and scenario-4 fishing rates are

$$\bar{v}_{k,a}^{15} = \frac{1}{15} \sum_{t=2011}^{2025} v_{k,t,a}, \quad \bar{v}_{k,a}^3 = \frac{1}{3} \sum_{t=2023}^{2025} v_{k,t,a}, \quad F_{k,y,a}^{(4)} = p_{k,T} F_{\text{MSY}}^{\text{proj}} \bar{v}_{k,a}^3. \quad (5)$$

With $Z_{y,a} = M_{T,a} + \sum_k F_{k,y,a}$, future biomass and catch follow the population forward from the fitted terminal age structure:

$$\begin{aligned} N_{y,1} &= R(B_{y-1})e^{\epsilon_y}, \\ N_{y+1,a+1} &= N_{y,a}e^{-Z_{y,a}} \quad (a < A-1), \\ N_{y+1,A} &= N_{y,A-1}e^{-Z_{y,A-1}} + N_{y,A}e^{-Z_{y,A}}, \\ B_y &= \sum_a N_{y,a}e^{-\tau Z_{y,a}} w_a^{\text{mat}}, \\ C_y &= \sum_{k \in s} \sum_a w_{k,T,a}^{\text{fsh}} N_{y,a} \frac{F_{k,y,a}}{Z_{y,a}} (1 - e^{-Z_{y,a}}). \end{aligned} \quad (6)$$

The recruitment lag is one year in these runs. The terminal stock-recruit regime supplies $R(\cdot)$ and `rec_dev_future` supplies ϵ_y . The population equations produce the finite trajectories `SSB_fut_4` and `Catch_fut_4`. The annex risk calculation compares the projected biomass distribution with the separate fixed base-model target, $P(B_y > \bar{B}_{\text{MSY}}^{\text{base}})$.

Scenarios 1–3 apply 1, 0.75 and 1.25 times the terminal fleet mean F with `sel_fut` when `fut_sel` > 0; scenario 6 sets $F = 0$. Scenario 5 uses the supplied TAC input and a **terminal-year selectivity** path distinct from scenario 4. The realized catches in the saved scenario differ from the supplied TAC; Section 4.2 records those values and their interpretation.

3.3.5 Source and interpretation checks

The equations above follow the 11 September `jjm.tpl` version's functions `get_future_sel`, `get_msy`, `projection_equilibrium`, `get_projection_msy`, `get_future_Fs`, `Future_projections`, `get_msy_robust`, `SRecruit` and `Requil`. The [validation record](#) identifies the source hash and independent numerical checks, including all ten years of scenario-4 biomass and catch. Earlier five-year and intermediate scenario-4 validation records describe superseded calculations; the ten-year manifest and its matching validation define the results presented here.

The saved base Model 1.06 assessment results used `jjm2`. The 11 September ten-year projections used a separately compiled `jjm` executable; their annual `msy_mt`, historical numbers at age and fitted fishing mortality reproduce the saved assessment values. The annual assessment script subsequently changed to call `jjm`. This document retains the saved outputs and the exact projection version identified by the manifest; a later executable path alone leaves their numerical provenance unchanged. The earlier `jjm2.tpl` replacement calculation is retained as historical context.

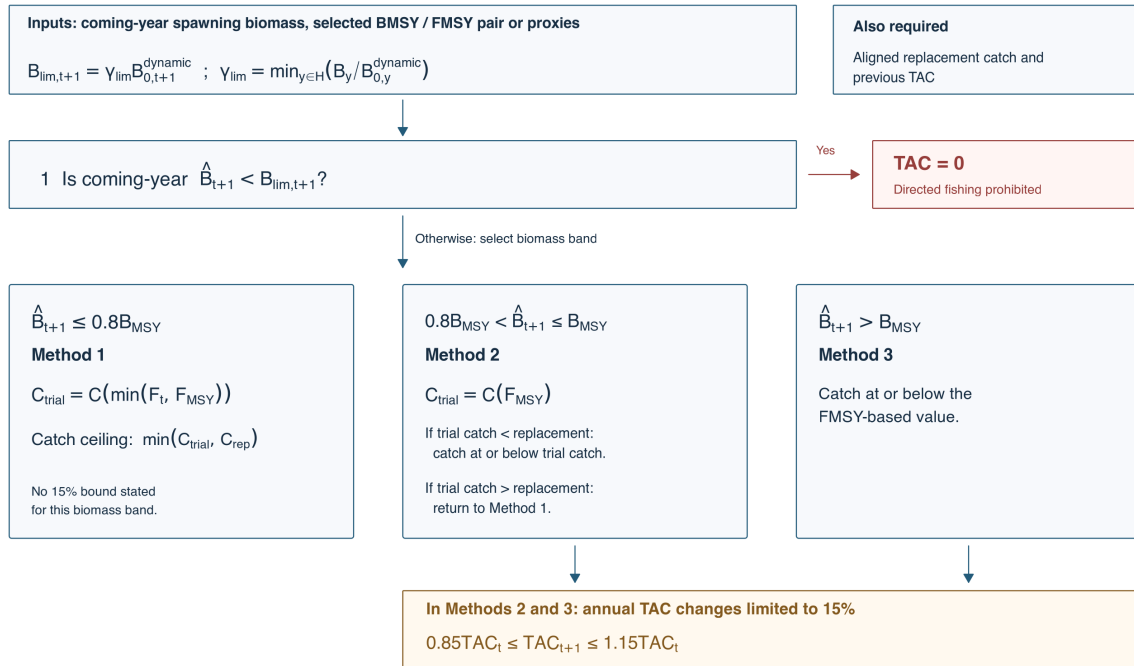
`Sp_Biom_NoFish` supports a separate dynamic unfished-biomass series. Multiplying that series by a selected B_{MSY}/B_0 ratio constructs a dynamic reference for comparison with biomass. The fixed target and scenario-4 population equations above retain their own definitions.

3.3.6 Applying the supplied 2022 Annex K guidance

The supplied Annex K proposal describes the jack mackerel harvest control rule as adjusted during SCW14 ([Proposed Jack Mackerel Harvest Control Rule, 2022, pp. 1–2](#)). It adds a **catch-decision step** to the reference-point and projection calculations above. The document records a proposal; its application in 2026 requires the agreed management record and an explicit reference-point convention.

2022 Annex K proposal: from model outputs to TAC

Decision layer supplied by SCW14 guidance; the saved projection scenarios do not execute this full rule.



Source: Proposed Jack Mackerel Harvest Control Rule (2022), Table 1 and explanation, pages 1-2.

Exact catch equality and conflicts between a biological ceiling and the 15% lower bound require an explicit implementation decision.

Figure 2: The 2022 proposal first compares coming-year spawning biomass with a dynamic limit, then uses three biomass bands to choose trial catch and replacement-yield safeguards. The upper two bands also limit annual TAC changes to 15%.

Decision year and biomass definition. All biomass quantities in this guidance are spawning biomass. The decision table uses estimated biomass in the **coming year**, $\hat{B}_{t+1}$: for a 2026 assessment this is $\hat{B}_{2027}$, rather than the terminal estimate

B_{2026} . The fishing assumption that generates that status forecast must accompany it. The saved scenario-4 forecast is one conditional forecast, rather than a uniquely specified HCR status input.

Dynamic limit. The proposal defines the biomass limit from the lowest historical ratio of spawning biomass to its corresponding unfished value:

$$\gamma_{\text{lim}} = \min_{y \in \mathcal{H}} \left(\frac{B_y}{B_{0,y}^{\text{dynamic}}} \right), \quad B_{\text{lim},t+1} = \gamma_{\text{lim}} B_{0,t+1}^{\text{dynamic}}. \quad (7)$$

Here $\mathcal{H}$ is the explicitly identified historical period. Unfished spawning biomass follows the estimated recruitment history, adjusted through the stock–recruit relationship under no fishing. In the saved template these quantities correspond to `Sp_Biom_NoFish` and `Sp_Biom_NoFishRatio`; the future no-fishing calculation supplies the coming-year denominator. This limit differs from both the fixed ten-year B_{MSY} target and the dynamic- B_{MSY} construction. An 8% limit follows only when the selected historical minimum equals 0.08. Calculating a numerical B_{lim} requires a matched historical minimum and coming-year unfished biomass.

Trial catch and biomass bands. Write $C_{t+1}(f)$ for the projected catch under trial fishing rate f , using a stated fleet allocation, selectivity, recruitment and weight convention. Define

$$C_1 = \min \left\{ C_{t+1}[\min(F_t, F_{\text{MSY}})], C_{t+1}^{\text{replacement}} \right\}, \quad C_F = C_{t+1}(F_{\text{MSY}}). \quad (8)$$

C_1 is the catch ceiling described by method 1. The comparison of F_t with F_{MSY} requires a common age and selectivity scaling. The annual F_{MSY} , scalar F_{MSY} and associated biomass references documented above should therefore retain their definitions when selecting the HCR inputs.

Table 4: The 2022 proposal’s biomass bands and catch calculations, with its first-row limit check given precedence over the overlapping low-biomass row. The source permits an identified BMSY/FMSY proxy.

Coming-year spawning-biomass condition	Catch guidance in the supplied proposal	TAC stability provision
$\widehat{B}_{t+1} < B_{\text{lim},t+1}$	Set TAC to zero; directed jack mackerel fishing prohibited under the proposed rule.	Closure is checked first.
Remaining cases with $\widehat{B}_{t+1} \leq 0.8B_{\text{MSY}}$	Trial catch at $\min(F_t, F_{\text{MSY}})$; catch at or below C_1 .	Stability bound unspecified for this row.
$0.8B_{\text{MSY}} < \widehat{B}_{t+1} \leq B_{\text{MSY}}$	Trial catch at F_{MSY} . If $C_F < C^{\text{replacement}}$, catch at or below C_F ; if $C_F > C^{\text{replacement}}$, return to method 1.	Annual TAC variation limited to 15%.
$\widehat{B}_{t+1} > B_{\text{MSY}}$	Catch at or below the value obtained using F_{MSY} .	Annual TAC variation limited to 15%.

The table places equality at 80% B_{MSY} in the lower band and equality at B_{MSY} in the middle band. The exact trial-catch/replacement-catch equality case remains an implementation choice to resolve. In the upper two bands, the stated annual stability interval is

$$0.85 \text{ TAC}_t \leq \text{TAC}_{t+1} \leq 1.15 \text{ TAC}_t. \quad (9)$$

That interval and the biological catch ceiling can conflict if the ceiling falls below 0.85 TAC_t . Reconciling these provisions requires an explicit implementation order, which remains unspecified in the proposal. The upper 15% bound is a constraint on annual variation.

For orientation, applying the **annex’s working fixed base-model B_{MSY} convention** gives the following 80% and 100% thresholds. Operational classification requires the agreed Annex K reference and a stated coming-year biomass forecast.

Table 5: Illustrative biomass bands using the fixed 2017–2026 mean from each base model.

Component	80% of fixed BMSY (Mt)	Fixed BMSY (Mt)
Single stock	5.495	6.869
South	5.219	6.524
Far North	1.044	1.305

Replacement-catch timing. The proposal defines replacement yield using consecutive spawning years: catch in year y leaves B_{y+1} equal to B_y . For a proposed catch in 2027, the aligned condition is

$$B_{2028}(C_{2027}^{\text{replacement}}) = B_{2027}, \quad (10)$$

with recruitment and fishing mortality during both spawning periods stated. The saved 11 September `jjm.tpl` version solves this consecutive-year condition with the projection `SRR`, `sel_fut`, terminal fleet shares and the same trial `F` in 2027 and 2028. Under the precautionary `.ls` configurations, replacement catches are 1,104.7 kt for the single stock, 895.5 kt for the southern component and 15.7 kt for Far North. All three attain a root. The earlier `jjm2.tpl` diagnostics used mean recruitment and a $B_{2028} = B_{2026}$ target; their saved values remain part of the original assessment output history.

What the saved projections implement. Scenario 4 applies a scalar F_{MSY} ; scenario 5 uses a supplied TAC input, with realized catches qualified in Section 4.2. A complete Annex K calculation additionally requires the dynamic-limit test, the 80% and 100% biomass branches, replacement-catch comparisons and annual 15% TAC constraints. A source-code comment mentions a 15% increase; implementing the stability provision requires an explicit calculation. The annex risk tables give $P(B_y > \bar{B}_{\text{MSY}}^{\text{base}})$; the limit-breach probability $P(B_y < B_{\text{lim},y})$ requires a separate analysis.

An operational comparison therefore needs a documented coming-year status forecast; the selected $B_{\text{MSY}}/F_{\text{MSY}}$ pair or proxies and their scaling; a matched dynamic limit; the stated replacement assumptions; the previous TAC; and explicit handling of the stability conflict, equality cases and stock-component allocation. These choices require reconciliation with the agreed management record. The 2022 proposal provides source guidance alongside the assessment and saved projection calculations.

The rule follows the supplied proposal (*Proposed Jack Mackerel Harvest Control Rule, 2022*, Table 1, pp. 1–2). The assessment calculations are traced through the saved `jjm.tpl` version’s functions `Calc_Dependent_Vars`, `Future_projections`, `get_future_Fs`, `repl_ssb` and `Get_Replacement_Yield`. The projection run manifest identifies the exact template hash and input settings.

4 Results

4.1 Current working stock estimates

The current estimates are given in Table 6. SSB and B_{MSY} are expressed in million tonnes (Mt); the underlying model biomass output is in thousand tonnes. The 2026 SSB estimate for the single-stock model is about 6.11 Mt, compared with 6.03 Mt in 2025. Its ratio to the fixed 2017–2026 mean B_{MSY} is about 0.89. The two-stock estimates differ, particularly for the Far North component. These are conditional estimates from the working configurations.

Table 6: Working 2026 assessment estimates. Mean BMSY is the fixed 2017–2026 mean of annual model estimates. F/FMSY is the 2026 annual ratio. These values use saved Model 1.06 results and retain preliminary assessment status.

Model	Component	SSB 2025 (Mt)	SSB 2026 (Mt)	Mean BMSY (Mt)	SSB / mean BMSY	F / FMSY
h1_1.06	Single stock	6.031	6.109	6.869	0.889	0.861
h2_1.06	South	6.529	6.661	6.524	1.021	0.705
h2_1.06	Far North	4.208	4.046	1.305	3.101	0.205

The North Chile acoustic-index decline from approximately 3.06 Mt in 2025 to 0.64 Mt in 2026 differs from the single-stock model’s slight increase in SSB. The survey measures a regional signal and has its own spatial coverage, timing

and availability effects. ENSO and changes in distribution remain hypotheses to examine alongside sampling and fishery behaviour. Inferring whole-stock biomass change requires integrating this regional signal with the other surveys, fishery data and model diagnostics. The model fits and index-influence analyses show how these observations affect the integrated assessment (*Jack Mackerel Assessment SC14, 2026*).

4.2 Projection risk summary

Table 7, Table 8 and Table 9 summarize the six fishing scenarios from the saved low-productivity projection runs. Spawning biomass (B) and catch are in **thousand tonnes (kt)**. The tables show biomass and the approximate probability of exceeding B_{MSY} at **2, 5 and 10 years after 2026: 2028, 2031 and 2036**, plus catches in 2027–2028. The projections cover 2027–2036 and were recalculated on **11 September 2026**, including ADMB standard errors, with the `j.jm.tpl` version identified in the run manifest. Later source revisions are outside this saved result set.

The biomass threshold is the **fixed 2017–2026 mean B_{MSY} from the corresponding base Model 1.06**, consistent with Table 6. All three components use steepness 0.65. The single stock and southern component use the 2001–2015 recruitment period; the Far North component retains 1970–2016. Probabilities follow the existing JJM risk-table method, using the projected biomass estimate and its reported ADMB standard error:

$$P(B_y > B_{MSY}) \simeq 1 - \Phi \left(\frac{B_{MSY} - \hat{B}_y}{SE(\hat{B}_y)} \right).$$

These are conditional normal approximations with the reference point held fixed. They measure exceedance in each stated year. The probability of remaining above a threshold throughout a period requires a separate calculation, as do the exploratory MCMC summaries.

Saved TAC scenario. Scenario 5 retains the 11 September implementation and its realized catches. With the supplied input of **1,674.975 kt**, the saved 2027 catch is **1,491.6 kt** under h1 and **1,656.65 kt** in total under h2 (1,489.38 kt southern and 167.27 kt Far North). These catches differ from the input and vary over the projection. This archived scenario therefore describes its recorded catches; use as a projection of a specified annual TAC requires a separately validated run that achieves that TAC in each intended year. The subsequent catch-solver revision is outside the saved results displayed here.

For the **1-stock table, both model runs contribute**: the ten-year extension of `h1_1.06.ls.ctrl` supplies projected biomass and its newly calculated ADMB standard error; `h1_1.06.ctrl` supplies the base-model B_{MSY} threshold through the original `h1_1.06_1_R.rep`. The precautionary run's own B_{MSY} estimate is separate from the threshold used here. For example, updated scenario 4 in 2028 gives $\hat{B} = 5774.30$ kt and $SE = 1147.25$ kt. With the base threshold 6869.029054 kt, the calculation is $1 - \Phi[(6869.029054 - 5774.30)/1147.25] = 0.169986$, displayed as **17%**. Uncertainty in the B_{MSY} threshold is excluded.

Calculation and run provenance: global MSY uses `sel_msy` (2011–2025), while scenario 4 both calculates and applies its own `Fmsy_proj` using `sel_fut` (2023–2025), the same future selectivity used in scenarios 1–3. Fitted assessment parameters were held at their saved values; future recruitment deviations were extended with zeros and the Hessian and derived standard errors were recomputed. The original fixed base biomass targets are retained. [Annual risk data](#) and the [run manifest](#) provide the values, controls, source hash and validation record. Earlier five-year validation files belong to the calculation history; the ten-year records support the tables below.

The supplied 2022 Annex K proposal (Section 3.3.6) also uses a dynamic limit, 80% and 100% biomass bands, replacement catch and specified TAC stability constraints. These tables describe fixed-BMSY exceedance under saved scenarios. Limit-breach probability and application of the complete harvest control rule require the additional definitions and calculations described in that section.

Table 7: Single-stock risk summary, `h1_1.06.ls`. B and catch are in kt; $P(B > B_{MSY})$ is the approximate annual probability that B exceeds fixed $B_{MSY} = 6869$ kt. Probabilities are rounded to whole percentages; values below 1% or above 99% are shown as <1 or >99.

Fishing scenario	B 2028	$P(B > B_{msy})$ in 2028	B 2031	$P(B > B_{msy})$ in 2031	B 2036	$P(B > B_{msy})$ in 2036	Catch 2027	Catch 2028
F = 0	7,884	77	10,880	96	14,110	98	0	0
$0.75 \times F_{2026}$	6,503	39	7,166	56	7,391	58	760	857

Fishing scenario	B 2028	P(B > B _{msy}) in 2028	B 2031	P(B > B _{msy}) in 2031	B 2036	P(B > B _{msy}) in 2036	Catch 2027	Catch 2028
1 × F2026	6,188	30	6,541	43	6,603	46	964	1,026
1.25 × F2026	5,910	23	6,024	32	5,975	34	1,152	1,169
FMSY scenario	5,774	17	5,782	26	5,688	28	1,247	1,236
TAC scenario (supplied TAC)	5,368	14	4,594	15	3,409	14	1,492	1,467

Table 8: Southern-component risk summary, h2_1.06.ls. B and catch are in kt; P(B > B_{MSY}) is the approximate annual probability that B exceeds fixed B_{MSY} = 6524 kt. Probabilities are rounded to whole percentages; values below 1% or above 99% are shown as <1 or >99.

Fishing scenario	B 2028	P(B > B _{msy}) in 2028	B 2031	P(B > B _{msy}) in 2031	B 2036	P(B > B _{msy}) in 2036	Catch 2027	Catch 2028
F = 0	8,439	90	10,944	98	13,275	98	0	0
0.75 × F2026	7,024	64	7,252	65	7,198	61	735	818
1 × F2026	6,701	55	6,677	53	6,524	50	934	973
1.25 × F2026	6,416	47	6,203	43	5,985	40	1,117	1,103
FMSY scenario	6,046	34	5,624	29	5,342	27	1,368	1,262
TAC scenario (supplied TAC)	5,742	30	4,377	16	2,111	8	1,489	1,492

Table 9: Far North-component risk summary, h2_1.06.ls. B and catch are in kt; P(B > B_{MSY}) is the approximate annual probability that B exceeds fixed B_{MSY} = 1305 kt. Probabilities are rounded to whole percentages; values below 1% or above 99% are shown as <1 or >99.

Fishing scenario	B 2028	P(B > B _{msy}) in 2028	B 2031	P(B > B _{msy}) in 2031	B 2036	P(B > B _{msy}) in 2036	Catch 2027	Catch 2028
F = 0	4,029	>99	4,142	>99	4,327	>99	0	0
0.75 × F2026	3,884	>99	3,750	>99	3,705	>99	86	98
1 × F2026	3,838	>99	3,632	>99	3,522	>99	113	128
1.25 × F2026	3,794	>99	3,520	>99	3,350	99	140	156
FMSY scenario	3,273	>99	2,385	97	1,703	83	489	447
TAC scenario (supplied TAC)	3,761	>99	3,482	99	3,306	97	167	167

4.3 Fits and retrospective patterns

The appended figures show fishery age and length fits, index fits, mean ages and lengths, selectivity patterns and summary trajectories for the current models. These provide the evidence for evaluating tensions among data sources. They replace the previous annex's claims about the fit of SC13 Model 1.14.

At each peel's final fitted year, the five-peel SSB Mohn's rho is **0.09** under the single-stock hypothesis, **0.3** for the southern component and **0.01** for the Far North component. The SSB figure titles use each peel's extra forecast year from the saved series, giving **0.26**, **0.43** and **-0.0086**, respectively. The two sets therefore describe different endpoints. The larger southern retrospective pattern under h2 warrants attention when comparing stock hypotheses and interpreting terminal estimates. Recruitment retrospectives are also shown. These diagnostics describe revision patterns within the fitted models; they provide a separate check from comparisons between successive annual assessments.

4.4 Leave-one-in and leave-one-out analyses

The September 2026 single-stock analyses compare the all-data base $h1_1.06$ with Models $h1_3.01$ – $h1_3.08$. The four tested index groups are Chile CPUE, Offshore CPUE, both Peruvian CPUE series together, and North Chile acoustics. LOI Models 3.01–3.04 retain one of these groups at its base weight; LOO Models 3.05–3.08 reduce the influence of each group in turn. In this implementation, exclusion means multiplying the selected index errors by 100. Catches and composition data remain in the likelihood.

LOI results retain the highest recent biomass when North Chile acoustics or the paired Peruvian CPUE series retain their base weights. Among the LOO runs, reducing the influence of Chile CPUE gives the largest increase in 2026 SSB, whereas reducing North Chile acoustics decreases it. Offshore and Peruvian CPUE exclusions have smaller effects. The comparisons diagnose sensitivity to index weighting and conflict among signals; they support discussion of the input uncertainties and model assumptions. The LOI/LOO report ([Leave-One-in and Leave-One-Out Analyses, 2026](#)) provides the common-axis comparisons and full run definitions.

4.5 Replacement-yield checks

The 11 September replacement calculation solves $B_{2028} = B_{2027}$, using the projection SRR, future selectivity, stock-specific fleet shares and the same trial fishing rate in both spawning years. It calculates catch in 2027 with the Baranov equation and retains survivors from the terminal plus group. All six model/component checks attain a root, with native reported absolute biomass residuals below 0.001 kt; independent reconstruction from rounded report values agrees within reporting precision. Under the precautionary $.1s$ assumptions, replacement catches are **1,104.7 kt** for the single stock, **895.5 kt** for the southern component and **15.7 kt** for Far North. Corresponding base-run values are **1,273.3**, **1,118.0** and **15.7 kt**. These consecutive-year results replace the earlier j_{jm2} diagnostics for a different $B_{2028} = B_{2026}$ target and mean-recruitment assumption. Those earlier diagnostics remain part of the assessment history. The consecutive-year calculation addresses the supplied Annex K replacement condition; application of the full rule still requires the other decision inputs in Section 3.3.6.

4.6 Exploratory uncertainty analysis

A separate ADNUTS analysis of $h1_1.06$ used three chains, 2,000 iterations per chain including 1,000 warmup iterations, and 3,000 retained draws. It completed without divergences or tree-depth saturation, but **366 of 1,593 active parameters** failed one or more diagnostic screens. Maximum R-hat was about 1.055, and minimum bulk and tail effective sample sizes were about 83 and 150. The derived summaries also contain diagnostic failures. Intervals, annual green probabilities and the 100 historical Kobe trajectories therefore remain exploratory.

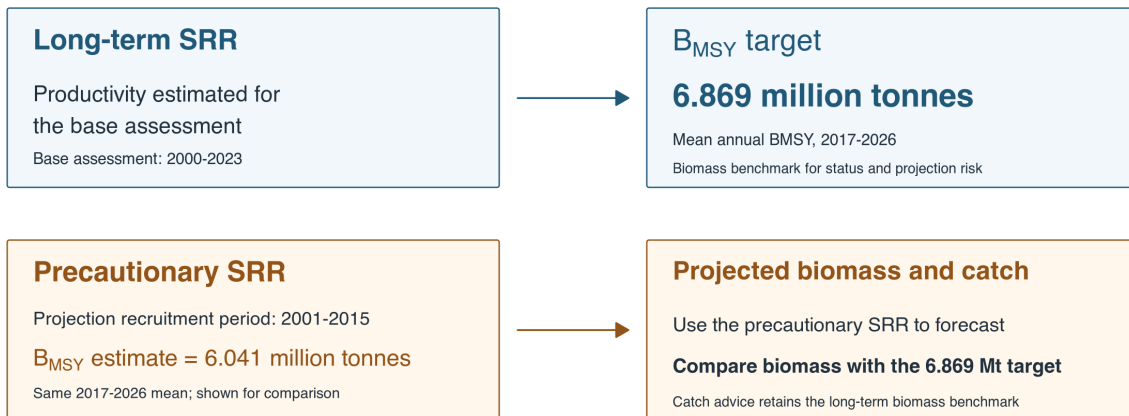
That Kobe display uses each draw's mean SSB_{MSY} for 2017–2026 and the matching annual F_{MSY} . Its biomass reference is the fixed ten-year mean. The outputs illustrate assessment uncertainty and mixing problems; definitive posterior probabilities require satisfactory sampling diagnostics ([Ianelli, 2026](#)).

5 Management advice and decision status

For the **1-stock hypothesis**, the B_{MSY} target is **6.869 million tonnes (Mt)**, based on the **long-term stock–recruitment relationship (SRR)** fitted over 2000–2023 in the base assessment. Following the assessment convention, projections use a **precautionary level of recruitment productivity**, fitted over 2001–2015, to evaluate future biomass and catch. The corresponding B_{MSY} estimate for the precautionary run is **6.041 Mt**. Both estimates use the 2017–2026 mean of annual equilibrium reference points, providing a comparison on the same averaging basis. **Projected biomass is assessed against the long-term target of 6.869 Mt.**

1-stock: a long-term target and precautionary projections

Model 1.06: the stock-recruitment relationship (SRR) has a different role in each calculation.



Both SRRs use steepness $h = 0.65$; the recruitment fitting period changes the productivity basis.

Both BMSY estimates are 2017-2026 means of annual equilibrium spawning biomass, in million tonnes (Mt).

Source: saved h1_1.06 and h1_1.06.ls outputs; the annual and scalar reference-point calculations are detailed in the text.

Figure 3: 1-stock hypothesis, Model 1.06: the long-term SRR supplies the BMSY target of 6.869 Mt; the precautionary SRR supplies biomass and catch projections. Its comparable BMSY estimate is 6.041 Mt, while projection risk is evaluated against the 6.869 Mt target. Both BMSY values are 2017–2026 means of annual equilibrium spawning biomass.

The estimates come from the saved h1_1.06 and h1_1.06.ls annual reference-point outputs (msy_mt). The 6.041 Mt estimate describes the precautionary run's equilibrium productivity; the annex retains 6.869 Mt as the target for stock status and the saved 11 September projection probabilities. The annual 2026 values and the separate scalar MSY calculation used to set the F_{MSY} projection scenario are distinguished in Section 3.3.1. Both SRRs use steepness $h = 0.65$. This carries forward the conservative recruitment convention described in the SC13 technical annex (SPRFMO Scientific Committee, 2025, para. 84) and the SC14 working assessment (Jack Mackerel Assessment SC14, 2026). The 2-stock hypothesis is discussed separately below.

This annex records the assessment evidence available through 11 September 2026. Final Committee conclusions and catch recommendations require the agreed text, date and report reference. The 2025 annex's TAC alternatives for 2026 remain part of the SC13 history (SPRFMO Scientific Committee, 2025); the present working assessment provides the updated evidence for the coming management period.

The supplied 2022 SCW14 Annex K proposal is incorporated in Section 3.3.6, including its coming-year biomass test, dynamic limit, trial/replacement-catch comparisons and specified 15% stability provisions. Its documented proposal status is retained pending the applicable agreed management record.

Advice for the coming management period requires confirmation of the annual assessment configuration, the biomass-reference convention and projection assumptions, followed by the agreed Committee catch recommendation. The recorded catches, recruitment assumptions and stock allocations should accompany any projection used in that decision.

5.1 1-stock

Under the single-stock hypothesis, h1_1.06 provides the base assessment and the long-term SRR basis for the biomass target. Its SRR uses **2000–2023**; the fixed 2017–2026 mean B_{MSY} is **6.869 Mt**. The current working stock estimates are in Table 6.

The h1_1.06.ls projections follow the precautionary SRR convention, using **2001–2015** recruitment productivity. Steepness remains **0.65** in the base and projection runs. Projected biomass and catch are summarized in Table 7, with biomass probabilities evaluated against the **base-model target**. The projection variant's own equilibrium B_{MSY} is a separate quantity.

For the supplied Annex K guidance, the decision input is a stated **2027 spawning-biomass forecast**. The corrected single-stock replacement calculation gives **1,104.7 kt** under precautionary .ls recruitment, with B_{2028} equal to B_{2027} under

the same trial F. The updated fixed-F and supplied-TAC scenarios provide projection evidence for the advice discussion; implementing the full rule also requires the dynamic limit, replacement-catch comparison and applicable TAC stability provisions described in Section 3.3.6.

5.2 2-stock

Under the two-stock hypothesis, $h2_1.06$ separates the **southern** and **Far North** components. Each component retains its own SRR, biomass target and projection results.

Southern component. The base SRR uses **2000–2023** and the precautionary $h2_1.06.1s$ SRR uses **2001–2015**, with steepness **0.65** in both. Its fixed base-model B_{MSY} target is **6.524 Mt**. The updated southern projection risks are shown in Table 8. The corrected precautionary replacement calculation gives **895.5 kt**, with a successful root for B_{2028} equal to B_{2027} .

Far North component. The base and $.1s$ configurations both retain the **1970–2016** SRR period and steepness **0.65**. Its fixed base-model B_{MSY} target is **1.305 Mt**. The component's updated projections and probabilities are shown separately in Table 9. The corrected consecutive-year replacement calculation attains a small positive catch of **15.7 kt**, replacing the earlier failed boundary diagnostic for a different target.

Applying the supplied HCR under this hypothesis requires a stated coming-year biomass forecast and matched dynamic limit for the relevant component, together with an explicit decision on how component-level criteria and fleet allocations translate into catch advice. The different recruitment assumptions and replacement diagnostics should accompany any comparison with the 1-stock hypothesis.

6 Assessment issues and follow-up

The benchmark and SC14 assessment analyses identify the following continuing priorities:

- **Abundance and availability:** explain the contrasting Chile CPUE, Offshore CPUE and North Chile acoustic signals using sampling coverage, standardization, survey footprint, targeting and environmental information. Evaluate the joint influence of the separate Peruvian CPUE series.
- **Biological and composition inputs:** review weight-at-age substitutions, length conventions, age-reading consistency and effective sample sizes, with attention to the FarNorth fleet and the recent Offshore data.
- **Stock structure and diagnostics:** assess the larger southern retrospective pattern under $h2$, resolve MCMC mixing problems, and preserve the distinction between penalized objective-function improvements and improvements to direct data fits.
- **Reference points and projections:** reconcile the dynamic- B_{MSY} implementation, retain the pairing of within-draw reference points, confirm future-selectivity and recruitment settings, review the corrected consecutive-year replacement calculation, resolve the interaction of biological catch ceilings and the 15% stability bounds, and review low-catch projection behavior.
- **Assessment and advice record:** incorporate the agreed SC14 paragraphs when available, including their qualifications and report reference; document the assessment configuration, projection scenario and reference-point conventions supporting the catch recommendation.

The benchmark report and SC14 working assessment provide the supporting discussion ([Jack Mackerel Assessment SC14, 2026](#); [SPRFMO Jack Mackerel Working Group, 2026a](#)). Completed analyses and formal Committee conclusions should each retain their source and date.

7 Assessment continuity and inherited methods

The SC13 technical annex records the assessment accepted in **2025**, including its data sources, model development and diagnostic findings. This section retains that background alongside the **2026 Model 1.06** assessment. Historical model labels and dates are stated explicitly; the current input tables, diagnostics, reference points and projections elsewhere in this annex describe the 2026 configurations ([SPRFMO Scientific Committee, 2025, pp. 1, 13–17](#)).

7.1 From the SC13 assessment to the 2026 benchmark

The 2025 assessment began from the SC12 Model 1.07 configuration, relabelled 0.00 for its data bridge. Models 0.01–0.10 progressively updated catches, compositions, weights at age and abundance indices. After the annual model update, SC13 examined the influence of the four principal abundance signals and alternative treatments of recent selectivity and composition data. Its final Model 1.14 incorporated additional 2025 selectivity flexibility for South-Central Chile and Offshore, increased weights on their 2025 age compositions, reduced influence of early North Chile acoustic observations, and revised historical weights for older FarNorth fish. This accepted SC13 configuration supplied the starting point for the 2026 benchmark (SPRFMO Scientific Committee, 2025, pp. 13–15; SPRFMO Scientific Committee, 2025, p. 42, Table 26).

The historical sequence explains why identical model numbers can describe different analyses in successive years. In SC13, Models 1.01–1.09 were mainly index-influence sensitivities. In the 2026 annual assessment, Model 1.06 identifies the current working configuration. The current model-development table records the separate 2026 progression.

SC13 also distinguished two retrospective comparisons. An analytical retrospective repeatedly removed the most recent year's data from one assessment configuration. A historical comparison assembled estimates from successive annual assessments, thereby incorporating changes in data, methods and assumptions. That distinction continues to apply to the five-peel diagnostics and the comparison of SC13 1.14, the 2026 workshop configuration and SC14 1.06 shown here (SPRFMO Scientific Committee, 2025, p. 15, paras. 74–76).

7.2 Construction and provenance of fishery compositions

Catch inputs aggregate Member submissions into four fleets defined by fishing area and gear. The composition inputs preserve a further distinction between observations processed into ages before assessment and lengths fitted directly by the assessment model. The SC13 record describes Chilean and older international length distributions converted with annual Chilean age–length keys. For more recent offshore submissions, quarterly EU, Russian or Chilean keys were applied as appropriate before aggregation. Peruvian and Ecuadorian length distributions were retained as lengths and compared with model predictions generated through the growth relationship (SPRFMO Scientific Committee, 2025, p. 3, paras. 14–16).

The offshore sampling protocol also forms part of the input history. Beginning with the 2021 data, self-sampling length distributions could supply quarters covered by self-sampling when observer sampling was absent, following the protocol described in SC10-JM03 and SC10-JM04. SC13 documented this as a supplement to observer coverage. The source and quarterly coverage of each submission therefore matter when interpreting subsequent annual revisions to offshore age compositions (SPRFMO Scientific Committee, 2025, p. 4, para. 17).

The 2022 ageing revision changed historical Chilean fishery and survey compositions and the associated biological inputs. SC13 identified consistency among ageing laboratories, older composition data, and mean weights at age as continuing review priorities. These concerns remain relevant to the 2026 input checks; the current composition and weight tables show the values actually used (SPRFMO Scientific Committee, 2025, pp. 3–4, 6–8; SPRFMO Scientific Committee, 2025, p. 18, paras. 90–91).

7.3 Historical development of abundance indices

The current 2026 CPUE specifications follow the changes described earlier in this annex. Their preceding history helps explain differences among archived assessment inputs:

- **South-Central Chile:** the SC13 account describes a trip-based GLM with year, quarter, zone and vessel hold capacity, using fishing days as effort. Earlier changes excluded trips with zero jack mackerel catch and introduced a catchability change associated with the year-2000 management changes. These describe the earlier series; the 2026 benchmark adopted a revised index and uncertainty treatment.
- **Peru:** before SC6, the industrial series was affected by shifts in fishing toward other pelagic species. The subsequent combined index used recovered artisanal and small-scale information with a gamma GAM and log link. SC13 recorded concerns about historical fleet identifiers. The 2026 separate industrial and artisanal series represent a later data and model specification.
- **Offshore:** separate fleet indices were progressively combined, first for EU and Russian data in SC6 and then across China, EU, Korea, Vanuatu and Russia in SC7. The SC13 record describes a negative-binomial GAM with a log-effort offset, temporal and environmental effects, and spatial smoothers; contracting party replaced vessel identity in the SC9 formulation. The assessment series began in 2008 (SPRFMO Scientific Committee, 2025, pp. 4–5, paras. 19–21).

SC13 documented interim efficiency corrections of 1% annually for the Chilean and Peruvian CPUE series and 2.5% for Offshore. These are historical specifications: the 2026 treatment of each series is given in the current data section. The earlier annex also identified search time as difficult to measure consistently and described weekly effort aggregation as one way of capturing more of that activity. Consequently, index revisions can reflect changes in effort definition, sampling, standardization and availability, alongside changes in abundance (SPRFMO Scientific Committee, 2025, p. 5, paras. 22–25).

Two survey records also explain the longer data history. The DEPM observations supplied a historical relative-abundance series and age compositions, while the Peruvian acoustic series represented a coastal survey footprint developed principally for anchoveta. Its later jack mackerel index used backscatter information and ended in 2013. Spatial coverage and changes in fish distribution were central qualifications in the SC13 account. Their 2026 development and removal from the annual Model 1.06 are recorded above; their historical observations remain documented in SC13 (SPRFMO Scientific Committee, 2025, p. 6, paras. 26–30; SPRFMO Scientific Committee, 2025, p. 34, Table 18).

7.4 Additional observation-model detail

The principal population equations are summarized in Table 29. Two further relationships make explicit how survey ages and fishery lengths connect to those population quantities, extending the model description in SC13 Table 20 (SPRFMO Scientific Committee, 2025, p. 36).

For a survey or index with age observations, expected sampled numbers are proportional to abundance surviving to the observation time and to index selectivity. Thus

$$p_{s,y,a}^{\text{index}} = \frac{N_{y,a} S_{s,y,a} \exp(-\Delta_s Z_{y,a})}{\sum_b N_{y,b} S_{s,y,b} \exp(-\Delta_s Z_{y,b})}. \quad (11)$$

Here Δ_s is the fraction of the year before observation. These are number proportions, so mean weight at age is absent from the expression. The currently disabled ageing-error option would apply its age-error matrix to this prediction when enabled. The formula follows the survey age-prediction calculation in both retained JJM source versions.

For the age-to-length conversion, mean length follows the recursive von Bertalanffy form, with a standard deviation proportional to mean length:

$$\begin{aligned} L_1 &= L_0, \\ L_a &= L_\infty(1 - e^{-k}) + e^{-k} L_{a-1} \quad (a > 1), \\ \sigma_a &= \text{CV}_L L_a. \end{aligned} \quad (12)$$

Let u_l^- and u_l^+ be the lower and upper boundaries of modeled length bin l , and let Φ denote the standard normal cumulative distribution function. The implemented conversion calculates finite-bin probabilities and normalizes them over the represented bins:

$$\begin{aligned} g_{l,a} &= \Phi\left(\frac{u_l^+ - L_a}{\sigma_a}\right) - \Phi\left(\frac{u_l^- - L_a}{\sigma_a}\right), \\ \Gamma_{l,a} &= \frac{g_{l,a}}{\sum_m g_{m,a}}, \\ \widehat{C}_{f,y,l} &= \sum_a \Gamma_{l,a} C_{f,y,a}. \end{aligned} \quad (13)$$

This normalization preserves predicted total numbers when age compositions are converted to length compositions. Expected length proportions then follow by dividing $\widehat{C}_{f,y,l}$ by their sum across lengths. The 2026 input convention uses 10–50 cm fork-length bins, with observations in the terminal category recorded as 50 cm plus. That data convention and the model’s normalized finite-bin probability calculation are separate parts of the treatment. Growth parameters and their stock/regime mapping follow the selected current controls.

7.5 Estimation and weighting history

The JJM estimates recruitment, annual fishing intensity, and fleet- or index-specific selectivity and catchability components through a penalized likelihood. The substantial parameter groups are annual recruitment deviations and the age and annual components of fishing mortality. Selectivity regularity penalties control variation among adjacent ages and changes over time; their active form, strength and estimation phase depend on the configuration. The 2026 selectivity-mapping and input-weight tables make these choices explicit alongside the fitted patterns (SPRFMO Scientific Committee, 2025, pp. 9–10; SPRFMO Scientific Committee, 2025, pp. 35–41, Tables 19–25).

SC13 records use of the Francis T1.8 composition-weighting method from 2018 and a subsequent weighting update at the 2022 benchmark. The 2025 update retained those benchmark weights, with specific adjustments to the partial-year South-Central Chile and Offshore age compositions. These effective sample sizes scale the composition likelihood; they differ from raw counts of sampled fish. The 2026 annual input vectors and coverage adjustments supersede the historical numerical weighting table for the calculations in this annex (SPRFMO Scientific Committee, 2025, p. 9, para. 47; SPRFMO Scientific Committee, 2025, p. 13, paras. 63, 66; SPRFMO Scientific Committee, 2025, p. 38, Table 22).

Finally, SC13 documented separate data-processing and assessment stages: preparing and raising catch compositions, assembling annual data bridges, defining sensitivities, running projections and generating results. Retaining those stages and their input vintage makes successive assessment estimates interpretable. The cumulative annex preserves that provenance while its numerical results continue to come from the saved 2026 assessment products (SPRFMO Scientific Committee, 2025, p. 14, para. 67).

References

- Bertalanffy, L. von. (1957). Quantitative laws in metabolism and growth. *The Quarterly Review of Biology*, 32(3), 217–231. <https://doi.org/10.1086/401873>
- Beverton, R. J. H., & Holt, S. J. (1957). *On the dynamics of exploited fish populations* (Vol. 19, pp. 1–533). Fisheries Investigations.
- Cerna, F. (2022). *Life history parameters of jack mackerel (Trachurus murphyi, Nichols 1920) based on the new validated ageing criteria* (Benchmark Workshop Working Document Nos. SCW14-WD04). South Pacific Regional Fisheries Management Organisation. <https://www.sprfmo.int/meetings/scientific-committee/sc-workshops/scw14-jack-mackerel>
- Fournier, D. A., Skaug, H. J., Ancheta, J., Ianelli, J., Magnusson, A., Maunder, M. N., Nielsen, A., & Sibert, J. (2012). AD Model Builder: Using automatic differentiation for statistical inference of highly parameterized complex nonlinear models. *Optimization Methods and Software*, 27(2), 233–249. <https://doi.org/10.1080/10556788.2011.597854>
- Francis, R. I. C. C. (2011). Data weighting in statistical fisheries stock assessment models. *Canadian Journal of Fisheries and Aquatic Sciences*, 68(7), 1124–1138. <https://doi.org/10.1139/f2011-025>
- Ianelli, J. (2026). *Jack mackerel base model: MCMC uncertainty* [Exploratory working report]. South Pacific Regional Fisheries Management Organisation. https://sprfmo.github.io/SC14_JM/papers/base-model-adnuts-2026/base-model-adnuts.html
- Jack mackerel assessment SC14 (draft)*. (2026). [Working assessment report]. South Pacific Regional Fisheries Management Organisation. https://sprfmo.github.io/SC14_JM/assessment/SC14.html
- Leal, E., Diaz, E., & Saavedra, J. C. (2012). *Reproductive timing and maturity at length and age of jack mackerel Trachurus murphyi in the Chilean coast* (Science Working Group Working Paper SWG-11-JM-07). South Pacific Regional Fisheries Management Organisation.
- Leave-one-in and leave-one-out analyses*. (2026). [Working assessment diagnostic report]. South Pacific Regional Fisheries Management Organisation. https://sprfmo.github.io/SC14_JM/assessment/loo.html
- Model developments* (Workshop Working Paper SCW17/Paper-01). (2026). South Pacific Regional Fisheries Management Organisation. <https://sprfmo.github.io/scw17/SCW17-Paper-01-ModelDevelopments.html>
- Proposed jack mackerel harvest control rule, 2022* (Proposal Annex K). (2022). South Pacific Regional Fisheries Management Organisation.
- SPRFMO Jack Mackerel Working Group. (2026a). *Report of the jack mackerel benchmark workshop (SCW16)* (Nos. SC14-JM02). South Pacific Regional Fisheries Management Organisation. https://sprfmo.github.io/JM_SCW_prep/JMWG-Benchmark-Meeting-report-2026.html
- SPRFMO Jack Mackerel Working Group. (2026b). *Report of the jack mackerel management strategy evaluation workshop (SCW17)* (Nos. SC14-JM03). South Pacific Regional Fisheries Management Organisation. <https://sprfmo.github.io/scw17/SCW17-MSE-Workshop-report.html>
- SPRFMO Scientific Committee. (2025). *Annex 11: Jack mackerel technical annex* (SC13 Report, Annex 11). South Pacific Regional Fisheries Management Organisation. https://sprfmo.int/assets/Meetings/02-SC/13th-SC-2025/SC13-Report-Annex-11_final.pdf

SPRFMO Secretariat. (2026). *Catch history of Trachurus murphyi* (Nos. SC14-JM01). South Pacific Regional Fisheries Management Organisation. <https://www.sprfmo.int/assets/Meetings/02-SC/14th-SC-2026/Jack-Mackerel/SC14-JM01-Catch-history-of-Trachurus-murphyi.pdf>

8 Tables

Country-level catch submissions are documented in ([SPRFMO Secretariat, 2026](#)). The table below reports the catch inputs in the current assessment, aggregated into the four model fleets; it preserves the distinction between preliminary 2025 catches and projected full-year 2026 catches.

Table 10: Input catch (thousand tonnes) by fleet for the stock assessment model. The 2025 catches remain preliminary and 2026 catches are projected full-year values.

Year	Fleet 1	Fleet 2	Fleet 3	Fleet 4
1970	101.69	10.31	4.71	1.00
1971	143.45	14.99	9.19	1.00
1972	64.46	22.55	18.78	5.50
1973	83.20	38.39	42.78	1.00
1974	164.76	28.75	129.21	1.00
1975	207.33	53.88	37.90	1.00
1976	257.70	84.57	54.15	1.04
1977	226.23	114.57	504.99	2.27
1978	398.41	188.27	386.79	51.29
1979	344.05	253.46	333.81	370.29
1980	288.81	273.45	414.30	339.80
1981	474.82	586.09	445.64	438.12
1982	789.91	704.77	143.72	733.20
1983	301.93	563.34	110.69	894.30
1984	727.00	699.30	200.67	1,059.93
1985	511.15	945.84	114.62	799.32
1986	55.21	1,129.11	51.03	837.50
1987	313.31	1,456.73	46.30	863.42
1988	325.46	1,812.79	244.23	863.22
1989	338.60	2,051.52	316.25	875.82
1990	323.09	2,148.79	370.82	872.06
1991	346.25	2,674.27	213.45	543.66
1992	304.24	2,907.82	111.68	37.93
1993	379.47	2,856.78	133.35	1.00
1994	222.25	3,819.19	233.35	1.00
1995	230.18	4,174.02	550.99	1.00
1996	278.44	3,604.89	495.52	1.00
1997	104.20	2,812.87	680.05	1.00
1998	30.27	1,582.64	412.85	1.00
1999	55.65	1,164.04	203.75	1.01
2000	118.73	1,115.57	303.70	2.32
2001	248.10	1,401.84	857.74	20.09
2002	108.73	1,410.27	154.82	76.26

Year	Fleet 1	Fleet 2	Fleet 3	Fleet 4
2003	143.28	1,278.02	217.73	158.20
2004	158.66	1,292.94	187.37	295.44
2005	165.63	1,264.81	80.66	243.58
2006	155.26	1,224.69	277.57	362.63
2007	172.70	1,130.08	255.36	438.83
2008	167.26	728.85	169.54	406.99
2009	134.02	700.90	76.63	371.92
2010	169.01	295.80	22.17	239.59
2011	30.82	216.47	326.39	60.89
2012	13.26	214.20	187.40	39.92
2013	16.36	215.00	80.59	41.18
2014	18.22	254.29	74.53	63.65
2015	34.89	250.33	22.45	86.67
2016	24.66	295.16	15.09	54.16
2017	35.00	311.86	8.87	49.11
2018	11.55	415.15	57.16	42.46
2019	11.88	432.45	135.78	51.44
2020	44.16	517.66	140.12	4.74
2021	61.36	567.27	123.64	55.30
2022	72.80	655.16	159.61	73.87
2023	68.05	740.20	222.29	95.33
2024	105.31	977.04	217.24	30.48
2025	174.04	1,067.68	200.38	52.43
2026	108.78	797.72	110.00	85.00

Table 11: Input catch at age for Fleet 1. Observed proportions are displayed as percentages; the model uses proportions summing to one within each year.

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1980	0	5	14	24	31	22	4	0	0	0	0	0
1981	1	7	13	21	33	19	5	1	0	0	0	0
1982	0	15	15	21	26	16	6	1	0	0	0	0
1983	1	9	17	27	28	15	3	0	0	0	0	0
1984	2	34	12	14	18	16	4	0	0	0	0	0
1985	1	18	26	30	18	5	1	0	0	0	0	0
1986	8	11	9	18	32	18	5	0	0	0	0	0
1987	15	68	11	3	2	1	0	0	0	0	0	0
1988	1	17	54	26	2	0	0	0	0	0	0	0
1989	0	9	42	39	8	1	0	0	0	0	0	0
1990	9	3	28	49	10	1	0	0	0	0	0	0
1991	11	33	8	18	24	6	1	0	0	0	0	0
1992	11	30	21	21	12	5	1	0	0	0	0	0
1993	15	72	8	4	1	0	0	0	0	0	0	0
1994	27	32	25	13	2	1	0	0	0	0	0	0
1995	5	69	18	6	2	0	0	0	0	0	0	0
1996	29	57	11	3	0	0	0	0	0	0	0	0
1997	36	60	3	0	0	0	0	0	0	0	0	0
1998	8	79	11	3	0	0	0	0	0	0	0	0
1999	9	84	5	2	0	0	0	0	0	0	0	0
2000	36	47	16	1	0	0	0	0	0	0	0	0
2001	51	48	1	0	0	0	0	0	0	0	0	0
2002	21	58	17	3	1	0	0	0	0	0	0	0
2003	21	72	4	2	1	0	0	0	0	0	0	0
2004	13	63	23	1	0	0	0	0	0	0	0	0
2005	40	44	11	4	1	0	0	0	0	0	0	0
2006	8	83	6	2	1	0	0	0	0	0	0	0
2007	12	69	13	3	2	0	0	0	0	0	0	0
2008	56	27	9	7	1	0	0	0	0	0	0	0
2009	20	68	4	8	0	0	0	0	0	0	0	0
2010	9	74	13	3	1	0	0	0	0	0	0	0
2011	77	20	2	1	0	0	0	0	0	0	0	0
2012	34	58	7	0	0	0	0	0	0	0	0	0
2013	31	66	1	1	1	0	0	0	0	0	0	0
2014	59	40	2	0	0	0	0	0	0	0	0	0
2015	14	60	15	6	4	1	0	0	0	0	0	0

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2016	10	20	13	19	19	7	10	1	0	0	0	0
2017	31	61	6	1	1	0	0	0	0	0	0	0
2018	100	0	0	0	0	0	0	0	0	0	0	0
2019	20	19	9	14	13	6	7	4	3	3	1	2
2020	0	27	25	23	15	8	2	0	0	0	0	0
2021	18	3	4	14	22	18	12	7	3	1	1	0
2022	2	1	1	11	26	29	19	6	3	1	0	0
2023	9	2	6	8	16	25	18	10	4	1	2	0
2024	0	1	10	14	13	19	21	14	7	1	0	0
2025	0	2	6	8	15	13	17	21	12	5	1	0
2026	0	2	1	4	14	12	17	26	15	7	3	0

Table 12: Input catch at age for Fleet 2. Observed proportions are displayed as percentages; the model uses proportions summing to one within each year.

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1980	2	23	40	26	8	1	0	0	0	0	0	0
1981	2	20	32	31	12	3	0	0	0	0	0	0
1982	2	27	37	25	8	1	0	0	0	0	0	0
1983	15	28	24	20	11	2	0	0	0	0	0	0
1984	7	50	8	14	12	6	2	0	0	0	0	0
1985	3	27	26	20	17	7	2	0	0	0	0	0
1986	4	11	24	27	21	12	2	0	0	0	0	0
1987	8	46	7	10	17	10	2	0	0	0	0	0
1988	12	38	29	7	8	6	1	0	0	0	0	0
1989	1	12	42	30	9	5	2	0	0	0	0	0
1990	0	1	6	26	33	18	12	3	0	0	0	0
1991	1	3	0	6	27	29	18	10	4	1	0	0
1992	1	7	6	6	8	21	22	16	9	4	0	0
1993	1	16	17	14	12	10	14	12	4	1	0	0
1994	0	6	17	18	13	11	17	13	4	1	0	0
1995	1	19	17	22	20	8	7	4	1	0	0	0
1996	4	22	19	17	15	10	6	3	1	0	0	0
1997	8	42	21	10	6	5	5	2	1	1	0	0
1998	9	58	14	6	3	3	4	2	1	0	0	0
1999	20	52	15	6	2	1	1	1	1	0	0	0
2000	10	49	24	10	3	1	1	1	1	0	0	0
2001	6	41	28	12	4	2	2	2	1	1	1	0
2002	7	34	23	16	6	4	3	2	2	2	1	0
2003	4	31	28	21	8	3	2	2	1	1	0	0
2004	2	22	29	26	11	5	3	2	1	0	0	0
2005	2	8	20	33	19	9	5	2	1	1	0	0
2006	1	6	9	20	25	14	11	7	3	2	1	1
2007	0	13	17	11	15	15	12	9	4	2	1	1
2008	3	1	6	22	20	16	11	9	5	3	2	2
2009	2	15	2	19	21	16	10	7	4	2	1	1
2010	1	32	20	10	11	6	9	6	2	1	1	0
2011	2	11	14	36	11	8	13	2	1	0	0	0
2012	0	8	25	27	29	7	3	1	0	0	0	0
2013	2	18	31	33	14	2	0	0	0	0	0	0
2014	1	13	24	26	21	12	3	1	0	0	0	0
2015	10	45	14	10	10	7	3	1	0	0	0	0

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2016	0	23	26	22	14	8	4	2	1	0	0	0
2017	3	21	16	16	16	11	7	4	3	1	0	1
2018	2	18	24	20	17	9	5	3	1	1	1	0
2019	0	9	17	22	24	14	8	4	1	0	0	0
2020	0	9	10	15	22	20	14	8	3	0	1	0
2021	0	4	15	18	24	18	11	6	2	1	0	0
2022	0	1	5	17	24	23	16	9	3	2	0	0
2023	0	2	9	17	23	23	15	7	3	1	0	0
2024	0	3	10	20	23	19	15	6	2	1	0	0
2025	0	0	6	20	29	19	12	8	4	1	0	0
2026	0	4	3	6	33	27	18	6	1	1	0	0

Table 13: Input catch at age for Fleet 4. Observed proportions are displayed as percentages; the model uses proportions summing to one within each year.

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2015	17	26	10	7	11	11	8	5	3	1	1	0
2016	6	14	17	25	22	7	3	2	1	1	0	0
2017	65	14	12	5	2	1	1	0	0	0	0	0
2018	15	21	7	12	18	15	8	3	1	0	0	0
2019	19	32	8	8	8	8	8	6	2	0	1	0
2020	14	53	24	4	1	1	1	0	0	0	0	0
2021	6	21	50	13	7	2	0	0	0	0	0	0
2022	1	67	8	7	10	3	2	1	1	0	0	0
2023	0	14	32	37	7	4	2	2	1	0	0	0
2024	1	10	24	36	20	6	2	0	0	0	0	0
2025	22	56	21	1	0	0	0	0	0	0	0	0
2026	2	28	48	18	3	1	0	0	0	0	0	0

Table 14: Input catch at fork length for Fleet 3, panel 1: 10–23 cm. Values are percentages; the model uses proportions summing to one across all length bins within each year.

Year	Fork length (cm)													
	10	11	12	13	14	15	16	17	18	19	20	21	22	23
1972	0	0	0	0	0	0	0	0	1	2	4	4	5	8
1973	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1974	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1975	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1976	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1977	0	0	0	0	0	0	0	0	0	0	1	2	2	1
1978	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1979	0	0	0	0	0	0	1	3	1	0	0	0	0	1
1980	1	0	0	1	1	3	2	2	0	0	1	1	1	1
1981	0	0	0	0	0	0	0	0	0	0	0	0	2	3
1982	0	0	0	0	1	1	2	3	4	4	1	0	0	0
1983	0	0	0	0	0	0	0	0	0	0	0	0	0	1
1984	0	0	0	0	0	0	0	0	0	0	0	0	1	2
1985	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1986	0	0	0	0	0	0	2	3	6	7	7	6	4	4
1987	0	0	0	0	1	5	7	4	1	1	3	4	5	9
1988	0	0	0	0	0	0	0	0	0	0	0	0	1	6
1989	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1990	0	0	0	0	0	0	0	0	0	0	0	1	1	1
1991	0	0	0	0	0	0	0	0	0	1	0	1	1	1
1992	0	0	0	0	0	0	0	0	0	0	0	1	1	3
1993	0	0	0	0	0	0	0	0	0	0	1	10	13	8
1994	0	0	0	0	0	0	0	0	0	0	0	1	1	1
1995	0	0	0	0	2	2	1	0	0	1	2	2	2	3
1996	0	0	0	1	1	2	1	0	0	1	2	6	10	11
1997	0	0	0	0	0	0	0	0	0	0	0	0	1	3
1998	0	0	0	0	1	0	0	1	4	5	4	4	6	15
1999	0	0	0	0	0	0	0	1	2	4	5	7	10	12
2000	0	0	0	0	0	0	0	0	2	5	8	10	9	7
2001	0	0	0	0	0	0	0	0	0	0	1	2	4	9
2002	0	0	0	0	0	0	0	0	0	0	0	1	3	3
2003	0	0	0	0	0	0	0	1	1	2	4	6	8	8
2004	0	0	0	0	0	0	0	0	0	0	0	0	1	1
2005	0	0	0	0	0	0	1	6	11	13	13	12	9	4
2006	0	0	0	0	0	0	0	1	5	12	17	18	16	12
2007	0	0	0	0	1	0	0	0	0	0	1	1	4	9

	Fork length (cm)													
Year	10	11	12	13	14	15	16	17	18	19	20	21	22	23
2008	0	0	0	0	0	0	0	0	0	0	0	0	0	1
2009	0	0	0	0	1	1	2	5	7	9	10	9	8	9
2010	0	0	0	0	0	0	0	1	3	3	3	6	19	24
2011	0	0	0	0	0	0	0	0	0	0	0	1	1	1
2012	0	0	0	0	0	0	0	0	0	0	1	1	1	1
2013	0	0	0	0	0	0	1	4	12	14	11	8	5	3
2014	0	0	0	0	0	0	0	1	3	7	8	6	6	6
2015	0	0	0	0	0	0	1	2	3	2	3	5	8	10
2016	0	0	0	0	0	0	0	0	0	0	1	1	3	5
2017	0	0	0	0	0	0	0	0	1	1	2	5	7	7
2018	0	0	0	0	0	0	1	1	1	2	2	2	2	4
2019	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2020	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2021	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2022	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2023	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2024	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2025	0	0	0	0	0	0	0	0	0	0	0	0	0	1
2026	0	0	0	0	0	0	0	0	0	0	1	2	3	3

Table 15: Input catch at fork length for Fleet 3, panel 2: 24–37 cm. Values are percentages of the same annual length composition.

Year	Fork length (cm)													
	24	25	26	27	28	29	30	31	32	33	34	35	36	37
1972	11	10	6	2	1	2	2	2	2	3	3	3	3	3
1973	0	0	1	2	7	12	11	9	5	4	3	3	3	4
1974	0	0	0	0	6	17	22	16	11	7	3	2	2	1
1975	0	1	1	1	3	4	5	9	13	16	15	11	7	4
1976	0	0	1	1	3	3	3	4	7	11	11	9	6	6
1977	2	3	9	15	11	7	7	6	6	5	4	3	2	2
1978	1	4	15	28	17	8	4	3	3	3	3	3	2	1
1979	2	4	5	7	10	12	9	7	5	4	4	3	4	4
1980	1	1	2	4	5	10	14	14	9	7	6	4	3	1
1981	3	4	5	9	10	10	12	10	8	6	4	4	3	3
1982	0	1	2	6	16	18	12	7	5	4	3	2	2	1
1983	1	3	3	3	5	11	16	16	13	10	7	4	2	1
1984	3	5	7	8	7	11	12	10	9	7	5	3	2	1
1985	0	2	3	8	10	10	11	10	8	6	5	4	4	4
1986	3	2	1	0	0	1	1	3	5	8	8	8	6	4
1987	12	12	11	8	4	3	2	1	1	1	1	1	1	1
1988	15	20	18	13	8	5	3	2	1	1	1	1	1	0
1989	1	3	6	11	13	12	11	9	8	7	5	4	3	2
1990	1	0	0	1	6	11	14	11	9	9	8	7	6	4
1991	2	4	4	5	6	9	13	13	10	9	7	6	4	3
1992	4	2	2	3	6	6	7	10	12	11	8	6	5	4
1993	12	9	10	9	4	4	4	4	4	3	2	1	0	0
1994	2	3	6	6	7	14	16	12	7	5	3	2	2	1
1995	3	4	7	8	10	12	14	10	6	3	2	1	1	1
1996	10	7	6	7	9	8	6	5	3	2	1	0	0	0
1997	9	11	7	5	10	17	19	11	4	2	1	0	0	0
1998	22	17	9	4	3	1	1	1	1	0	0	0	0	0
1999	12	12	13	10	5	2	2	1	1	0	0	0	0	0
2000	9	11	9	11	10	6	2	1	0	0	0	0	0	0
2001	13	15	18	19	12	6	2	0	0	0	0	0	0	0
2002	3	2	3	10	19	20	17	11	4	2	1	0	0	0
2003	6	7	9	10	8	5	4	4	4	5	4	2	1	0
2004	1	1	3	6	8	9	11	13	13	10	8	7	5	3
2005	2	1	1	1	1	0	0	0	0	0	0	1	3	6
2006	8	4	2	1	1	1	1	1	0	0	0	0	0	0
2007	10	7	7	5	4	7	10	11	9	6	4	2	1	0

	Fork length (cm)													
Year	24	25	26	27	28	29	30	31	32	33	34	35	36	37
2008	1	1	2	4	12	21	22	16	9	4	2	1	1	1
2009	9	5	3	2	2	3	2	1	1	0	0	1	1	2
2010	17	11	7	4	2	1	1	0	0	0	0	0	0	0
2011	4	9	15	18	18	18	12	3	0	0	0	0	0	0
2012	1	1	1	1	1	0	2	9	18	19	19	15	6	2
2013	2	2	2	3	4	5	4	2	2	1	0	0	0	1
2014	5	5	5	4	6	11	13	7	3	2	1	1	1	0
2015	12	13	11	9	7	5	4	2	1	1	1	0	0	0
2016	7	7	9	9	8	8	7	8	6	6	5	4	2	1
2017	8	10	13	13	9	6	5	3	3	2	2	1	1	1
2018	8	14	20	18	9	4	3	3	2	2	1	1	1	0
2019	1	2	4	3	4	8	11	16	17	13	8	5	3	2
2020	0	0	0	1	2	3	4	7	10	14	17	16	11	7
2021	0	0	0	0	0	1	1	1	2	4	6	7	9	9
2022	0	0	0	1	1	2	3	3	3	4	5	7	7	7
2023	0	0	0	0	1	2	3	5	6	6	5	4	4	4
2024	0	0	0	0	1	0	0	1	1	1	2	3	5	5
2025	1	1	1	1	1	1	1	1	1	2	2	3	3	3
2026	4	5	6	7	5	4	3	2	2	2	2	2	2	1

Table 16: Input catch at fork length for Fleet 3, panel 3: 38–50 cm. Values are percentages of the same annual length composition; the terminal observation category is 50 cm plus.

	Fork length (cm)												
Year	38	39	40	41	42	43	44	45	46	47	48	49	50
1972	2	2	2	2	2	2	1	1	1	1	1	1	5
1973	4	5	5	4	3	3	2	2	2	1	1	1	4
1974	1	1	1	0	1	1	1	0	0	0	0	0	5
1975	2	1	0	0	0	0	0	0	0	1	0	1	5
1976	5	4	4	3	3	2	2	2	2	2	1	1	3
1977	2	1	1	1	1	1	0	0	0	0	0	0	1
1978	1	1	1	0	0	0	0	0	0	0	0	0	0
1979	3	2	2	2	1	1	1	0	0	0	0	0	1
1980	1	1	1	1	1	1	1	0	0	0	0	0	0
1981	2	1	0	0	0	0	0	0	0	0	0	0	0
1982	1	1	0	0	0	0	0	0	0	0	0	0	0
1983	1	0	0	0	0	0	0	0	0	0	0	0	0
1984	1	1	1	1	1	1	1	0	0	0	0	0	0
1985	3	2	2	1	1	1	1	1	1	0	0	0	0
1986	2	2	1	1	1	1	1	0	0	0	0	0	0
1987	0	0	0	0	0	0	0	0	0	0	0	0	0
1988	1	0	0	0	0	0	0	0	0	0	0	0	0
1989	1	1	0	0	0	0	0	0	0	0	0	0	0
1990	3	2	1	1	0	0	0	0	0	0	0	0	0
1991	1	1	0	0	0	0	0	0	0	0	0	0	0
1992	3	3	2	1	1	0	0	0	0	0	0	0	1
1993	0	0	0	0	0	0	0	0	0	0	0	0	0
1994	1	1	1	1	1	0	1	1	0	0	0	0	2
1995	0	0	0	0	0	0	0	0	0	0	0	0	0
1996	0	0	0	0	0	0	0	0	0	0	0	0	0
1997	0	0	0	0	0	0	0	0	0	0	0	0	0
1998	0	0	0	0	0	0	0	0	0	0	0	0	0
1999	0	0	0	0	0	0	0	0	0	0	0	0	0
2000	0	0	0	0	0	0	0	0	0	0	0	0	0
2001	0	0	0	0	0	0	0	0	0	0	0	0	0
2002	0	0	0	0	0	0	0	0	0	0	0	0	0
2003	0	0	0	0	0	0	0	0	0	0	0	0	0
2004	1	1	0	0	0	0	0	0	0	0	0	0	0
2005	6	4	2	1	0	0	0	0	0	0	0	0	0
2006	0	0	0	0	0	0	0	0	0	0	0	0	0
2007	0	0	0	0	0	0	0	0	0	0	0	0	0

	Fork length (cm)												
Year	38	39	40	41	42	43	44	45	46	47	48	49	50
2008	1	0	0	0	0	0	0	0	0	0	0	0	0
2009	3	1	1	0	0	0	0	0	0	0	0	0	0
2010	0	0	0	0	0	0	0	0	0	0	0	0	0
2011	0	0	0	0	0	0	0	0	0	0	0	0	0
2012	1	0	0	0	0	0	0	0	0	0	0	0	0
2013	3	5	3	1	0	0	0	0	0	0	0	0	0
2014	0	0	0	0	0	0	0	0	0	0	0	0	0
2015	0	0	0	0	0	0	0	0	0	0	0	0	0
2016	1	0	0	0	0	0	0	0	0	0	0	0	0
2017	0	0	0	0	0	0	0	0	0	0	0	0	0
2018	0	0	0	0	0	0	0	0	0	0	0	0	0
2019	1	1	0	0	0	0	0	0	0	0	0	0	0
2020	4	2	1	0	0	0	0	0	0	0	0	0	0
2021	9	11	11	10	7	5	3	1	1	0	0	0	0
2022	6	5	6	6	6	6	6	5	3	2	2	1	1
2023	4	5	5	5	5	5	5	5	5	4	4	3	6
2024	5	6	5	6	6	6	6	5	5	4	4	4	14
2025	4	4	5	6	7	8	10	9	7	6	3	2	3
2026	1	1	2	3	3	4	4	5	6	5	4	3	4

Table 17: Observed abundance-index inputs for model 1.06. Each series retains its own input units and scaling. Blank cells indicate years without observations.

Year	Chile_AcousCS	Chile_AcousN	Chile_CPUE	Peru_Artis_CPUE	Peru_Ind_CPUE	Offshore_CPUE
1970						
1971						
1972						
1973						
1974						
1975						
1976						
1977						
1978						
1979						
1980						
1981						
1982						
1983						
1984						
1985						
1986						
1987						
1988						
1989						
1990						
1991						
1992						
1993						
1994						
1995						
1996						
1997	3,530					
1998	3,200		102.00			
1999	4,100		89.89			
2000	5,600		126.20			
2001	5,950		128.80			
2002	3,700		117.30	0.16320	0.2055	
2003	2,640		131.30	0.17840	0.3464	
2004	2,640		154.00	0.21430	0.3693	
2005	4,110		129.90	0.20520	0.2132	
2006	3,192	112.0	120.00	0.26510	0.3483	

Year	Chile_AcousCS	Chile_AcousN	Chile_CPUE	Peru_Artis_CPUE	Peru_Ind_CPUE	Offshore_CPUE
2007	3,140	275.0	99.35	0.32690	0.4358	
2008	487	259.0	112.90	0.17800	0.4122	1,295
2009	328	18.0	107.90	0.07891	0.2300	1,145
2010		440.0	93.93	0.15660		680
2011		432.0	58.94	0.32990	0.3716	541
2012		230.0	59.27	0.29510	0.4428	454
2013		144.0	63.82	0.16300	0.2335	562
2014		87.0	60.14	0.18840	0.2820	543
2015		459.0	60.95	0.19250		804
2016		587.2	69.84	0.16120		527
2017		610.5	74.73	0.20330		632
2018		374.1	70.18	0.25580	0.4160	548
2019		1,487.0	104.30	0.41330	0.4432	688
2020		1,728.0	97.27	0.46820	0.4730	669
2021		1,870.0	93.99	0.46760	0.4860	966
2022			91.32	0.57080	0.4933	1,394
2023		2,502.0	100.20	0.61660	0.4486	1,410
2024		2,788.0	82.42	0.67220	0.4226	490
2025		3,059.0	96.67	0.49460	0.3975	1,030
2026		642.1	104.40	0.56740	0.2574	

The current six-index configuration contains two Chilean acoustic series, Chilean CPUE, separate Peruvian artisanal and industrial CPUE, and offshore CPUE. Data coverage and uncertainty are listed below.

Table 18: Input catch at age for acoustic surveys in southern Chile. Observed proportions are displayed as percentages; the model uses proportions summing to one within each year.

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2001	1	56	10	17	6	4	2	1	1	1	0	0
2002	2	45	27	13	5	5	2	1	0	0	0	0
2003	1	29	32	22	7	4	2	1	1	1	0	0
2004	1	13	19	25	17	10	9	4	1	0	0	0
2005	1	12	20	41	16	5	2	1	1	0	0	0
2006	0	0	13	34	32	8	6	4	2	1	0	0
2007	0	0	2	14	19	21	18	13	8	2	2	1
2008	0	0	0	12	33	25	13	9	4	2	1	2
2009	0	0	0	0	1	30	24	16	17	6	3	3

Table 19: Input catch at age for acoustic surveys in northern Chile. Observed proportions are displayed as percentages; the model uses proportions summing to one within each year.

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2006	30	69	1	0	0	0	0	0	0	0	0	0
2007	8	60	23	8	2	0	0	0	0	0	0	0
2009	68	31	1	0	0	0	0	0	0	0	0	0
2013	45	13	21	15	5	1	0	0	0	0	0	0
2014	95	2	0	1	1	0	0	0	0	0	0	0
2015	72	21	4	2	1	0	0	0	0	0	0	0
2016	73	19	4	2	1	0	0	0	0	0	0	0
2017	76	16	6	1	0	0	0	0	0	0	0	0
2018	93	5	1	0	0	0	0	0	0	0	0	0
2019	16	59	20	4	1	1	0	0	0	0	0	0
2020	23	8	25	31	11	2	0	0	0	0	0	0
2021	63	0	13	14	5	3	2	0	0	0	0	0
2023	5	18	23	22	14	11	6	1	0	0	0	0
2024	66	2	5	9	8	5	4	1	0	0	0	0
2025	62	0	8	12	11	4	2	1	0	0	0	0
2026	11	71	4	2	4	3	2	1	1	0	0	0

Table 20: Jack mackerel sexual maturity by age used in the JJM models.

Stock	1	2	3	4	5	6	7	8	9	10	11	12
Single / Southern	0.52	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Far North	0.00	0.37	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 21: Growth and natural mortality parameters in the current assessment. Growth lengths are fork length; natural mortality is annual instantaneous mortality.

Stock	Linf (cm)	k (per year)	L0 (cm)	M (per year)
Single stock	73.560	0.160	13.560	0.280
Two-stock: south	73.560	0.160	13.560	0.280
Two-stock: Far North	73.560	0.160	13.560	0.330

L_0 is the mean fork length at the recruitment age (1 year).

Table 22: Ageing error matrix of jack mackerel. Rows represent observed ages and columns represent true ages. The current assessment disables the ageing-error option.

Age	1	2	3	4	5	6	7	8	9	10	11	12
1	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	0.000	0.762	0.244	0.022	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	0.000	0.222	0.512	0.229	0.024	0.001	0.000	0.000	0.000	0.000	0.000	0.000
4	0.000	0.016	0.225	0.499	0.232	0.027	0.001	0.000	0.000	0.000	0.000	0.000
5	0.000	0.000	0.018	0.229	0.487	0.234	0.031	0.001	0.000	0.000	0.000	0.000
6	0.000	0.000	0.000	0.021	0.232	0.475	0.236	0.034	0.002	0.000	0.000	0.000
7	0.000	0.000	0.000	0.000	0.024	0.234	0.464	0.238	0.037	0.002	0.000	0.000
8	0.000	0.000	0.000	0.000	0.001	0.027	0.236	0.454	0.239	0.041	0.003	0.000
9	0.000	0.000	0.000	0.000	0.000	0.001	0.031	0.238	0.444	0.240	0.044	0.003
10	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.034	0.239	0.434	0.241	0.047
11	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.037	0.240	0.424	0.242
12	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.002	0.043	0.288	0.708

Table 23: Input mean body mass (kg) at age over time assumed for Fleet 1.

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1971	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1972	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1973	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1974	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1975	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1976	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1977	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1978	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1979	0.108	0.155	0.247	0.314	0.388	0.510	0.653	0.780	0.949	1.153	1.349	1.575
1980	0.181	0.209	0.263	0.338	0.434	0.462	0.499	1.062	0.984	1.144	1.350	1.551
1981	0.205	0.235	0.287	0.353	0.424	0.500	0.601	0.735	0.984	1.144	1.350	1.551
1982	0.164	0.192	0.236	0.301	0.354	0.422	0.460	0.490	0.739	1.144	1.350	1.551
1983	0.122	0.191	0.245	0.309	0.376	0.429	0.485	0.486	0.529	1.144	1.350	1.551
1984	0.157	0.174	0.241	0.323	0.387	0.443	0.510	0.559	0.984	1.144	1.350	1.551
1985	0.209	0.206	0.275	0.344	0.406	0.482	0.553	0.595	0.984	1.144	1.350	1.551
1986	0.083	0.166	0.279	0.358	0.432	0.523	0.559	0.716	0.984	1.144	1.350	1.551
1987	0.159	0.163	0.237	0.335	0.432	0.491	0.542	0.545	0.984	1.144	1.350	1.551
1988	0.178	0.214	0.251	0.296	0.382	0.581	0.690	0.852	0.984	1.144	1.350	1.551
1989	0.137	0.194	0.252	0.297	0.354	0.496	0.579	0.547	0.701	1.144	1.350	1.551
1990	0.116	0.238	0.293	0.338	0.386	0.490	0.688	0.677	0.950	1.144	1.350	1.551
1991	0.144	0.165	0.283	0.344	0.388	0.440	0.739	0.855	0.775	1.144	1.350	1.551
1992	0.116	0.159	0.259	0.285	0.342	0.394	0.471	0.673	0.984	1.144	1.350	1.551
1993	0.133	0.130	0.258	0.334	0.379	0.432	0.520	0.638	0.815	0.935	0.960	1.551
1994	0.105	0.132	0.282	0.322	0.379	0.436	0.508	0.584	0.984	1.144	1.350	1.551
1995	0.072	0.147	0.270	0.340	0.394	0.394	0.632	0.852	0.984	1.144	1.350	1.551
1996	0.117	0.151	0.203	0.230	0.284	0.388	1.320	0.852	0.984	1.144	1.350	1.551
1997	0.162	0.150	0.225	0.291	0.425	0.556	0.704	0.852	0.984	1.144	1.350	1.551
1998	0.183	0.124	0.246	0.285	0.339	0.411	0.704	0.852	0.984	1.144	1.350	1.551
1999	0.111	0.095	0.266	0.306	0.370	0.612	0.704	0.852	0.984	1.144	1.350	1.551
2000	0.061	0.173	0.232	0.256	0.456	0.556	0.704	0.852	0.984	1.144	1.350	1.551
2001	0.117	0.116	0.208	0.289	0.400	0.556	0.704	0.852	0.984	1.144	1.350	1.551
2002	0.097	0.133	0.240	0.324	0.389	0.483	0.704	0.852	0.984	1.144	1.350	1.551
2003	0.095	0.112	0.234	0.314	0.422	0.478	0.510	0.852	0.984	1.144	1.350	1.551
2004	0.140	0.182	0.199	0.246	0.360	0.556	0.704	0.852	0.984	1.144	1.350	1.551
2005	0.084	0.134	0.223	0.265	0.294	0.660	0.739	0.852	0.984	1.144	1.350	1.551
2006	0.077	0.127	0.202	0.386	0.457	0.529	0.636	0.852	0.984	1.144	1.350	1.551

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	0.116	0.140	0.217	0.296	0.401	0.539	0.658	0.852	0.984	1.144	1.350	1.551
2008	0.053	0.100	0.246	0.290	0.389	0.592	0.629	0.761	0.984	1.144	1.350	1.551
2009	0.088	0.126	0.246	0.286	0.334	0.534	0.704	0.852	0.984	1.144	1.350	1.551
2010	0.056	0.102	0.231	0.287	0.401	0.602	0.701	0.852	0.984	1.144	1.350	1.551
2011	0.064	0.109	0.275	0.325	0.382	0.556	0.704	0.852	0.984	1.144	1.350	1.551
2012	0.047	0.179	0.252	0.349	0.361	0.556	0.704	0.852	0.984	1.144	1.350	1.551
2013	0.064	0.064	0.284	0.390	0.441	0.657	0.520	0.852	0.984	1.144	1.350	1.551
2014	0.052	0.105	0.218	0.283	0.430	0.553	0.704	0.852	0.984	1.144	1.350	1.551
2015	0.037	0.185	0.279	0.369	0.437	0.608	0.779	0.852	0.984	1.144	1.350	1.551
2016	0.164	0.175	0.233	0.294	0.366	0.499	0.718	0.860	0.658	1.144	1.350	1.551
2017	0.049	0.158	0.192	0.259	0.332	0.451	0.600	0.528	0.984	1.144	1.350	1.551
2018	0.034	0.129	0.248	0.277	0.279	0.556	0.704	0.852	0.984	1.144	1.350	1.551
2019	0.025	0.179	0.284	0.471	0.501	0.528	0.732	0.853	0.888	1.280	1.695	1.806
2020	0.068	0.211	0.242	0.303	0.397	0.474	0.770	1.066	1.351	1.568	1.350	2.294
2021	0.042	0.147	0.457	0.452	0.536	0.655	0.745	0.923	1.002	1.098	1.193	1.804
2022	0.027	0.244	0.404	0.520	0.606	0.684	0.792	0.816	1.065	1.120	1.193	2.659
2023	0.010	0.212	0.318	0.506	0.590	0.674	0.805	0.830	0.864	1.229	1.224	1.489
2024	0.135	0.243	0.306	0.375	0.507	0.686	0.842	0.972	1.055	1.252	1.314	1.489
2025	0.100	0.288	0.397	0.417	0.505	0.632	0.854	0.955	1.045	1.135	1.278	1.264
2026	0.120	0.186	0.280	0.413	0.462	0.570	0.930	1.092	1.086	1.216	1.188	1.614

Table 24: Input mean body mass (kg) at age over time assumed for Fleet 2.

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1971	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1972	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1973	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1974	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1975	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1976	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1977	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1978	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1979	0.157	0.202	0.271	0.346	0.444	0.570	0.709	0.867	1.076	1.313	1.579	1.826
1980	0.203	0.201	0.237	0.275	0.328	0.375	0.504	0.861	0.995	1.159	1.397	1.534
1981	0.164	0.187	0.238	0.268	0.308	0.368	0.464	0.796	0.995	1.159	1.397	1.534
1982	0.183	0.201	0.233	0.261	0.295	0.344	0.402	0.447	0.995	1.159	1.397	1.534
1983	0.120	0.166	0.249	0.284	0.330	0.418	0.497	0.606	0.995	1.159	1.397	1.534
1984	0.151	0.148	0.243	0.289	0.342	0.421	0.499	0.567	0.995	1.159	1.397	1.534
1985	0.192	0.204	0.233	0.299	0.366	0.452	0.537	0.627	0.695	1.159	1.397	1.534
1986	0.136	0.212	0.273	0.313	0.408	0.475	0.550	0.687	1.000	1.159	1.397	1.534
1987	0.126	0.137	0.218	0.335	0.407	0.455	0.492	0.564	0.824	1.159	1.397	1.534
1988	0.182	0.197	0.221	0.340	0.444	0.490	0.539	0.801	1.108	1.159	1.397	1.534
1989	0.211	0.224	0.257	0.310	0.436	0.536	0.579	0.625	0.948	1.159	1.397	1.534
1990	0.110	0.271	0.318	0.380	0.457	0.572	0.675	0.752	0.797	1.485	1.397	1.534
1991	0.170	0.136	0.295	0.418	0.469	0.538	0.657	0.761	0.829	0.921	0.966	1.211
1992	0.147	0.186	0.230	0.296	0.470	0.545	0.605	0.712	0.844	0.968	1.334	1.534
1993	0.162	0.177	0.246	0.320	0.389	0.533	0.684	0.820	0.925	1.117	1.827	1.534
1994	0.195	0.226	0.287	0.347	0.454	0.614	0.783	0.884	1.014	1.178	1.581	1.534
1995	0.174	0.190	0.266	0.339	0.425	0.563	0.797	1.012	1.187	1.425	1.797	1.534
1996	0.189	0.193	0.281	0.362	0.512	0.704	0.954	1.182	1.356	1.445	2.008	1.534
1997	0.174	0.196	0.266	0.360	0.518	0.699	0.887	1.084	1.287	1.529	1.786	1.779
1998	0.151	0.165	0.251	0.343	0.539	0.794	1.025	1.218	1.404	1.584	1.933	2.526
1999	0.161	0.167	0.259	0.338	0.494	0.789	1.039	1.235	1.397	1.654	1.841	1.952
2000	0.188	0.199	0.262	0.357	0.486	0.801	1.058	1.159	1.310	1.454	1.656	2.052
2001	0.183	0.202	0.266	0.336	0.455	0.614	0.868	1.119	1.395	1.568	1.813	1.929
2002	0.182	0.201	0.265	0.330	0.449	0.638	0.860	1.093	1.312	1.499	1.665	2.073
2003	0.174	0.192	0.249	0.305	0.403	0.588	0.786	1.026	1.261	1.504	1.734	1.861
2004	0.195	0.204	0.259	0.311	0.396	0.520	0.685	0.857	1.065	1.395	1.517	1.772
2005	0.083	0.234	0.280	0.318	0.396	0.506	0.642	0.751	0.920	1.160	1.324	1.606
2006	0.114	0.186	0.289	0.349	0.413	0.512	0.618	0.760	0.938	1.041	1.312	1.725

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	0.124	0.187	0.230	0.333	0.431	0.513	0.625	0.777	0.909	1.056	1.228	1.542
2008	0.033	0.215	0.287	0.336	0.421	0.525	0.620	0.726	0.880	1.016	1.160	1.479
2009	0.138	0.139	0.273	0.346	0.418	0.539	0.624	0.759	0.892	1.007	1.138	1.398
2010	0.095	0.182	0.236	0.321	0.414	0.539	0.651	0.796	1.056	1.374	1.560	1.778
2011	0.198	0.202	0.296	0.360	0.478	0.640	0.806	1.025	1.261	1.450	1.874	1.981
2012	0.201	0.213	0.297	0.349	0.491	0.650	0.827	1.062	0.968	1.835	2.222	2.796
2013	0.218	0.245	0.312	0.381	0.448	0.580	0.714	0.926	1.292	1.751	2.082	2.512
2014	0.192	0.265	0.418	0.544	0.643	0.785	0.913	1.002	1.345	1.592	2.407	2.971
2015	0.214	0.214	0.282	0.480	0.610	0.746	0.884	0.990	1.049	1.239	1.130	1.483
2016	0.236	0.258	0.316	0.377	0.483	0.584	0.791	0.872	1.132	1.284	1.544	2.045
2017	0.182	0.226	0.295	0.368	0.444	0.549	0.676	0.922	1.096	1.391	1.741	1.583
2018	0.105	0.241	0.304	0.376	0.493	0.594	0.771	0.922	1.342	1.627	1.792	2.549
2019	0.019	0.268	0.305	0.393	0.482	0.578	0.683	0.759	0.888	1.339	1.978	2.906
2020	0.062	0.230	0.302	0.424	0.560	0.686	0.813	1.014	1.204	1.366	1.408	2.801
2021	0.231	0.272	0.318	0.405	0.562	0.695	0.809	0.956	1.115	1.404	1.484	1.693
2022	0.025	0.355	0.437	0.560	0.690	0.806	0.925	1.057	1.231	1.128	1.415	1.620
2023	0.196	0.275	0.404	0.511	0.613	0.736	0.935	1.057	1.244	1.368	1.540	2.128
2024	0.196	0.325	0.349	0.416	0.505	0.608	0.800	0.941	1.182	1.311	1.483	1.895
2025	0.147	0.379	0.406	0.432	0.481	0.557	0.690	0.879	1.070	1.177	1.368	1.564
2026	0.159	0.168	0.312	0.506	0.555	0.640	0.706	0.902	1.002	1.307	1.374	1.714

Table 25: Input mean body mass (kg) at age over time assumed for Fleet 3.

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	0.045	0.171	0.377	0.642	0.945	1.265	1.587	1.900	2.196	2.470	2.721	2.946
1971	0.045	0.171	0.377	0.643	0.946	1.266	1.588	1.902	2.198	2.472	2.723	2.949
1972	0.030	0.130	0.306	0.548	0.835	1.148	1.470	1.789	2.095	2.382	2.647	2.887
1973	0.037	0.147	0.330	0.568	0.842	1.134	1.430	1.718	1.991	2.246	2.478	2.688
1974	0.038	0.147	0.326	0.558	0.825	1.108	1.393	1.671	1.934	2.178	2.402	2.603
1975	0.034	0.136	0.310	0.540	0.808	1.095	1.387	1.674	1.946	2.201	2.434	2.645
1976	0.044	0.160	0.340	0.567	0.822	1.087	1.351	1.606	1.845	2.065	2.266	2.446
1977	0.032	0.130	0.294	0.510	0.760	1.028	1.300	1.566	1.818	2.054	2.270	2.465
1978	0.032	0.129	0.295	0.516	0.774	1.050	1.332	1.608	1.872	2.117	2.343	2.547
1979	0.036	0.138	0.304	0.518	0.762	1.020	1.280	1.532	1.770	1.991	2.193	2.375
1980	0.036	0.136	0.298	0.506	0.743	0.994	1.245	1.490	1.721	1.934	2.130	2.306
1981	0.041	0.148	0.314	0.524	0.758	1.003	1.247	1.481	1.702	1.905	2.089	2.255
1982	0.039	0.144	0.309	0.519	0.755	1.002	1.249	1.488	1.712	1.920	2.108	2.278
1983	0.042	0.138	0.280	0.451	0.638	0.828	1.014	1.191	1.356	1.507	1.643	1.764
1984	0.044	0.156	0.328	0.541	0.778	1.024	1.267	1.501	1.719	1.921	2.103	2.267
1985	0.040	0.149	0.322	0.541	0.789	1.048	1.308	1.558	1.794	2.012	2.211	2.389
1986	0.042	0.151	0.323	0.539	0.781	1.033	1.285	1.527	1.755	1.965	2.156	2.327
1987	0.034	0.132	0.294	0.504	0.745	1.001	1.260	1.512	1.751	1.973	2.176	2.359
1988	0.038	0.145	0.315	0.533	0.780	1.041	1.302	1.554	1.793	2.013	2.215	2.396
1989	0.044	0.158	0.337	0.561	0.812	1.074	1.334	1.585	1.821	2.038	2.236	2.413
1990	0.042	0.150	0.320	0.532	0.769	1.017	1.263	1.499	1.722	1.927	2.113	2.280
1991	0.039	0.142	0.305	0.511	0.743	0.985	1.227	1.461	1.680	1.883	2.068	2.234
1992	0.040	0.148	0.318	0.534	0.776	1.031	1.286	1.531	1.763	1.976	2.171	2.346
1993	0.039	0.147	0.323	0.549	0.807	1.080	1.354	1.620	1.871	2.104	2.317	2.508
1994	0.036	0.147	0.335	0.584	0.874	1.186	1.503	1.813	2.109	2.385	2.638	2.867
1995	0.038	0.146	0.318	0.540	0.792	1.058	1.325	1.583	1.827	2.053	2.260	2.446
1996	0.038	0.145	0.317	0.537	0.788	1.053	1.318	1.576	1.820	2.045	2.251	2.436
1997	0.045	0.152	0.312	0.506	0.720	0.940	1.155	1.361	1.553	1.729	1.889	2.031
1998	0.040	0.140	0.294	0.483	0.693	0.911	1.126	1.333	1.526	1.703	1.864	2.008
1999	0.037	0.146	0.324	0.557	0.824	1.107	1.394	1.673	1.938	2.183	2.408	2.611
2000	0.035	0.145	0.336	0.592	0.893	1.218	1.550	1.877	2.189	2.481	2.750	2.994
2001	0.033	0.139	0.324	0.572	0.864	1.180	1.504	1.822	2.127	2.412	2.674	2.912
2002	0.036	0.145	0.330	0.576	0.861	1.167	1.478	1.783	2.074	2.344	2.593	2.817
2003	0.040	0.154	0.341	0.584	0.862	1.157	1.454	1.743	2.017	2.272	2.504	2.714
2004	0.038	0.149	0.333	0.574	0.852	1.148	1.447	1.740	2.017	2.275	2.511	2.724
2005	0.037	0.150	0.341	0.595	0.890	1.206	1.527	1.842	2.142	2.422	2.678	2.911
2006	0.038	0.152	0.347	0.606	0.907	1.230	1.558	1.880	2.187	2.473	2.735	2.973

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	0.038	0.149	0.335	0.579	0.861	1.161	1.465	1.762	2.044	2.306	2.546	2.763
2008	0.036	0.146	0.334	0.585	0.876	1.190	1.510	1.823	2.122	2.400	2.656	2.888
2009	0.038	0.150	0.337	0.582	0.865	1.167	1.474	1.773	2.057	2.321	2.563	2.782
2010	0.039	0.150	0.332	0.567	0.837	1.123	1.411	1.691	1.956	2.203	2.428	2.631
2011	0.031	0.143	0.351	0.644	1.000	1.395	1.806	2.217	2.614	2.990	3.337	3.655
2012	0.032	0.145	0.349	0.632	0.971	1.344	1.731	2.115	2.485	2.834	3.156	3.449
2013	0.030	0.136	0.329	0.600	0.928	1.291	1.671	2.052	1.922	2.165	2.387	2.588
2014	0.029	0.137	0.339	0.625	0.976	1.368	1.781	2.196	1.922	2.165	2.387	2.588
2015	0.033	0.140	0.328	0.584	0.889	1.223	1.568	1.910	1.922	2.165	2.387	2.588
2016	0.037	0.143	0.320	0.553	0.821	1.109	1.402	1.689	1.922	2.165	2.387	2.588
2017	0.029	0.132	0.318	0.578	0.892	1.240	1.602	1.966	1.922	2.165	2.387	2.588
2018	0.031	0.136	0.326	0.591	0.910	1.262	1.628	1.995	1.922	2.165	2.387	2.588
2019	0.030	0.134	0.322	0.586	0.904	1.256	1.623	1.991	1.922	2.165	2.387	2.588
2020	0.029	0.131	0.319	0.584	0.905	1.262	1.637	2.012	1.922	2.165	2.387	2.588
2021	0.031	0.134	0.315	0.564	0.861	1.186	1.523	1.858	1.922	2.165	2.387	2.588
2022	0.032	0.135	0.315	0.559	0.847	1.162	1.486	1.808	1.922	2.165	2.387	2.588
2023	0.034	0.137	0.311	0.540	0.808	1.096	1.390	1.680	1.922	2.165	2.387	2.588
2024	0.038	0.145	0.323	0.553	0.820	1.104	1.392	1.674	1.922	2.165	2.387	2.588
2025	0.033	0.136	0.315	0.556	0.841	1.151	1.470	1.785	1.922	2.165	2.387	2.588
2026	0.035	0.141	0.323	0.566	0.850	1.158	1.474	1.785	1.922	2.165	2.387	2.588

Table 26: Input mean body mass (kg) at age over time assumed for Fleet 4.

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1971	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1972	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1973	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1974	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1975	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1976	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1977	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1978	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1979	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1980	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1981	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1982	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1983	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1984	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1985	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1986	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1987	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1988	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1989	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1990	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1991	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1992	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1993	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1994	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1995	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1996	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1997	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1998	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
1999	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2000	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2001	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2002	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2003	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2004	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2005	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2006	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2008	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2009	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2010	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2011	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2012	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2013	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2014	0.157	0.223	0.329	0.429	0.613	0.741	0.835	0.935	1.049	1.145	1.308	1.543
2015	0.228	0.248	0.295	0.434	0.655	0.818	0.933	1.098	1.214	1.326	1.270	1.823
2016	0.311	0.383	0.399	0.428	0.481	0.610	0.837	0.883	0.985	1.094	1.535	1.265
2017	0.059	0.192	0.470	0.549	0.659	0.703	0.739	0.922	0.962	1.094	1.359	1.543
2018	0.066	0.146	0.305	0.388	0.507	0.606	0.649	0.634	0.778	0.868	1.051	1.680
2019	0.127	0.136	0.244	0.510	0.790	0.927	1.040	1.042	1.128	1.263	1.249	1.405
2020	0.152	0.234	0.259	0.265	0.588	0.778	0.811	1.029	1.228	1.226	1.382	1.543
2021	0.103	0.204	0.251	0.277	0.279	0.343	0.544	0.670	0.617	0.966	1.032	0.979
2022	0.154	0.141	0.218	0.251	0.278	0.544	0.785	0.933	1.090	1.252	1.362	1.490
2023	0.151	0.187	0.211	0.224	0.310	0.397	0.513	0.653	0.758	0.821	0.912	1.163
2024	0.209	0.228	0.272	0.310	0.349	0.405	0.470	0.482	0.561	0.977	1.297	1.163
2025	0.154	0.199	0.242	0.266	0.361	0.494	0.624	0.753	0.851	1.049	1.197	1.268
2026	0.154	0.199	0.242	0.266	0.361	0.494	0.624	0.753	0.851	1.049	1.197	1.268

Table 27: Years represented by each data source in the current 1.06 input file. Age and Length columns refer to composition data; a dash indicates no observations of that type. The terminal-year fishery information is partial-year information and the full-year catch is projected.

Source	Type	Catch	Index	Age	Length
N_Chile	Fishery	1970–2026	—	1980–2026	—
SC_Chile_PS	Fishery	1970–2026	—	1980–2026	—
FarNorth	Fishery	1970–2026	—	—	1972–2026
Offshore_Trawl	Fishery	1970–2026	—	2015–2026	—
Chile_AcousCS	Index	—	1997–2009	2001–2009	—
Chile_AcousN	Index	—	2006–2021; 2023–2026	2006–2007; 2009; 2013–2021; 2023–2026	—
Chile_CPUE	Index	—	1998–2026	—	—
Peru_Artis_CPUE	Index	—	2002–2026	—	—
Peru_Ind_CPUE	Index	—	2002–2009; 2011–2014; 2018–2026	—	—
Offshore_CPUE	Index	—	2008–2025	—	—

Table 28: Definitions for the principal population and observation equations.

Symbol	Definition
y, a, f, s	Year, age, fishery and index; the fitted model spans 1970–2026 and ages 1–12 with a terminal plus group.
$N_{y,a}$	Numbers at age at the start of the year.
$M_a, F_{f,y,a}, Z_{y,a}$	Natural, fleet-specific fishing and total instantaneous mortality; $Z_{y,a} = M_a + \sum_f F_{f,y,a}$.
$w_{f,y,a}, m_a$	Mean weight at age and maturity at age.
$q_{s,y}, S_{s,y,a}, \Delta_s$	Index catchability, selectivity and fraction of the year before observation.
$\Gamma_{l,a}$	Probability that a fish of age a occurs in fork-length bin l ; finite-bin probabilities are normalized across the modeled length range.

Table 29: Principal JJM population and observation equations. Catch uses total mortality in the denominator of the Baranov equation. Recruitment, stock allocation, selectivity constraints and estimation phases follow the selected control file.

Component	Equation
Catch in numbers	$C_{f,y,a} = N_{y,a} \frac{F_{f,y,a}}{Z_{y,a}} (1 - e^{-Z_{y,a}})$
Catch biomass	$Y_{f,y} = \sum_a C_{f,y,a} w_{f,y,a}$
Survival to the next age	$N_{y+1,a+1} = N_{y,a} e^{-Z_{y,a}}$ for ages below the plus group, before any configured movement.
Plus group A	$N_{y+1,A} = N_{y,A-1} e^{-Z_{y,A-1}} + N_{y,A} e^{-Z_{y,A}}$, before any configured movement.
Biomass index	$\hat{I}_{s,y} = q_{s,y} \sum_a N_{y,a} S_{s,y,a} w_{s,y,a} e^{-\Delta_s Z_{y,a}}$
Fishery age composition	$p_{f,y,a} = C_{f,y,a} / \sum_a C_{f,y,a}$
Fishery length composition	$C_{f,y,l} = \sum_a \Gamma_{l,a} C_{f,y,a}$ and $p_{f,y,l} = C_{f,y,l} / \sum_l C_{f,y,l}$
Spawning biomass	$SSB_y = \sum_a N_{y,a} w_{y,a}^{sp} m_a e^{-\Delta_{sp} Z_{y,a}}$

Table 30: Components of the model objective. Numerical weights and estimation phases are read from the current input and control files rather than inherited from the previous annex.

Component	Role in the penalized objective function
Catch and abundance-index observations	Log-scale residual contributions weighted by the specified observation uncertainty.
Age and length compositions	Multinomial composition contributions weighted by effective sample sizes.
Recruitment	Contributions from the selected stock–recruitment relationship, recruitment deviations and their specified uncertainty.
Selectivity and catchability	Regularity penalties, change constraints and priors specified in the selected control file.
Biological parameters	Fixed values or prior contributions and estimation phases specified in the selected control file.
Total	Sum of the active likelihood and penalty contributions.

Table 31: Ranges of annual catch and index CVs and effective composition sample sizes in model 1.06. Catch CVs are supplied directly; index CVs are the supplied standard errors divided by the corresponding observed index values, including the input adjustments for observation weight. A dash indicates no observations. These quantities specify observation precision and composition weight; they are neither counts of fish measured nor posterior uncertainty.

Source	Catch/index CV	Age effective N	Length effective N
N_Chile	0.050–0.050	2.387–23.866	—
SC_Chile_PS	0.050–0.050	6.428–64.277	—
FarNorth	0.050–0.050	—	2.157–79.739
Offshore_Trawl	0.050–0.050	1.258–12.576	—
Chile_AcousCS	0.300–0.300	6.804–6.804	—
Chile_AcousN	0.300–0.300	12.439–12.439	—
Chile_CPUE	0.051–0.424	—	—
Peru_Artis_CPUE	0.194–1.279	—	—
Peru_Ind_CPUE	0.194–1.279	—	—
Offshore_CPUE	0.200–0.200	—	—

Table 32: Selectivity mapping in the single-stock 1.06 control file. A shared selectivity source means that the source profile is used for that fleet or index. Change years and curvature weights refer to the source profile; negative estimation phases indicate a fixed profile.

Source	Stock	Selectivity source	Change years	Estimation phase	Curvature weight
N_Chile	Stock 1	N_Chile	1984–2026	2	1.0
SC_Chile_PS	Stock 1	SC_Chile_PS	1984–2026	3	25.0
FarNorth	Stock 1	FarNorth	1980; 1999; 2010; 2020–2021	4	12.5
Offshore_Trawl	Stock 1	Offshore_Trawl	1980–1991; 2000–2003; 2006–2026	3	12.5
Chile_AcousCS	Stock 1	Chile_AcousCS	2005	2	100.0
Chile_AcousN	Stock 1	Chile_AcousN	2012; 2016	5	100.0
Chile_CPUE	Stock 1	SC_Chile_PS	1984–2026	3	25.0
Peru_Artis_CPUE	Stock 1	FarNorth	1980; 1999; 2010; 2020–2021	4	12.5
Peru_Ind_CPUE	Stock 1	FarNorth	1980; 1999; 2010; 2020–2021	4	12.5
Offshore_CPUE	Stock 1	Offshore_Trawl	1980–1991; 2000–2003; 2006–2026	3	12.5

Table 33: Selectivity mapping for the southern component of the two-stock 1.06 control file.

Source	Stock	Selectivity source	Change years	Estimation phase	Curvature weight
N_Chile	Stock 1	N_Chile	1984–2026	2	1.0
SC_Chile_PS	Stock 1	SC_Chile_PS	1984–2026	3	25.0
Offshore_Trawl	Stock 1	Offshore_Trawl	1980–1991; 2000–2003; 2006–2026	3	12.5
Chile_AcousCS	Stock 1	Chile_AcousCS	2005	2	100.0
Chile_AcousN	Stock 1	Chile_AcousN	2012; 2016	5	100.0
Chile_CPUE	Stock 1	SC_Chile_PS	1984–2026	3	25.0

Source	Stock	Selectivity source	Change years	Estimation phase	Curvature weight
Offshore_CPUE	Stock 1	Offshore_Trawl	1980–1991; 2000–2003; 2006–2026	3	12.5

Table 34: Selectivity mapping for the Far North component of the two-stock 1.06 control file.

Source	Stock	Selectivity source	Change years	Estimation phase	Curvature weight
FarNorth	Stock 2	FarNorth	1980; 1999; 2010; 2020	4	12.5
Peru_Artis_CPUE	Stock 2	FarNorth	1980; 1999; 2010; 2020	4	12.5
Peru_Ind_CPUE	Stock 2	FarNorth	1980; 1999; 2010; 2020	4	12.5

Table 35: SC14 model progression, using the assessment run register with the 1.04 weight window and 1.06 catch adjustment reconciled to the active input files. Model 0.00 is the SCW17 0.16 configuration; model 1.06 is the current 2026 assessment configuration shown in this annex.

Model	Description
Models 0.x	Data introductions
0.00	Exact 2026 (single stock h1 and two-stock h2) model and data set (model 0.16) from SCW17.
0.01	As 0.00 but with revised catches through 2025 (currently still estimates)
0.02	As 0.01 but with updated 2025 fishery age composition data for N_Chile, SC_Chile, and Offshore_Trawl, and updated 2025 fishery length composition data for FarNorth
0.03	As 0.02 but with updated 2025 weight at age data for all fisheries and their associated CPUE indices
0.04	As 0.03 but replaced offshore CPUE up to 2025
0.05	As 0.04 but with 2026 catch projections
0.06	As 0.05 but with updated 2026 fishery age composition data for N_Chile, SC_Chile, and Offshore_Trawl, and updated 2026 fishery length composition data for FarNorth
0.07	As 0.06 but with updated 2026 weight at age data for N_Chile, SC_Chile, and FarNorth fleets, and for their associated CPUE indices
0.08	As 0.07 but replaced SC_Chile_CPUE index (updated index from SC14-JM10)
0.09	As 0.08 but replaced Peru_CPUE index
0.10	As 0.09 but updated 2026 AcousN index, with associated age composition and weight at age
Models 1.x	Updated Model and Sensitivities
1.00	As 0.10 but with updated model (selectivity changes, recruitment) to 2026; 0.10 data file
1.01	As 1.00 but removing early years (pre-2006) of Acoustic North survey data
1.02	As 1.01 but replaced SC_Chile_CPUE with one from fishing trip database (SC14-JM15)
1.03	As 1.02 but removing the 2026 data point for Chile CPUE
1.04	As 1.01 but replacing offshore-fleet weights at age for 2025 and 2026 with the 2020–2024 mean.
1.05	As 1.04 but adding 22799 tons from Chile high seas to N_Chile fleet for 2025
1.06	As 1.05 but adding 10,000 tonnes to the 2026 Far North catch.
Models 1.xx.yy	Base Model Projections
1.xx.ls	As 1.xx but low steepness and short recruitment time series (2001-2015)

Table 36: Spawning biomass (thousand tonnes) from the archived SC13 model 1.14, SCW17 model 0.16 and current SC14 model 1.06, under the single-stock hypothesis. These are successive assessment configurations with different data and assumptions, rather than retrospective peels.

Year	h1_1.06	h1_0.16	h1_1.14
1970	13,315	13,650	15,472
1971	12,525	12,812	14,391
1972	12,080	12,328	13,411
1973	11,610	11,824	12,436
1974	10,827	11,010	11,402
1975	10,330	10,486	10,540
1976	10,369	10,499	9,889
1977	10,409	10,515	9,410
1978	10,522	10,604	9,206
1979	10,660	10,722	9,061
1980	10,868	10,911	9,079
1981	10,641	10,667	8,858
1982	9,991	10,002	8,397
1983	10,739	10,736	9,377
1984	10,973	10,957	9,737
1985	11,334	11,309	10,252
1986	14,768	14,740	13,835
1987	19,580	19,554	18,422
1988	21,360	21,339	20,188
1989	19,905	19,892	19,005
1990	18,135	18,128	17,486
1991	16,515	16,511	16,284
1992	15,116	15,112	15,293
1993	13,328	13,325	13,662
1994	10,463	10,461	11,048
1995	7,459	7,457	8,051
1996	5,263	5,260	5,866
1997	3,719	3,718	4,528
1998	3,738	3,749	4,518
1999	4,827	4,846	5,494
2000	6,433	6,459	6,672
2001	6,726	6,762	6,977
2002	7,236	7,280	7,547
2003	7,146	7,196	7,500
2004	6,395	6,445	6,817
2005	5,986	6,021	6,159
2006	5,153	5,169	5,059

Year	h1_1.06	h1_0.16	h1_1.14
2007	3,548	3,553	3,333
2008	2,375	2,381	2,245
2009	1,811	1,828	1,851
2010	1,667	1,694	1,813
2011	1,517	1,550	1,731
2012	1,512	1,547	1,752
2013	1,631	1,669	1,870
2014	1,894	1,937	2,191
2015	2,219	2,270	2,629
2016	2,640	2,714	3,375
2017	3,334	3,478	4,840
2018	4,297	4,598	7,145
2019	5,125	5,575	9,474
2020	5,640	6,156	11,169
2021	6,171	6,716	12,906
2022	6,530	7,092	14,336
2023	6,345	6,913	14,691
2024	5,973	6,668	14,745
2025	6,031	6,834	15,183
2026	6,109		

Table 37: Estimated begin-year numbers at age (millions of fish; Model h1_1.06; single-stock hypothesis).

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	5,072	4,755	3,805	3,022	2,425	1,949	1,552	1,235	974	763	594	2,158
1971	6,513	3,832	3,584	2,855	2,251	1,784	1,431	1,155	926	732	574	2,069
1972	8,265	4,919	2,884	2,677	2,110	1,632	1,290	1,056	862	694	549	1,981
1973	5,826	6,243	3,703	2,150	1,989	1,558	1,205	962	792	648	522	1,901
1974	4,794	4,399	4,687	2,728	1,581	1,454	1,140	894	720	594	486	1,815
1975	8,973	3,618	3,291	3,354	1,964	1,123	1,031	831	662	535	441	1,710
1976	12,118	6,770	2,704	2,398	2,405	1,367	780	744	613	492	397	1,598
1977	12,633	9,135	5,034	1,935	1,684	1,633	928	555	544	453	363	1,473
1978	16,048	9,508	6,659	3,018	1,260	1,130	1,107	658	401	392	326	1,322
1979	15,951	12,057	6,904	4,171	1,920	777	699	744	462	284	277	1,167
1980	16,021	11,945	8,719	4,385	2,650	1,181	476	453	493	305	188	954
1981	16,857	11,994	8,533	5,449	2,867	1,689	752	314	302	324	201	752
1982	18,940	12,541	8,288	4,938	3,213	1,639	978	464	198	188	202	594
1983	27,543	13,972	8,538	4,879	2,610	1,497	756	510	255	110	104	442
1984	21,565	20,323	9,723	5,347	2,893	1,474	829	422	280	138	59	295
1985	24,519	15,704	13,412	5,952	3,008	1,332	633	383	201	133	66	169
1986	56,865	17,969	10,724	8,445	3,428	1,481	632	319	199	105	70	123
1987	56,452	41,923	12,774	7,112	5,265	1,884	766	347	177	110	58	106
1988	23,866	41,305	28,652	8,503	4,496	2,881	920	397	186	96	60	89
1989	14,566	17,350	28,343	18,500	5,491	2,744	1,609	494	212	99	51	80
1990	16,338	10,678	12,141	17,898	11,585	3,423	1,627	892	262	111	52	69
1991	19,192	11,999	7,576	7,818	11,464	7,018	1,944	868	469	137	59	64
1992	23,778	14,088	8,440	4,940	5,098	6,915	3,760	914	377	209	65	59
1993	14,931	17,570	9,909	5,563	3,214	3,188	3,840	1,668	290	114	80	48
1994	13,690	10,900	11,793	6,262	3,462	1,926	1,819	1,878	574	81	36	41
1995	17,885	10,011	7,372	7,033	3,610	1,857	933	662	501	123	17	16
1996	13,185	12,926	5,694	3,067	3,032	1,439	716	262	114	46	7	2
1997	16,380	9,137	6,674	1,711	1,030	1,049	465	171	34	5	1	0
1998	17,444	11,343	3,429	1,530	531	364	337	98	15	1	0	0
1999	21,307	12,383	4,883	981	661	253	162	112	18	1	0	0
2000	24,822	15,201	6,585	2,447	527	376	139	75	34	2	0	0
2001	16,786	17,953	8,863	3,538	1,429	323	227	75	30	8	0	0
2002	19,150	11,636	9,349	4,060	1,913	825	181	109	25	4	0	0
2003	9,143	13,757	7,105	5,492	2,325	1,105	455	84	34	4	0	0
2004	9,741	6,460	8,316	4,045	3,150	1,372	642	232	31	7	0	0
2005	13,433	6,844	3,770	4,616	2,207	1,786	782	331	92	7	1	0
2006	7,846	9,354	4,224	2,134	2,457	1,169	976	398	136	22	1	0

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	2,570	5,415	5,422	2,304	1,120	1,155	556	430	145	32	3	0
2008	5,242	1,666	2,838	2,680	1,181	511	448	191	117	28	4	0
2009	7,915	3,135	779	1,313	1,228	504	198	154	55	28	7	1
2010	4,684	5,143	1,437	335	451	355	146	57	39	14	9	2
2011	3,363	2,997	2,568	676	146	164	132	54	23	18	7	5
2012	3,501	2,360	1,825	1,208	332	64	72	54	24	11	8	6
2013	3,517	2,573	1,546	984	689	153	30	35	26	11	5	6
2014	5,125	2,577	1,731	909	574	404	88	16	18	12	4	4
2015	4,931	3,752	1,760	1,069	559	341	238	49	8	8	4	3
2016	6,696	3,625	2,541	1,162	694	340	199	133	24	3	2	2
2017	9,586	4,989	2,571	1,706	746	423	198	111	66	10	1	1
2018	11,923	7,100	3,572	1,780	1,133	455	240	103	50	24	3	1
2019	9,622	8,892	5,146	2,430	1,172	694	260	123	45	18	6	1
2020	7,925	7,219	6,456	3,509	1,625	709	386	131	54	17	6	2
2021	10,559	5,964	5,364	4,642	2,396	1,031	393	185	52	17	4	2
2022	6,490	7,929	4,422	3,868	3,212	1,533	594	193	73	15	3	1
2023	5,071	4,874	5,858	3,202	2,672	2,112	939	320	83	21	2	1
2024	8,332	3,806	3,600	4,181	2,134	1,690	1,302	534	153	25	2	0
2025	14,314	6,271	2,829	2,567	2,735	1,262	973	715	245	36	1	0
2026	5,560	10,750	4,635	2,001	1,582	1,508	651	491	317	60	1	0

Table 38: Estimated begin-year numbers at age (millions of fish; Model h2_1.06; two-stock hypothesis; southern stock).

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	6,881	5,470	4,378	3,485	2,752	2,158	1,684	1,306	1,008	774	593	2,033
1971	6,496	5,198	4,124	3,289	2,600	2,027	1,587	1,254	980	759	583	1,976
1972	6,073	4,906	3,914	3,089	2,437	1,891	1,471	1,174	938	736	570	1,922
1973	5,441	4,587	3,697	2,937	2,304	1,804	1,399	1,099	882	706	554	1,875
1974	5,408	4,109	3,450	2,762	2,176	1,690	1,324	1,041	824	663	531	1,826
1975	6,984	4,083	3,087	2,570	2,027	1,557	1,206	971	775	617	496	1,765
1976	8,416	5,270	3,055	2,279	1,856	1,417	1,087	874	719	578	460	1,687
1977	10,825	6,345	3,923	2,229	1,612	1,257	959	775	642	533	429	1,593
1978	13,420	8,158	4,710	2,845	1,567	1,090	852	684	569	476	395	1,499
1979	13,769	10,089	5,983	3,322	1,898	969	674	577	486	411	344	1,371
1980	13,977	10,312	7,343	4,170	2,188	1,168	593	440	388	329	279	1,162
1981	15,371	10,468	7,500	5,119	2,769	1,387	741	394	298	264	224	981
1982	16,935	11,438	7,358	4,867	3,062	1,567	797	460	253	193	171	781
1983	28,879	12,479	7,776	4,448	2,558	1,395	711	419	258	144	110	542
1984	24,011	21,275	8,672	4,984	2,628	1,427	768	400	233	142	79	359
1985	24,803	17,469	14,140	5,529	2,785	1,179	602	357	193	114	69	214
1986	56,516	18,169	12,009	9,120	3,167	1,345	551	305	187	102	60	150
1987	49,925	41,664	12,949	8,094	5,718	1,728	689	302	170	104	57	118
1988	21,887	36,502	28,520	8,727	5,162	3,159	842	357	163	93	57	97
1989	12,378	15,894	25,043	18,909	5,684	3,174	1,776	452	191	88	51	84
1990	16,430	9,074	11,151	16,202	11,923	3,556	1,890	988	241	101	47	72
1991	20,216	12,084	6,514	7,653	10,475	7,258	2,033	1,020	527	128	55	64
1992	21,856	14,861	8,595	4,499	5,038	6,311	3,912	968	454	242	63	58
1993	13,119	16,133	10,511	5,844	2,940	3,152	3,494	1,756	317	144	96	48
1994	13,493	9,544	10,828	6,884	3,678	1,764	1,797	1,699	615	92	47	47
1995	10,771	9,824	6,448	6,775	4,037	1,986	849	646	444	132	19	19
1996	12,679	7,706	5,673	3,287	3,095	1,650	773	239	112	41	7	2
1997	13,926	8,663	3,762	2,317	1,260	1,143	559	196	34	6	1	0
1998	14,334	9,564	3,536	1,298	858	495	400	132	22	1	0	0
1999	15,997	10,130	4,441	1,700	662	454	244	154	31	2	0	0
2000	18,155	11,288	5,575	2,497	994	400	268	126	57	6	0	0
2001	18,794	13,011	6,720	3,237	1,526	634	252	155	58	15	1	0
2002	17,828	13,084	7,608	3,721	1,871	926	378	134	62	12	1	0
2003	12,175	12,837	8,285	4,564	2,160	1,109	538	194	50	12	1	0
2004	7,735	8,670	8,116	5,000	2,653	1,292	664	294	82	12	1	0
2005	8,873	5,450	5,458	4,818	2,805	1,528	751	359	130	22	1	0
2006	5,660	6,095	3,428	3,298	2,649	1,527	846	389	155	35	2	0

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	2,466	3,844	3,681	2,084	1,835	1,315	749	379	145	39	4	0
2008	5,735	1,542	2,052	2,017	1,122	881	548	268	105	30	5	1
2009	4,658	3,277	755	1,071	986	507	371	205	82	28	8	1
2010	3,244	2,802	1,549	358	399	310	158	111	55	24	9	3
2011	3,061	1,885	1,137	695	156	150	117	56	42	24	11	6
2012	2,960	2,102	1,268	731	375	73	67	47	26	22	12	9
2013	3,346	2,163	1,485	839	425	179	34	32	24	13	11	11
2014	5,133	2,442	1,501	964	492	241	101	18	17	11	6	9
2015	4,980	3,745	1,716	1,007	607	288	136	55	9	7	4	6
2016	6,113	3,651	2,548	1,156	655	366	163	73	26	4	2	3
2017	7,633	4,547	2,594	1,728	740	396	209	88	35	11	1	2
2018	10,456	5,631	3,242	1,798	1,141	446	218	105	38	13	3	1
2019	9,033	7,786	4,108	2,257	1,186	694	251	109	44	13	4	1
2020	7,570	6,777	5,765	2,946	1,521	715	383	125	47	17	5	2
2021	11,428	5,699	5,052	4,224	2,072	969	395	182	50	16	5	2
2022	7,911	8,586	4,229	3,654	2,968	1,341	555	192	71	15	3	1
2023	5,606	5,945	6,358	3,082	2,581	1,995	818	294	82	21	2	1
2024	8,296	4,212	4,409	4,599	2,136	1,703	1,242	460	138	25	2	0
2025	13,968	6,248	3,142	3,196	3,156	1,327	997	679	206	33	1	0
2026	5,312	10,495	4,634	2,261	2,110	1,890	715	511	297	49	1	0

Table 39: Estimated begin-year numbers at age (millions of fish; Model h2_1.06; two-stock hypothesis; northern stock).

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	2,681	2,583	2,036	1,342	917	643	457	326	233	167	120	306
1971	3,348	1,928	1,856	1,458	963	659	463	329	235	168	120	306
1972	4,486	2,407	1,385	1,324	1,044	691	473	332	236	169	121	306
1973	3,149	3,225	1,727	974	941	749	497	340	239	170	121	307
1974	2,200	2,264	2,308	1,183	683	673	537	357	245	172	122	308
1975	4,111	1,581	1,608	1,462	798	483	481	385	256	176	123	309
1976	5,366	2,954	1,132	1,108	1,028	570	347	346	277	184	126	310
1977	3,646	3,856	2,110	757	768	733	409	249	248	199	132	314
1978	4,162	2,608	2,584	709	370	502	511	290	178	178	142	319
1979	3,627	2,982	1,788	1,114	393	250	354	364	207	127	127	330
1980	3,604	2,599	2,050	792	627	266	176	252	260	149	91	328
1981	3,418	2,578	1,683	684	464	428	187	125	180	186	106	300
1982	3,648	2,442	1,619	449	378	313	299	133	89	129	133	291
1983	2,519	2,618	1,688	874	299	267	223	214	95	64	92	304
1984	1,161	1,809	1,828	980	594	212	191	160	154	68	46	285
1985	2,042	833	1,245	956	648	418	151	136	115	110	49	237
1986	3,497	1,466	580	706	645	458	299	108	98	82	79	206
1987	9,135	2,511	1,030	353	485	459	328	214	78	70	59	205
1988	3,527	6,562	1,774	652	245	346	329	235	154	56	51	190
1989	3,162	2,529	4,468	855	422	172	246	235	169	110	40	172
1990	1,794	2,270	1,761	2,535	577	299	123	176	169	121	79	152
1991	1,460	1,286	1,526	772	1,599	402	212	88	126	121	87	166
1992	2,839	1,047	874	727	498	1,119	286	151	63	91	87	181
1993	2,854	2,037	722	464	482	351	797	204	108	45	65	192
1994	2,468	2,046	1,379	333	296	337	249	570	146	78	32	185
1995	5,567	1,768	1,364	569	207	205	238	178	408	105	56	155
1996	1,736	3,952	967	132	240	130	140	166	126	290	75	150
1997	3,097	1,228	1,997	52	48	146	87	97	117	89	205	159
1998	3,162	2,177	541	40	15	27	95	60	68	83	63	257
1999	5,335	2,157	500	0	3	6	15	60	40	46	57	219
2000	7,722	3,808	956	128	0	2	4	11	43	29	33	197
2001	2,022	5,523	1,936	328	74	0	1	3	8	31	20	164
2002	5,143	1,438	1,940	301	150	46	0	1	2	6	22	131
2003	548	3,691	921	1,089	201	105	33	0	1	1	4	110
2004	3,104	393	2,248	464	705	140	74	23	0	1	1	82
2005	6,979	2,228	250	1,241	309	494	99	53	17	0	0	59
2006	3,234	5,011	1,465	148	844	218	352	71	38	12	0	43

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	1,019	2,319	3,027	726	96	586	154	251	51	27	9	31
2008	1,095	731	1,487	1,704	485	67	416	110	180	37	20	28
2009	8,615	786	471	844	1,142	341	48	298	79	129	26	34
2010	2,324	6,185	516	278	572	806	243	34	214	57	93	43
2011	1,319	1,670	4,386	359	197	409	578	175	25	154	41	98
2012	3,018	946	1,093	2,516	235	137	289	412	125	18	110	100
2013	2,232	2,166	626	645	1,670	164	97	206	296	90	13	151
2014	2,383	1,603	1,479	398	441	1,176	117	69	148	212	65	117
2015	2,147	1,712	1,106	963	275	312	838	84	50	106	153	131
2016	3,273	1,543	1,214	770	683	196	224	602	60	36	76	203
2017	6,362	2,352	1,099	853	548	489	141	161	433	43	26	201
2018	5,351	4,573	1,682	780	610	393	351	101	115	311	31	163
2019	4,415	3,845	3,209	1,140	548	435	281	252	73	83	224	140
2020	4,958	3,171	2,654	2,091	788	387	310	202	181	52	60	261
2021	4,711	3,561	2,268	1,869	1,406	526	272	222	145	130	38	230
2022	2,227	3,385	2,550	1,605	1,277	956	371	195	159	104	93	192
2023	2,243	1,599	2,420	1,795	1,078	851	671	266	140	114	75	205
2024	3,033	1,611	1,140	1,685	1,163	690	591	479	191	100	82	201
2025	3,222	2,178	1,148	793	1,088	742	478	421	344	137	72	203
2026	3,206	2,313	1,550	794	503	680	511	341	302	247	98	198

Table 40: Estimated total fishing mortality at age (per year; Model h1_1.06; single-stock hypothesis).

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	0.000	0.003	0.007	0.014	0.027	0.029	0.016	0.008	0.005	0.005	0.005	0.005
1971	0.001	0.004	0.012	0.022	0.041	0.044	0.024	0.012	0.008	0.008	0.008	0.008
1972	0.001	0.004	0.014	0.017	0.023	0.023	0.013	0.008	0.006	0.006	0.006	0.006
1973	0.001	0.007	0.026	0.027	0.034	0.033	0.018	0.011	0.009	0.008	0.008	0.008
1974	0.001	0.010	0.055	0.049	0.062	0.064	0.035	0.020	0.017	0.017	0.017	0.017
1975	0.002	0.011	0.036	0.052	0.082	0.084	0.046	0.024	0.017	0.017	0.017	0.017
1976	0.003	0.016	0.055	0.073	0.107	0.108	0.060	0.032	0.024	0.023	0.023	0.023
1977	0.004	0.036	0.232	0.149	0.119	0.109	0.064	0.045	0.049	0.049	0.049	0.049
1978	0.006	0.040	0.188	0.173	0.203	0.200	0.117	0.073	0.066	0.065	0.065	0.065
1979	0.009	0.044	0.174	0.174	0.205	0.211	0.155	0.133	0.135	0.134	0.134	0.134
1980	0.009	0.056	0.190	0.145	0.170	0.171	0.134	0.126	0.138	0.137	0.137	0.137
1981	0.016	0.090	0.267	0.248	0.279	0.266	0.203	0.183	0.194	0.192	0.192	0.192
1982	0.024	0.105	0.250	0.357	0.484	0.494	0.371	0.317	0.310	0.309	0.309	0.309
1983	0.024	0.083	0.188	0.243	0.291	0.311	0.304	0.322	0.338	0.337	0.337	0.337
1984	0.037	0.136	0.211	0.295	0.496	0.566	0.491	0.462	0.461	0.460	0.460	0.460
1985	0.031	0.101	0.183	0.272	0.429	0.465	0.403	0.376	0.373	0.370	0.370	0.370
1986	0.025	0.061	0.131	0.192	0.319	0.378	0.321	0.311	0.315	0.313	0.313	0.313
1987	0.032	0.101	0.127	0.179	0.323	0.436	0.377	0.341	0.334	0.330	0.330	0.330
1988	0.039	0.097	0.157	0.157	0.214	0.302	0.342	0.348	0.349	0.344	0.344	0.344
1989	0.030	0.077	0.180	0.188	0.193	0.243	0.311	0.356	0.369	0.362	0.362	0.362
1990	0.029	0.063	0.160	0.165	0.221	0.285	0.348	0.361	0.370	0.356	0.356	0.356
1991	0.029	0.072	0.148	0.147	0.225	0.344	0.475	0.553	0.527	0.456	0.456	0.456
1992	0.023	0.072	0.137	0.150	0.189	0.308	0.533	0.869	0.913	0.677	0.677	0.677
1993	0.035	0.119	0.179	0.194	0.232	0.281	0.435	0.786	0.995	0.867	0.867	0.867
1994	0.033	0.111	0.237	0.271	0.343	0.444	0.731	1.042	1.259	1.301	1.301	1.301
1995	0.045	0.284	0.597	0.561	0.640	0.672	0.991	1.481	2.097	2.567	2.567	2.567
1996	0.087	0.381	0.922	0.811	0.782	0.850	1.152	1.751	2.751	3.657	3.657	3.657
1997	0.087	0.700	1.193	0.891	0.759	0.855	1.281	2.125	3.593	4.946	4.946	4.946
1998	0.063	0.563	0.971	0.560	0.459	0.531	0.822	1.414	2.475	3.544	3.544	3.544
1999	0.058	0.352	0.411	0.341	0.284	0.318	0.488	0.899	1.736	2.650	2.650	2.650
2000	0.044	0.259	0.341	0.258	0.210	0.225	0.339	0.625	1.235	1.962	1.962	1.962
2001	0.086	0.373	0.501	0.335	0.270	0.301	0.453	0.830	1.631	2.539	2.539	2.539
2002	0.051	0.213	0.252	0.277	0.269	0.314	0.489	0.884	1.642	2.552	2.552	2.552
2003	0.067	0.223	0.283	0.276	0.247	0.263	0.395	0.725	1.363	2.124	2.124	2.124
2004	0.073	0.259	0.309	0.326	0.288	0.283	0.383	0.640	1.179	1.906	1.906	1.906
2005	0.082	0.203	0.289	0.351	0.355	0.324	0.396	0.611	1.144	2.116	2.116	2.116
2006	0.091	0.265	0.326	0.365	0.474	0.463	0.540	0.728	1.177	1.850	1.850	1.850

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	0.153	0.366	0.425	0.388	0.506	0.667	0.789	1.018	1.364	1.865	1.865	1.865
2008	0.234	0.480	0.490	0.501	0.573	0.666	0.785	0.968	1.151	1.136	1.136	1.136
2009	0.151	0.500	0.563	0.788	0.962	0.955	0.972	1.085	1.056	0.911	0.911	0.911
2010	0.166	0.415	0.475	0.554	0.732	0.709	0.726	0.618	0.530	0.467	0.467	0.467
2011	0.074	0.216	0.474	0.430	0.538	0.546	0.619	0.508	0.507	0.491	0.491	0.491
2012	0.028	0.143	0.338	0.281	0.495	0.492	0.445	0.428	0.495	0.555	0.555	0.555
2013	0.031	0.116	0.251	0.258	0.254	0.275	0.321	0.383	0.505	0.642	0.642	0.642
2014	0.032	0.101	0.202	0.206	0.240	0.251	0.296	0.403	0.575	0.784	0.784	0.784
2015	0.028	0.110	0.135	0.152	0.219	0.262	0.302	0.424	0.655	0.922	0.922	0.922
2016	0.014	0.064	0.118	0.164	0.215	0.261	0.305	0.416	0.640	0.911	0.911	0.911
2017	0.020	0.054	0.088	0.130	0.214	0.289	0.375	0.504	0.719	0.933	0.933	0.933
2018	0.013	0.042	0.105	0.138	0.209	0.281	0.385	0.553	0.774	1.056	1.056	1.056
2019	0.007	0.040	0.103	0.122	0.223	0.307	0.407	0.551	0.710	0.875	0.875	0.875
2020	0.004	0.017	0.050	0.102	0.174	0.311	0.458	0.641	0.851	1.088	1.088	1.088
2021	0.006	0.019	0.047	0.088	0.166	0.272	0.432	0.653	0.959	1.417	1.417	1.417
2022	0.006	0.023	0.043	0.090	0.139	0.210	0.340	0.556	0.953	1.711	1.711	1.711
2023	0.007	0.023	0.057	0.126	0.178	0.204	0.285	0.459	0.930	2.004	2.004	2.004
2024	0.004	0.017	0.058	0.144	0.245	0.272	0.320	0.500	1.154	3.295	3.295	3.295
2025	0.006	0.022	0.067	0.204	0.316	0.381	0.404	0.534	1.122	3.079	3.079	3.079
2026	0.009	0.021	0.049	0.125	0.240	0.295	0.344	0.469	0.899	1.921	1.921	1.921

Table 41: Estimated total fishing mortality at age (per year; Model h2_1.06; two-stock hypothesis; southern stock).

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	0.000	0.002	0.006	0.013	0.026	0.027	0.014	0.007	0.004	0.004	0.004	0.004
1971	0.001	0.004	0.009	0.020	0.039	0.041	0.021	0.010	0.006	0.006	0.006	0.006
1972	0.001	0.003	0.007	0.013	0.021	0.021	0.012	0.006	0.004	0.004	0.004	0.004
1973	0.001	0.005	0.011	0.020	0.030	0.029	0.016	0.008	0.005	0.005	0.005	0.005
1974	0.001	0.006	0.015	0.030	0.055	0.057	0.030	0.015	0.009	0.009	0.009	0.009
1975	0.002	0.010	0.024	0.045	0.078	0.080	0.042	0.021	0.013	0.013	0.013	0.013
1976	0.002	0.015	0.035	0.066	0.109	0.110	0.058	0.029	0.019	0.019	0.019	0.019
1977	0.003	0.018	0.041	0.072	0.111	0.109	0.058	0.030	0.020	0.019	0.019	0.019
1978	0.005	0.030	0.069	0.125	0.201	0.201	0.111	0.061	0.044	0.043	0.043	0.043
1979	0.009	0.038	0.081	0.138	0.205	0.211	0.146	0.116	0.110	0.109	0.109	0.109
1980	0.009	0.038	0.081	0.130	0.176	0.176	0.128	0.109	0.106	0.105	0.105	0.105
1981	0.016	0.073	0.153	0.234	0.289	0.274	0.196	0.163	0.157	0.154	0.154	0.154
1982	0.025	0.106	0.223	0.363	0.506	0.510	0.363	0.300	0.286	0.282	0.282	0.282
1983	0.026	0.084	0.165	0.246	0.304	0.317	0.296	0.307	0.319	0.317	0.317	0.317
1984	0.038	0.128	0.170	0.302	0.522	0.583	0.487	0.446	0.439	0.436	0.436	0.436
1985	0.031	0.095	0.159	0.277	0.448	0.481	0.401	0.366	0.357	0.353	0.353	0.353
1986	0.025	0.059	0.115	0.187	0.326	0.388	0.321	0.303	0.304	0.300	0.300	0.300
1987	0.033	0.099	0.115	0.170	0.313	0.439	0.378	0.336	0.323	0.316	0.316	0.316
1988	0.040	0.097	0.131	0.149	0.206	0.296	0.341	0.344	0.338	0.328	0.328	0.328
1989	0.030	0.074	0.155	0.181	0.189	0.238	0.306	0.351	0.359	0.348	0.348	0.348
1990	0.027	0.051	0.096	0.156	0.216	0.279	0.337	0.349	0.350	0.336	0.336	0.336
1991	0.028	0.061	0.090	0.138	0.227	0.338	0.462	0.529	0.497	0.430	0.430	0.430
1992	0.024	0.066	0.106	0.145	0.189	0.311	0.521	0.837	0.867	0.643	0.643	0.643
1993	0.038	0.119	0.143	0.183	0.231	0.282	0.441	0.769	0.958	0.835	0.835	0.835
1994	0.037	0.112	0.189	0.254	0.336	0.451	0.744	1.062	1.259	1.307	1.307	1.307
1995	0.055	0.269	0.394	0.503	0.615	0.663	0.989	1.476	2.094	2.620	2.620	2.620
1996	0.101	0.437	0.616	0.679	0.716	0.802	1.094	1.677	2.685	3.726	3.726	3.726
1997	0.096	0.616	0.784	0.713	0.654	0.771	1.165	1.924	3.350	4.876	4.876	4.876
1998	0.067	0.487	0.453	0.394	0.356	0.427	0.672	1.166	2.127	3.286	3.286	3.286
1999	0.069	0.317	0.296	0.257	0.223	0.249	0.382	0.713	1.448	2.457	2.457	2.457
2000	0.053	0.239	0.264	0.213	0.170	0.182	0.270	0.499	1.031	1.871	1.871	1.871
2001	0.082	0.257	0.311	0.268	0.219	0.238	0.350	0.629	1.260	2.321	2.321	2.321
2002	0.048	0.177	0.231	0.264	0.243	0.264	0.388	0.705	1.403	2.498	2.498	2.498
2003	0.060	0.179	0.225	0.262	0.234	0.233	0.324	0.582	1.174	2.082	2.082	2.082
2004	0.070	0.183	0.241	0.298	0.272	0.263	0.335	0.535	1.030	1.864	1.864	1.864
2005	0.096	0.184	0.224	0.318	0.328	0.311	0.377	0.561	1.041	2.091	2.091	2.091
2006	0.107	0.224	0.218	0.306	0.420	0.432	0.524	0.706	1.092	1.789	1.789	1.789

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	0.190	0.348	0.322	0.339	0.454	0.595	0.747	1.000	1.299	1.799	1.799	1.799
2008	0.280	0.434	0.370	0.435	0.513	0.584	0.704	0.904	1.047	1.058	1.058	1.058
2009	0.228	0.469	0.465	0.708	0.877	0.888	0.931	1.034	0.960	0.813	0.813	0.813
2010	0.263	0.621	0.522	0.553	0.699	0.698	0.761	0.679	0.559	0.486	0.486	0.486
2011	0.096	0.116	0.162	0.337	0.482	0.529	0.624	0.470	0.392	0.366	0.366	0.366
2012	0.034	0.067	0.134	0.262	0.461	0.476	0.445	0.402	0.406	0.424	0.424	0.424
2013	0.035	0.085	0.152	0.253	0.288	0.294	0.334	0.381	0.455	0.530	0.530	0.530
2014	0.035	0.073	0.119	0.182	0.256	0.295	0.320	0.412	0.540	0.672	0.672	0.672
2015	0.031	0.105	0.115	0.150	0.226	0.290	0.345	0.457	0.644	0.827	0.827	0.827
2016	0.016	0.062	0.109	0.165	0.224	0.278	0.339	0.458	0.632	0.811	0.811	0.811
2017	0.024	0.058	0.087	0.135	0.226	0.314	0.412	0.553	0.735	0.856	0.856	0.856
2018	0.015	0.035	0.082	0.136	0.217	0.298	0.416	0.591	0.786	0.972	0.972	0.972
2019	0.007	0.020	0.052	0.115	0.226	0.314	0.419	0.559	0.685	0.790	0.790	0.790
2020	0.004	0.014	0.031	0.072	0.171	0.314	0.464	0.641	0.817	0.984	0.984	0.984
2021	0.006	0.018	0.044	0.073	0.155	0.277	0.441	0.658	0.932	1.305	1.305	1.305
2022	0.006	0.020	0.036	0.068	0.117	0.215	0.355	0.569	0.936	1.585	1.585	1.585
2023	0.006	0.019	0.044	0.087	0.136	0.194	0.296	0.477	0.923	1.859	1.859	1.859
2024	0.004	0.013	0.042	0.096	0.196	0.255	0.324	0.524	1.164	3.066	3.066	3.066
2025	0.006	0.019	0.049	0.135	0.233	0.339	0.389	0.546	1.151	2.897	2.897	2.897
2026	0.009	0.018	0.036	0.081	0.168	0.236	0.307	0.445	0.862	1.775	1.775	1.775

Table 42: Estimated total fishing mortality at age (per year; Model h2_1.06; two-stock hypothesis; far north stock).

Year	Age group (years)											
	1	2	3	4	5	6	7	8	9	10	11	12
1970	0.000	0.000	0.004	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1971	0.000	0.001	0.008	0.004	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1972	0.000	0.002	0.022	0.011	0.003	0.001	0.000	0.000	0.000	0.000	0.000	0.000
1973	0.000	0.004	0.048	0.024	0.006	0.002	0.001	0.000	0.000	0.000	0.000	0.000
1974	0.001	0.012	0.127	0.064	0.016	0.005	0.002	0.001	0.001	0.001	0.001	0.001
1975	0.000	0.004	0.043	0.022	0.005	0.002	0.001	0.000	0.000	0.000	0.000	0.000
1976	0.000	0.007	0.072	0.036	0.009	0.003	0.001	0.001	0.000	0.000	0.000	0.000
1977	0.005	0.070	0.760	0.385	0.095	0.030	0.014	0.008	0.005	0.005	0.005	0.005
1978	0.003	0.047	0.511	0.259	0.064	0.020	0.009	0.005	0.004	0.004	0.004	0.004
1979	0.003	0.045	0.485	0.246	0.060	0.019	0.009	0.005	0.003	0.003	0.003	0.003
1980	0.005	0.105	0.767	0.204	0.050	0.021	0.011	0.006	0.004	0.004	0.004	0.004
1981	0.006	0.135	0.991	0.263	0.065	0.028	0.014	0.008	0.006	0.006	0.006	0.006
1982	0.002	0.039	0.287	0.076	0.019	0.008	0.004	0.002	0.002	0.002	0.002	0.002
1983	0.001	0.029	0.213	0.057	0.014	0.006	0.003	0.002	0.001	0.001	0.001	0.001
1984	0.002	0.043	0.318	0.084	0.021	0.009	0.005	0.003	0.002	0.002	0.002	0.002
1985	0.001	0.032	0.238	0.063	0.016	0.007	0.003	0.002	0.001	0.001	0.001	0.001
1986	0.001	0.023	0.166	0.044	0.011	0.005	0.002	0.001	0.001	0.001	0.001	0.001
1987	0.001	0.017	0.127	0.034	0.008	0.004	0.002	0.001	0.001	0.001	0.001	0.001
1988	0.003	0.055	0.400	0.106	0.026	0.011	0.006	0.003	0.002	0.002	0.002	0.002
1989	0.001	0.032	0.237	0.063	0.016	0.007	0.003	0.002	0.001	0.001	0.001	0.001
1990	0.003	0.067	0.494	0.131	0.033	0.014	0.007	0.004	0.003	0.003	0.003	0.003
1991	0.003	0.056	0.411	0.109	0.027	0.011	0.006	0.003	0.002	0.002	0.002	0.002
1992	0.002	0.041	0.304	0.081	0.020	0.008	0.004	0.002	0.002	0.002	0.002	0.002
1993	0.003	0.060	0.443	0.118	0.029	0.012	0.006	0.004	0.002	0.002	0.002	0.002
1994	0.003	0.076	0.555	0.147	0.037	0.016	0.008	0.005	0.003	0.003	0.003	0.003
1995	0.013	0.273	2.002	0.531	0.132	0.056	0.029	0.016	0.011	0.011	0.011	0.011
1996	0.016	0.353	2.586	0.686	0.170	0.072	0.037	0.021	0.015	0.015	0.015	0.015
1997	0.023	0.489	3.588	0.952	0.236	0.100	0.051	0.029	0.020	0.020	0.020	0.020
1998	0.052	1.140	8.361	2.219	0.550	0.233	0.120	0.068	0.047	0.047	0.047	0.047
1999	0.007	0.483	1.034	0.307	0.097	0.046	0.023	0.012	0.008	0.008	0.008	0.008
2000	0.005	0.346	0.740	0.220	0.069	0.033	0.016	0.008	0.005	0.005	0.005	0.005
2001	0.011	0.716	1.532	0.454	0.144	0.068	0.034	0.017	0.011	0.011	0.011	0.011
2002	0.002	0.116	0.247	0.073	0.023	0.011	0.005	0.003	0.002	0.002	0.002	0.002
2003	0.002	0.166	0.354	0.105	0.033	0.016	0.008	0.004	0.003	0.003	0.003	0.003
2004	0.002	0.123	0.264	0.078	0.025	0.012	0.006	0.003	0.002	0.002	0.002	0.002
2005	0.001	0.089	0.190	0.056	0.018	0.008	0.004	0.002	0.001	0.001	0.001	0.001
2006	0.003	0.174	0.372	0.110	0.035	0.017	0.008	0.004	0.003	0.003	0.003	0.003

	Age group (years)											
Year	1	2	3	4	5	6	7	8	9	10	11	12
2007	0.002	0.114	0.245	0.073	0.023	0.011	0.005	0.003	0.002	0.002	0.002	0.002
2008	0.002	0.111	0.237	0.070	0.022	0.011	0.005	0.003	0.002	0.002	0.002	0.002
2009	0.001	0.091	0.195	0.058	0.018	0.009	0.004	0.002	0.001	0.001	0.001	0.001
2010	0.000	0.014	0.033	0.013	0.006	0.003	0.001	0.000	0.000	0.000	0.000	0.000
2011	0.002	0.094	0.226	0.091	0.038	0.018	0.007	0.003	0.002	0.002	0.002	0.002
2012	0.002	0.082	0.198	0.080	0.033	0.016	0.006	0.003	0.002	0.002	0.002	0.002
2013	0.001	0.051	0.124	0.050	0.021	0.010	0.004	0.002	0.001	0.001	0.001	0.001
2014	0.001	0.041	0.099	0.040	0.017	0.008	0.003	0.001	0.001	0.001	0.001	0.001
2015	0.000	0.014	0.033	0.013	0.005	0.003	0.001	0.000	0.000	0.000	0.000	0.000
2016	0.000	0.009	0.022	0.009	0.004	0.002	0.001	0.000	0.000	0.000	0.000	0.000
2017	0.000	0.005	0.013	0.005	0.002	0.001	0.000	0.000	0.000	0.000	0.000	0.000
2018	0.001	0.024	0.059	0.024	0.010	0.005	0.002	0.001	0.000	0.000	0.000	0.000
2019	0.001	0.041	0.098	0.040	0.017	0.008	0.003	0.001	0.001	0.001	0.001	0.001
2020	0.001	0.005	0.021	0.067	0.074	0.023	0.005	0.001	0.001	0.001	0.001	0.001
2021	0.001	0.004	0.016	0.050	0.056	0.017	0.004	0.001	0.001	0.001	0.001	0.001
2022	0.001	0.005	0.021	0.068	0.076	0.023	0.005	0.001	0.001	0.001	0.001	0.001
2023	0.001	0.008	0.032	0.104	0.116	0.036	0.007	0.002	0.001	0.001	0.001	0.001
2024	0.001	0.009	0.033	0.107	0.120	0.037	0.008	0.002	0.001	0.001	0.001	0.001
2025	0.001	0.010	0.039	0.125	0.140	0.043	0.009	0.003	0.001	0.001	0.001	0.001
2026	0.001	0.007	0.028	0.091	0.102	0.031	0.006	0.002	0.001	0.001	0.001	0.001

Table 43: Summary of results for Model h1_1.06 (single-stock hypothesis). Note that MSY values are a function of time-varying selectivity and average weight.

Year	Landings ('000 t)	SSB ('000 t)	Recruitment (age 1, millions)	Reported F (per year)	F_MS _Y	SSB_MS _Y ('000 t)
1970	118	13,315	5,072	0.01	0.18	5,750
1971	169	12,525	6,513	0.02	0.18	5,717
1972	111	12,080	8,265	0.01	0.17	5,758
1973	165	11,610	5,826	0.02	0.16	5,726
1974	324	10,827	4,794	0.03	0.16	5,653
1975	300	10,330	8,973	0.03	0.17	5,751
1976	397	10,369	12,118	0.05	0.17	5,632
1977	848	10,409	12,633	0.08	0.14	5,859
1978	1,025	10,522	16,048	0.11	0.16	5,810
1979	1,302	10,660	15,951	0.14	0.18	6,028
1980	1,316	10,868	16,021	0.13	0.18	6,108
1981	1,945	10,641	16,857	0.19	0.18	5,953
1982	2,372	9,991	18,940	0.30	0.19	6,019
1983	1,870	10,739	27,543	0.26	0.21	6,304
1984	2,687	10,973	21,565	0.38	0.21	6,293
1985	2,371	11,334	24,519	0.31	0.22	6,091
1986	2,073	14,768	56,865	0.25	0.23	6,210
1987	2,680	19,580	56,452	0.27	0.22	6,220
1988	3,246	21,360	23,866	0.25	0.23	6,290
1989	3,582	19,905	14,566	0.25	0.24	6,115
1990	3,715	18,135	16,338	0.26	0.25	6,156
1991	3,778	16,515	19,192	0.32	0.29	5,999
1992	3,362	15,116	23,778	0.44	0.33	6,585
1993	3,371	13,328	14,931	0.49	0.28	7,152
1994	4,276	10,463	13,690	0.70	0.32	6,849
1995	4,956	7,459	17,885	1.26	0.27	7,392
1996	4,380	5,263	13,185	1.70	0.28	7,281
1997	3,598	3,719	16,380	2.19	0.28	7,464
1998	2,027	3,738	17,444	1.54	0.24	8,022
1999	1,424	4,827	21,307	1.07	0.29	7,985
2000	1,540	6,433	24,822	0.79	0.30	7,694
2001	2,528	6,726	16,786	1.03	0.26	8,039
2002	1,750	7,236	19,150	1.00	0.33	8,104
2003	1,797	7,146	9,143	0.85	0.29	8,141
2004	1,934	6,395	9,741	0.79	0.29	7,513
2005	1,755	5,986	13,433	0.84	0.33	7,183
2006	2,020	5,153	7,846	0.83	0.29	6,935

Year	Landings ('000 t)	SSB ('000 t)	Recruitment (age 1, millions)	Reported F (per year)	F_MSY	SSB_MSY ('000 t)
2007	1,997	3,548	2,570	0.94	0.26	6,757
2008	1,473	2,375	5,242	0.77	0.20	6,632
2009	1,283	1,811	7,915	0.81	0.19	6,136
2010	727	1,667	4,684	0.53	0.16	6,285
2011	635	1,517	3,363	0.45	0.17	6,557
2012	455	1,512	3,501	0.40	0.19	7,016
2013	353	1,631	3,517	0.36	0.21	7,335
2014	411	1,894	5,125	0.39	0.25	7,290
2015	394	2,219	4,931	0.42	0.36	6,390
2016	389	2,640	6,696	0.41	0.36	6,780
2017	405	3,334	9,586	0.43	0.34	7,276
2018	526	4,297	11,923	0.47	0.30	8,100
2019	632	5,125	9,622	0.42	0.33	7,314
2020	707	5,640	7,925	0.49	0.50	6,949
2021	808	6,171	10,559	0.57	0.68	6,650
2022	961	6,530	6,490	0.62	0.90	6,407
2023	1,126	6,345	5,071	0.69	0.79	6,797
2024	1,330	5,973	8,332	1.05	1.08	6,682
2025	1,495	6,031	14,314	1.02	0.94	6,180
2026	1,101	6,109	5,560	0.68	0.79	6,335

Table 44: Summary of results for Model h2_1.06 (two-stock hypothesis; south stock). Note that MSY values are a function of time-varying selectivity and average weight.

Year	Landings ('000 t)	SSB ('000 t)	Recruitment (age 1, millions)	Reported F (per year)	F_MSY	SSB_MSY ('000 t)
1970	113	14,154	6,881	0.01	0.18	5,430
1971	159	13,504	6,496	0.01	0.18	5,420
1972	93	12,898	6,073	0.01	0.17	5,299
1973	123	12,252	5,441	0.01	0.17	5,315
1974	195	11,531	5,408	0.02	0.18	5,307
1975	262	10,904	6,984	0.03	0.17	5,298
1976	343	10,461	8,416	0.04	0.17	5,340
1977	343	10,348	10,825	0.04	0.17	5,328
1978	638	10,382	13,420	0.08	0.17	5,330
1979	968	10,375	13,769	0.12	0.19	5,677
1980	902	10,524	13,977	0.11	0.19	5,660
1981	1,499	10,343	15,371	0.17	0.18	5,554
1982	2,228	9,549	16,935	0.29	0.19	5,613
1983	1,760	10,344	28,879	0.25	0.21	5,899
1984	2,486	10,987	24,011	0.37	0.21	5,845
1985	2,256	11,626	24,803	0.31	0.22	5,650
1986	2,022	15,090	56,516	0.24	0.23	5,681
1987	2,633	19,434	49,925	0.26	0.22	5,795
1988	3,001	20,872	21,887	0.24	0.23	5,896
1989	3,266	19,384	12,378	0.24	0.24	5,752
1990	3,344	17,778	16,430	0.24	0.26	5,715
1991	3,564	16,400	20,216	0.31	0.31	5,544
1992	3,250	14,979	21,856	0.42	0.33	6,181
1993	3,237	13,030	13,119	0.47	0.28	6,782
1994	4,042	10,218	13,493	0.70	0.33	6,425
1995	4,405	7,102	10,771	1.24	0.28	7,055
1996	3,884	4,893	12,679	1.67	0.29	6,944
1997	2,918	3,738	13,926	2.06	0.30	7,072
1998	1,614	3,823	14,334	1.33	0.26	7,786
1999	1,221	4,552	15,997	0.94	0.29	7,642
2000	1,237	5,617	18,155	0.71	0.31	7,244
2001	1,670	6,466	18,794	0.88	0.29	7,665
2002	1,595	7,248	17,828	0.93	0.34	7,596
2003	1,580	7,490	12,175	0.79	0.30	7,707
2004	1,747	6,881	7,735	0.74	0.31	6,999
2005	1,674	6,044	8,873	0.81	0.33	6,795
2006	1,743	4,957	5,660	0.78	0.31	6,472

Year	Landings ('000 t)	SSB ('000 t)	Recruitment (age 1, millions)	Reported F (per year)	F_MSY	SSB_MSY ('000 t)
2007	1,742	3,429	2,466	0.89	0.28	6,205
2008	1,303	2,416	5,735	0.70	0.20	6,090
2009	1,207	1,657	4,658	0.75	0.19	5,707
2010	704	1,226	3,244	0.57	0.15	5,871
2011	308	1,199	3,061	0.36	0.19	6,148
2012	267	1,308	2,960	0.33	0.20	6,516
2013	273	1,463	3,346	0.32	0.21	6,806
2014	336	1,773	5,133	0.35	0.26	6,572
2015	372	2,134	4,980	0.40	0.34	5,946
2016	374	2,540	6,113	0.39	0.34	6,243
2017	396	3,064	7,633	0.43	0.32	6,748
2018	469	3,841	10,456	0.46	0.30	7,465
2019	496	4,650	9,033	0.40	0.37	6,612
2020	567	5,214	7,570	0.46	0.48	6,699
2021	684	5,868	11,428	0.54	0.63	6,412
2022	802	6,479	7,911	0.59	0.92	6,024
2023	904	6,563	5,606	0.65	0.81	6,587
2024	1,113	6,365	8,296	0.98	1.08	6,573
2025	1,294	6,529	13,968	0.96	1.01	6,048
2026	991	6,661	5,312	0.62	0.89	6,074

Table 45: Summary of results for Model h2_1.06 (two-stock hypothesis; far north stock). Note that MSY values are a function of time-varying selectivity and average weight.

Year	Landings ('000 t)	SSB ('000 t)	Recruitment (age 1, millions)	Reported F (per year)	F_MSY	SSB_MSY ('000 t)
1970	5	4,657	2,681	0.00	0.09	1,250
1971	9	4,688	3,348	0.00	0.09	1,250
1972	19	4,628	4,486	0.00	0.09	1,250
1973	43	4,607	3,149	0.01	0.09	1,250
1974	129	4,589	2,200	0.02	0.09	1,250
1975	38	4,488	4,111	0.01	0.09	1,250
1976	54	4,348	5,366	0.01	0.09	1,250
1977	505	4,037	3,646	0.12	0.09	1,250
1978	387	3,776	4,162	0.08	0.09	1,250
1979	334	3,501	3,627	0.07	0.09	1,250
1980	414	3,219	3,604	0.10	0.09	1,236
1981	446	2,877	3,418	0.13	0.09	1,236
1982	144	2,750	3,648	0.04	0.09	1,236
1983	111	2,758	2,519	0.03	0.09	1,236
1984	201	2,758	1,161	0.04	0.09	1,236
1985	115	2,682	2,042	0.03	0.09	1,236
1986	51	2,559	3,497	0.02	0.09	1,236
1987	46	2,541	9,135	0.02	0.09	1,236
1988	244	2,715	3,527	0.05	0.09	1,236
1989	316	3,175	3,162	0.03	0.09	1,236
1990	371	3,216	1,794	0.06	0.09	1,236
1991	213	3,161	1,460	0.05	0.09	1,236
1992	112	3,012	2,839	0.04	0.09	1,236
1993	133	2,820	2,854	0.06	0.09	1,236
1994	233	2,648	2,468	0.07	0.09	1,236
1995	551	2,224	5,567	0.26	0.09	1,236
1996	496	1,871	1,736	0.33	0.09	1,236
1997	680	1,423	3,097	0.46	0.09	1,236
1998	413	1,062	3,162	1.08	0.09	1,236
1999	204	888	5,335	0.17	0.09	1,298
2000	304	878	7,722	0.12	0.09	1,298
2001	858	840	2,022	0.25	0.09	1,298
2002	155	1,012	5,143	0.04	0.09	1,298
2003	218	1,208	548	0.06	0.09	1,298
2004	187	1,475	3,104	0.04	0.09	1,298
2005	81	1,571	6,979	0.03	0.09	1,298
2006	278	1,832	3,234	0.06	0.09	1,298

Year	Landings ('000 t)	SSB ('000 t)	Recruitment (age 1, millions)	Reported F (per year)	F_MSY	SSB_MSY ('000 t)
2007	255	2,224	1,019	0.04	0.09	1,298
2008	170	2,377	1,095	0.04	0.09	1,298
2009	77	2,350	8,615	0.03	0.09	1,298
2010	22	2,507	2,324	0.01	0.09	1,267
2011	326	2,964	1,319	0.04	0.09	1,307
2012	187	2,981	3,018	0.04	0.09	1,307
2013	81	2,940	2,232	0.02	0.09	1,307
2014	75	2,965	2,383	0.02	0.09	1,307
2015	22	2,968	2,147	0.01	0.09	1,307
2016	15	2,980	3,273	0.00	0.09	1,307
2017	9	3,002	6,362	0.00	0.09	1,307
2018	57	3,195	5,351	0.01	0.09	1,307
2019	136	3,615	4,415	0.02	0.09	1,307
2020	140	3,979	4,958	0.02	0.11	1,309
2021	124	4,281	4,711	0.01	0.11	1,309
2022	160	4,564	2,227	0.02	0.11	1,309
2023	222	4,665	2,243	0.03	0.11	1,309
2024	217	4,472	3,033	0.03	0.11	1,273
2025	200	4,208	3,222	0.03	0.11	1,309
2026	110	4,046	3,206	0.02	0.11	1,309

The earlier five-year (2027–2031) $\pm 1\sigma$ projection outputs are shown in Figure 45, Figure 46, Figure 47 and Figure 48. The updated ten-year (2027–2036) risk summaries are in Table 7, Table 8 and Table 9. Their conditional normal-approximation probabilities use the reported projection standard errors and the fixed biomass reference points described in Section 4.2.

9 Figures

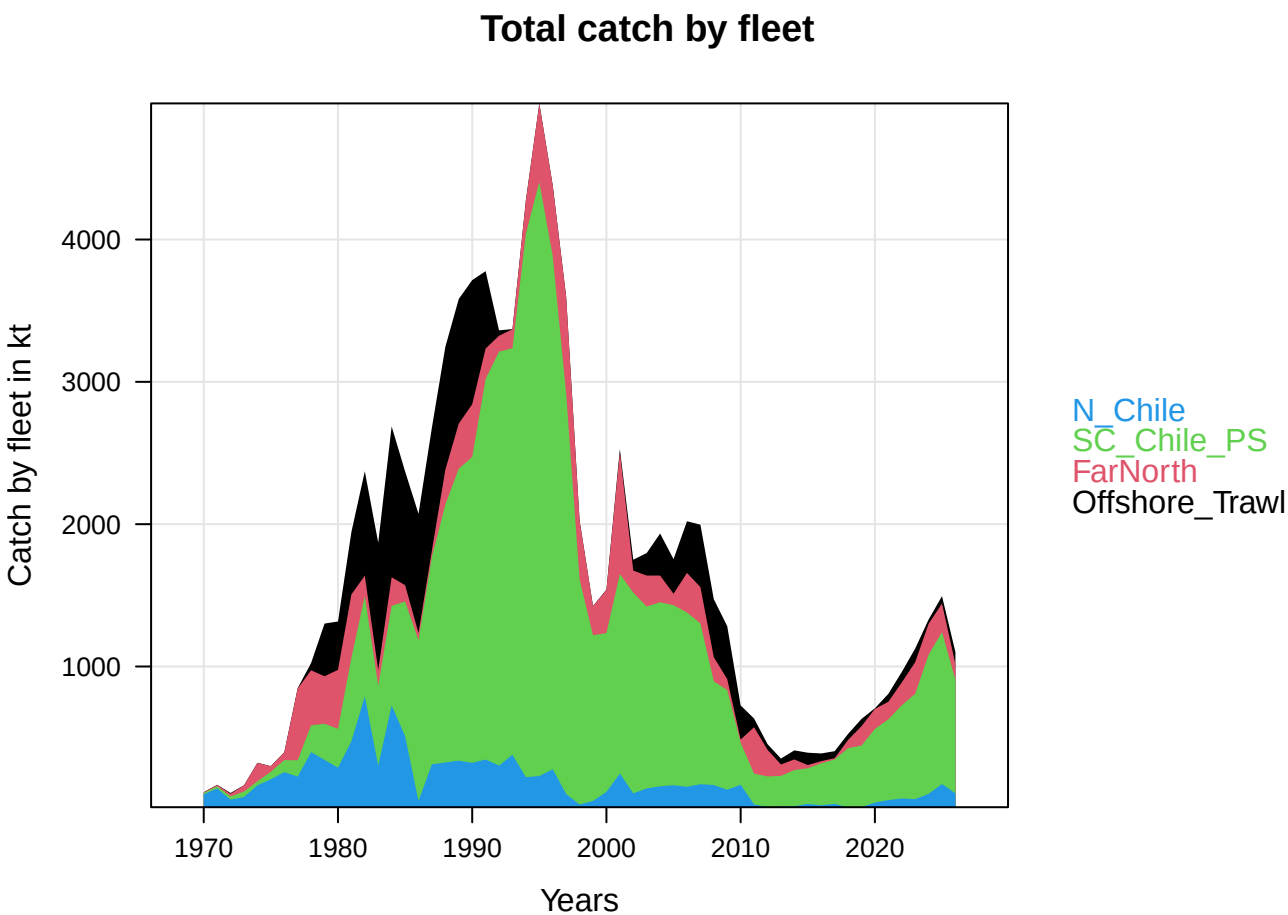


Figure 4: Catch of jack mackerel by fleet. Blue is the northern Chilean fleet, green is the south-central Chilean fleet, red is the far north fleet, and black is the offshore trawl fleet.

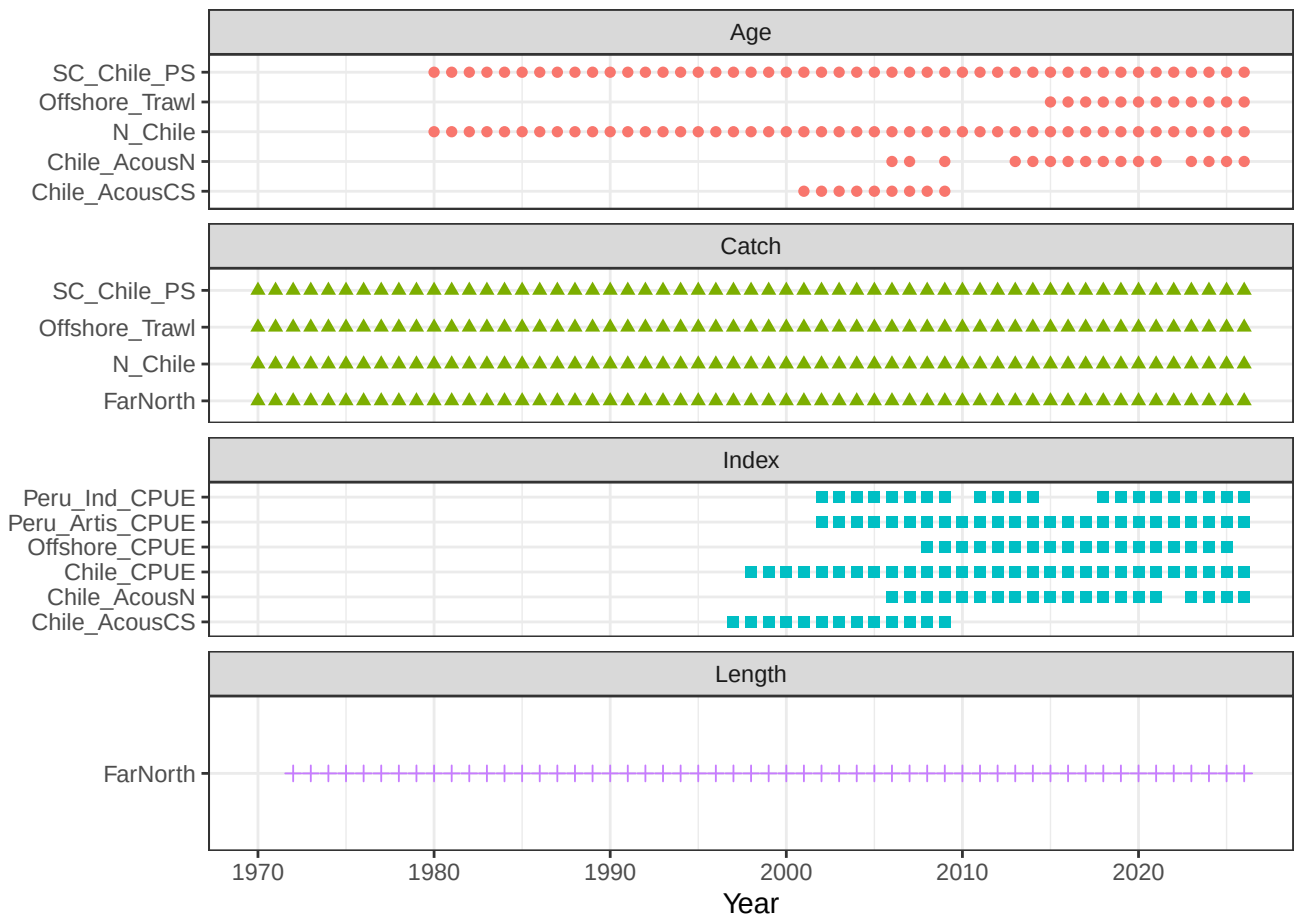


Figure 5: Observation years by data source and type, read from the current 1.06 input file. Catch spans all model years, with the 2026 full-year values supplied as projections.

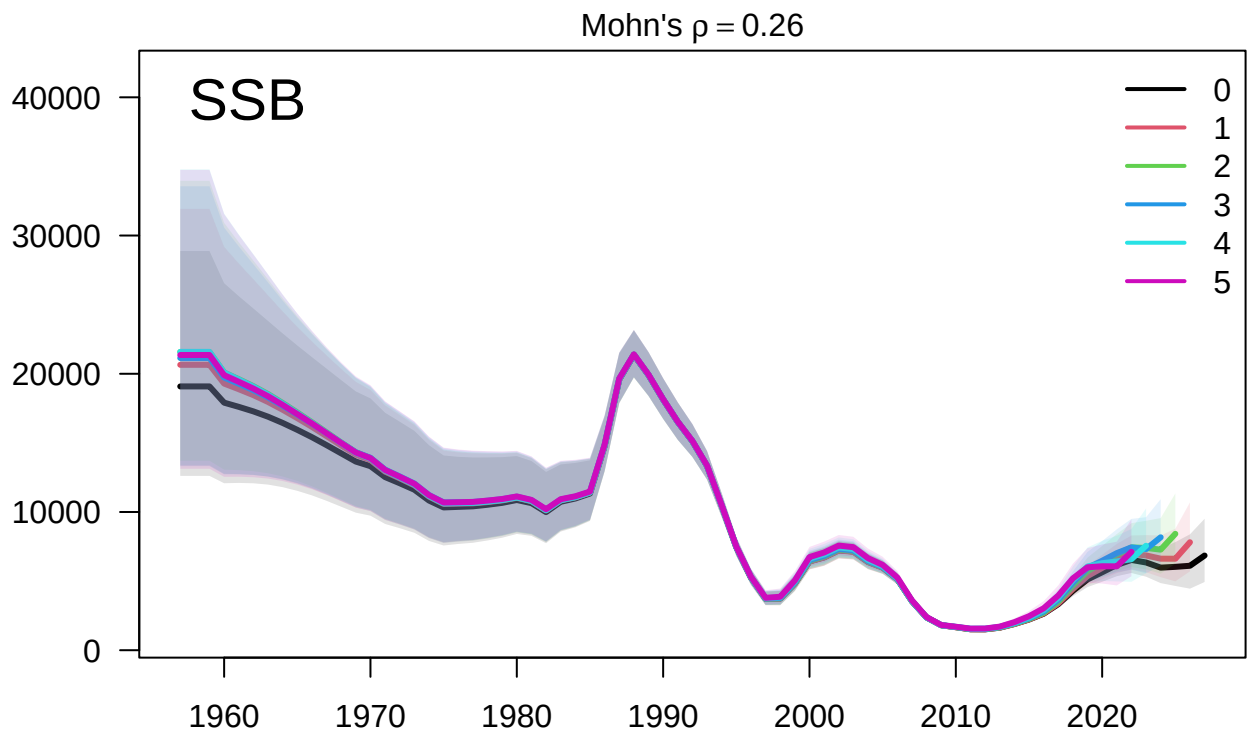


Figure 6: Spawning biomass (thousand tonnes) from the full assessment and five retrospective peels, based on Model h1_1.06 (single-stock hypothesis). Legend 0 is the full assessment; 1–5 are successive terminal-year peels. Shading shows saved model uncertainty bounds. Title rho uses each peel's one-year-ahead SSB endpoint.

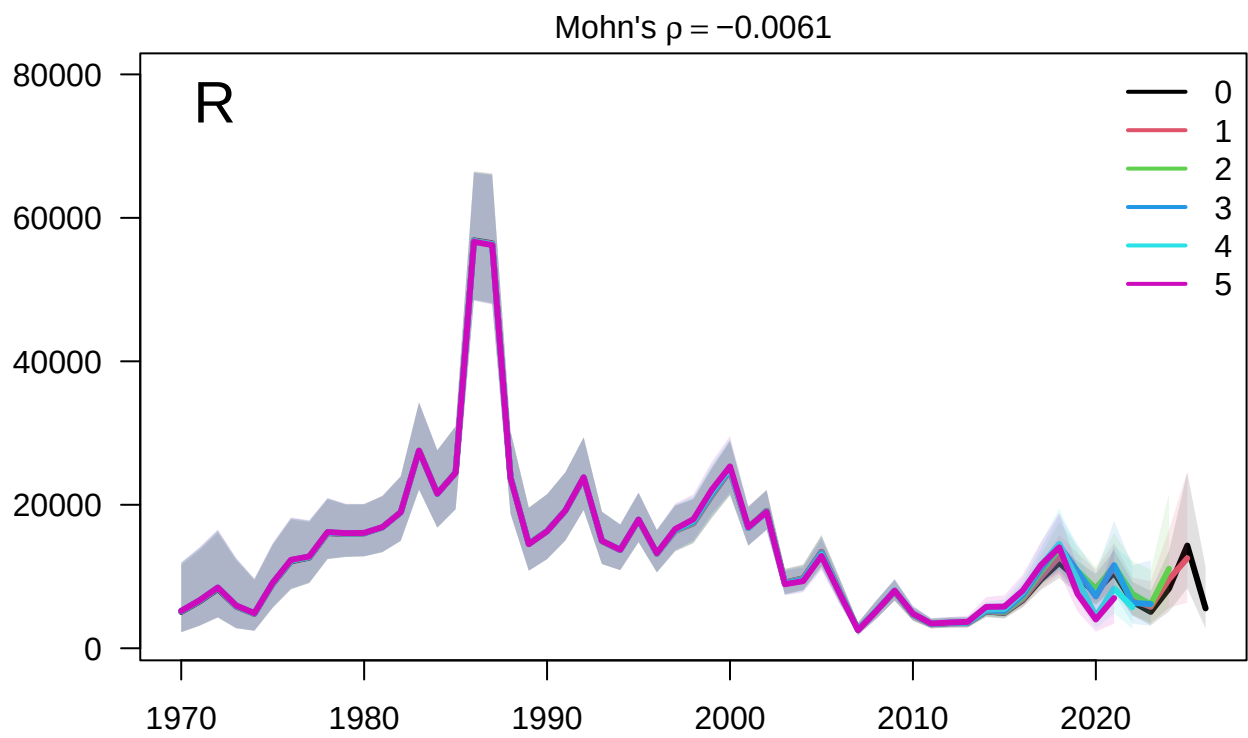


Figure 7: Recruitment (millions of fish) from the full assessment and five retrospective peels, based on Model h1_1.06 (single-stock hypothesis). Legend 0 is the full assessment; 1–5 are successive terminal-year peels. Shading shows saved model uncertainty bounds.

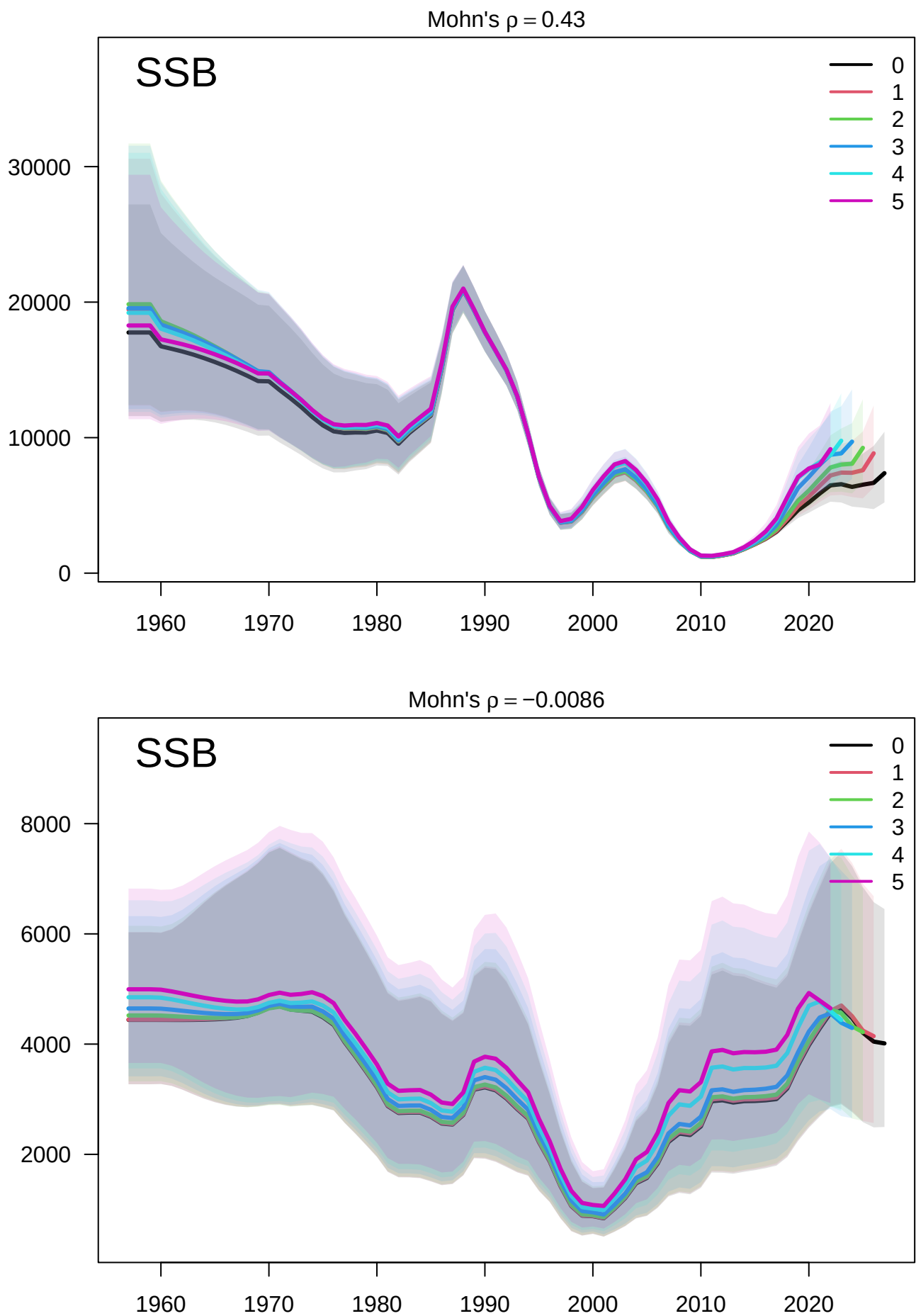


Figure 8: Spawning biomass (thousand tonnes) from the full assessment and five retrospective peels for the southern stock (top) and far north stock (bottom), based on Model h2_1.06 (two-stock hypothesis). Legend 0 is the full assessment; 1–5 are successive terminal-year peels. Shading shows saved model uncertainty bounds. Title rho uses each peel's one-year-ahead SSB endpoint.

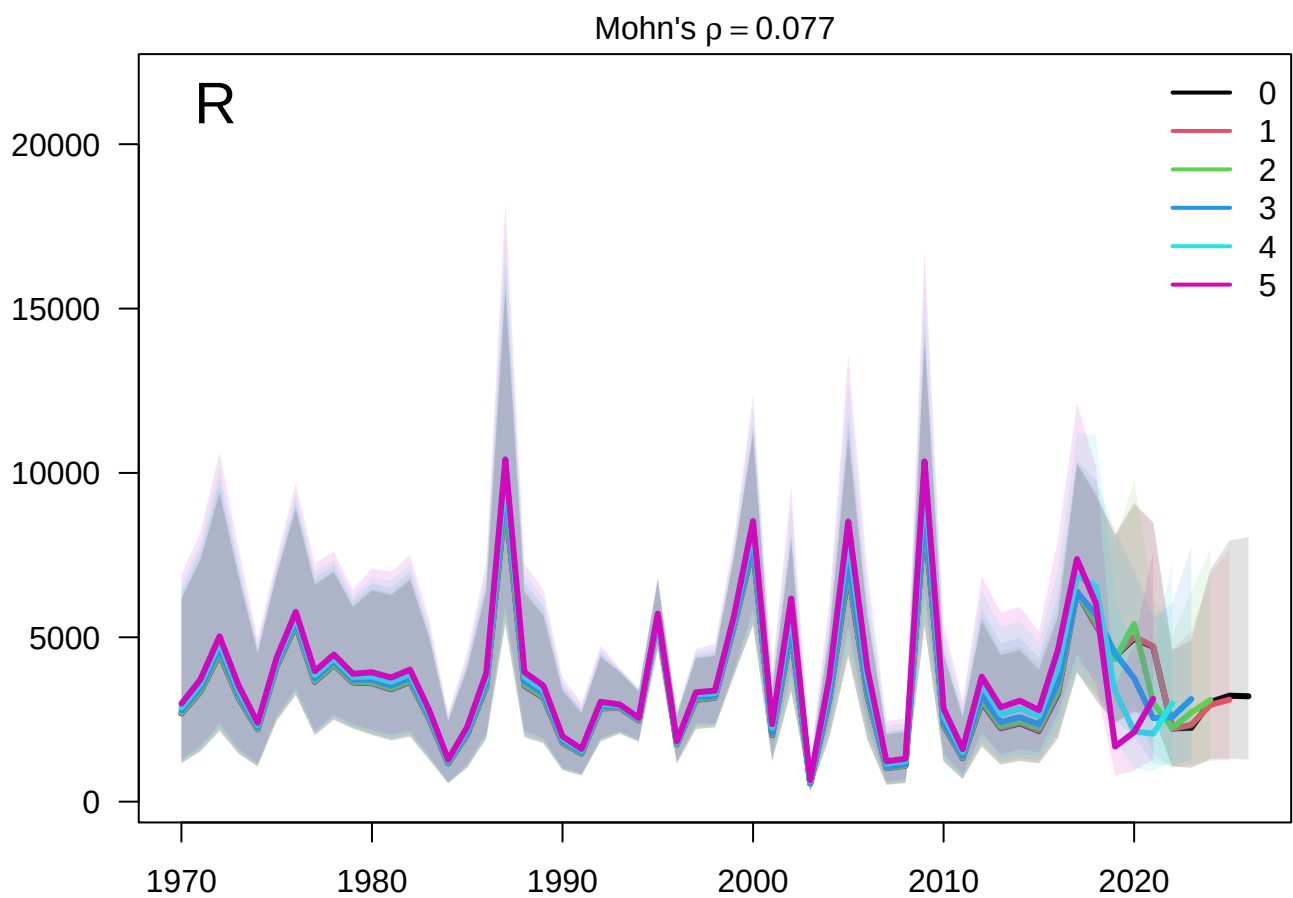
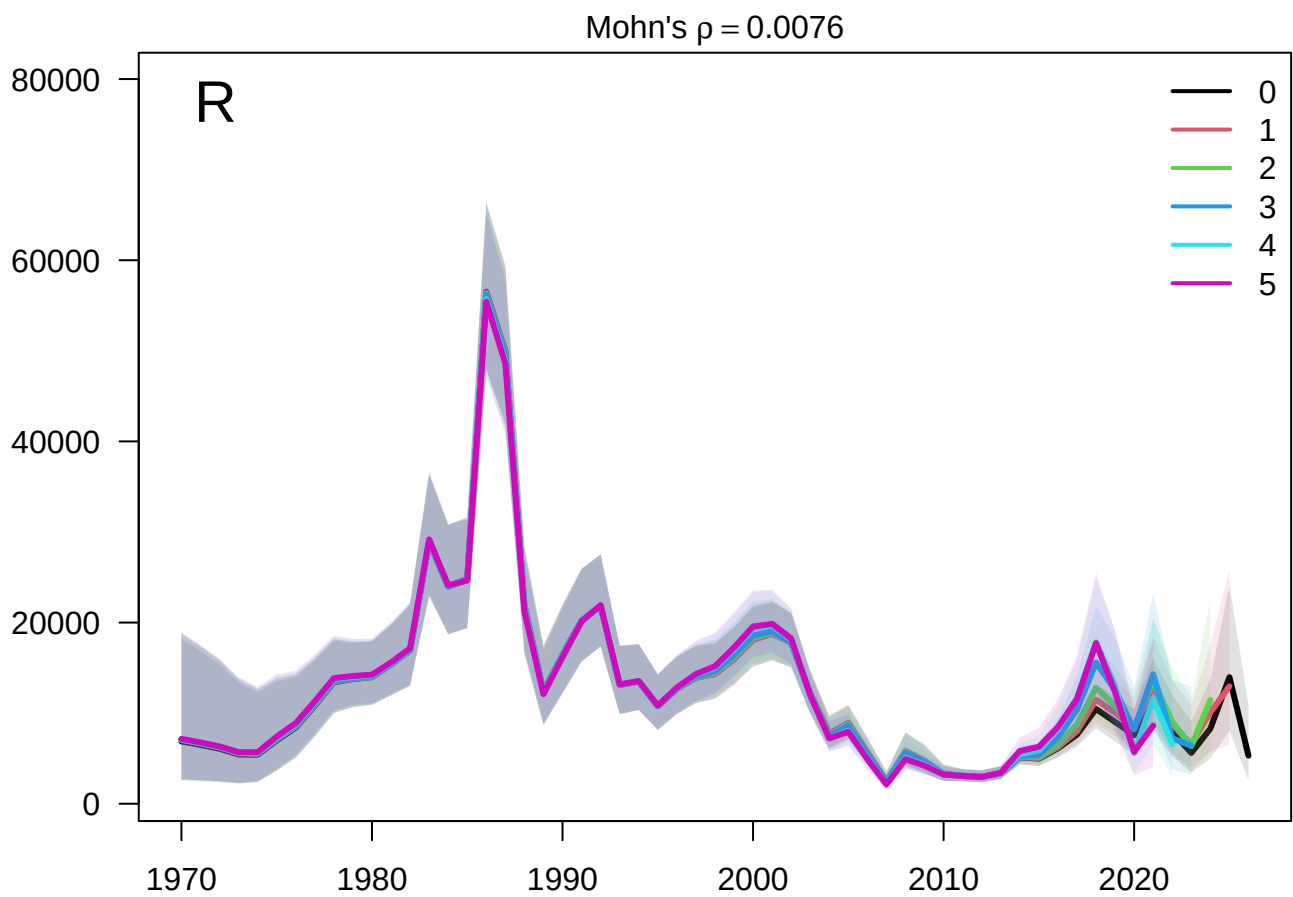


Figure 9: Recruitment (millions of fish) from the full assessment and five retrospective peels for the southern stock (top) and far north stock (bottom), based on Model h2_1.06 (two-stock hypothesis). Legend 0 is the full assessment; 1–5 are successive terminal-year peels. Shading shows saved model uncertainty bounds.

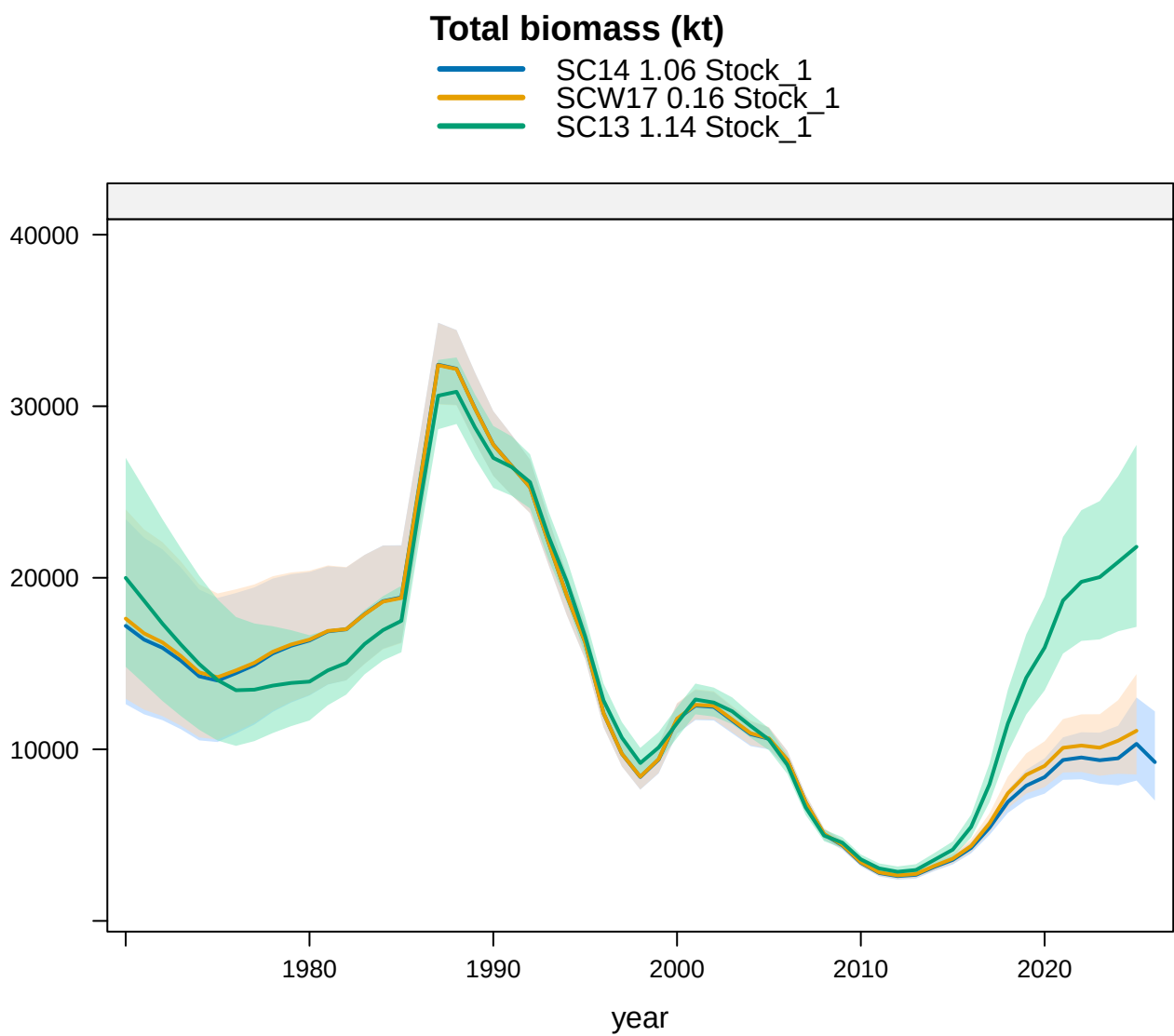


Figure 10: Total biomass (thousand tonnes) comparison for the archived SC13 1.14, SCW17 0.16 and current SC14 1.06 single-stock configurations. Differences include successive input updates and changes in model assumptions.

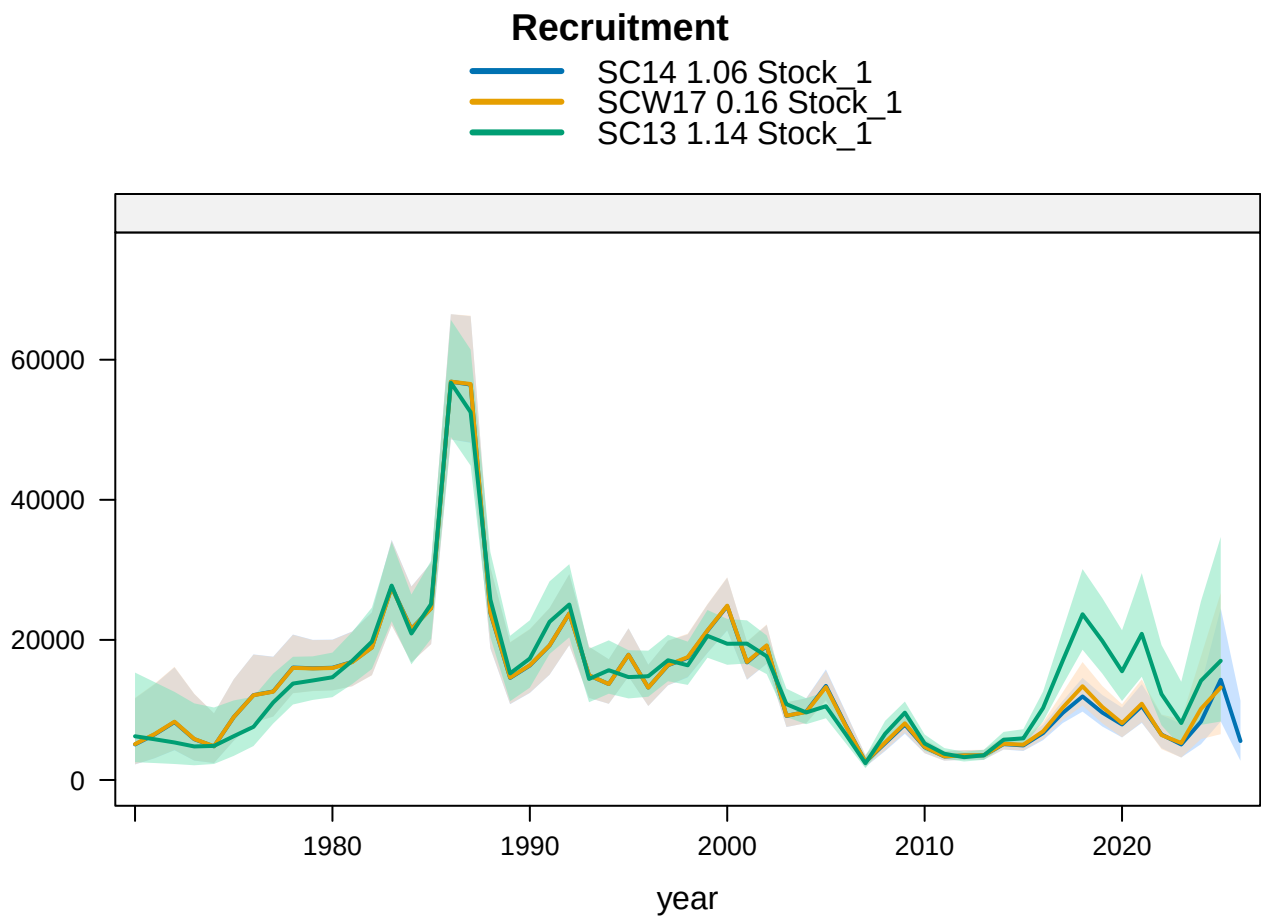


Figure 11: Recruitment (millions of fish) comparison for the archived SC13 1.14, SCW17 0.16 and current SC14 1.06 single-stock configurations.

Weight at age in the fishery

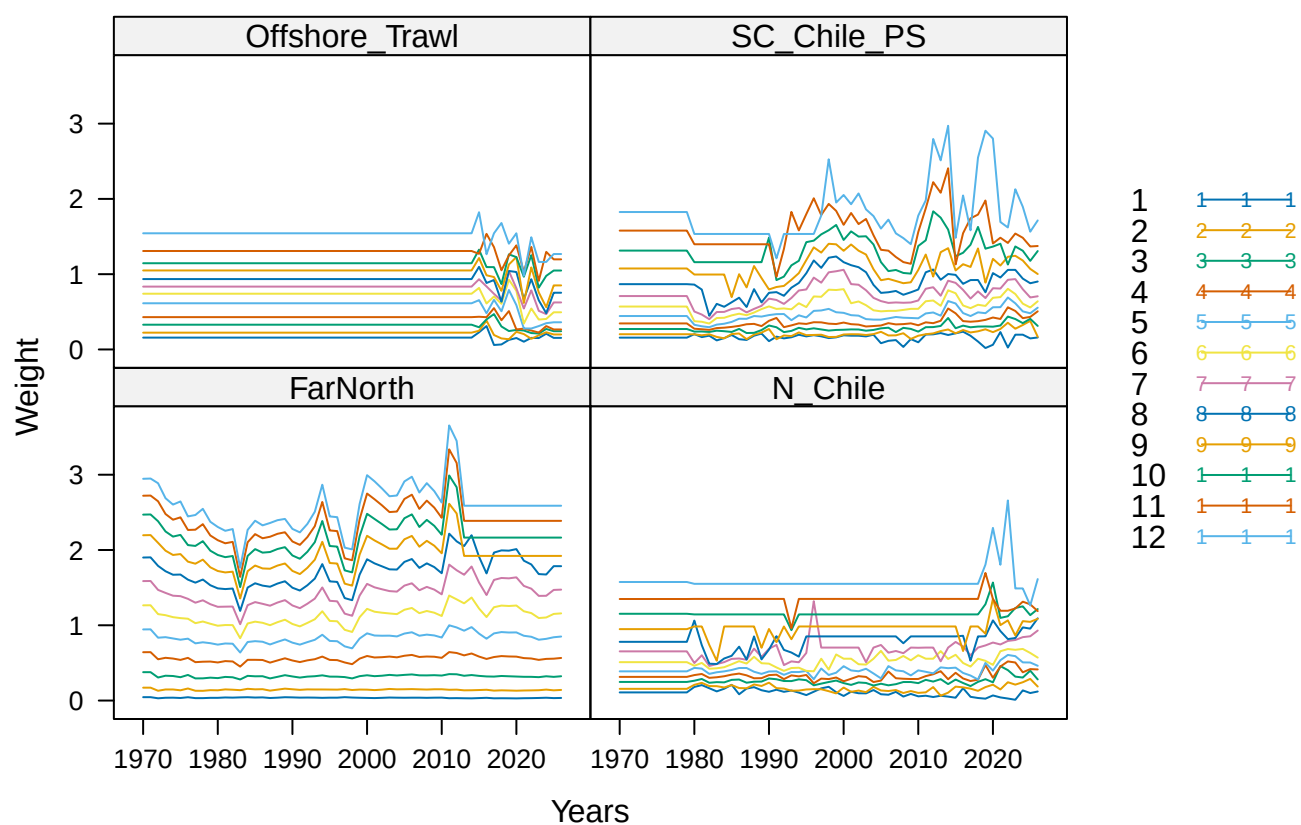


Figure 12: Mean weights-at-age (kg) over time used for the fisheries in the JJM models. Each line represents an age from 1 to 12.

Weight at age in the survey

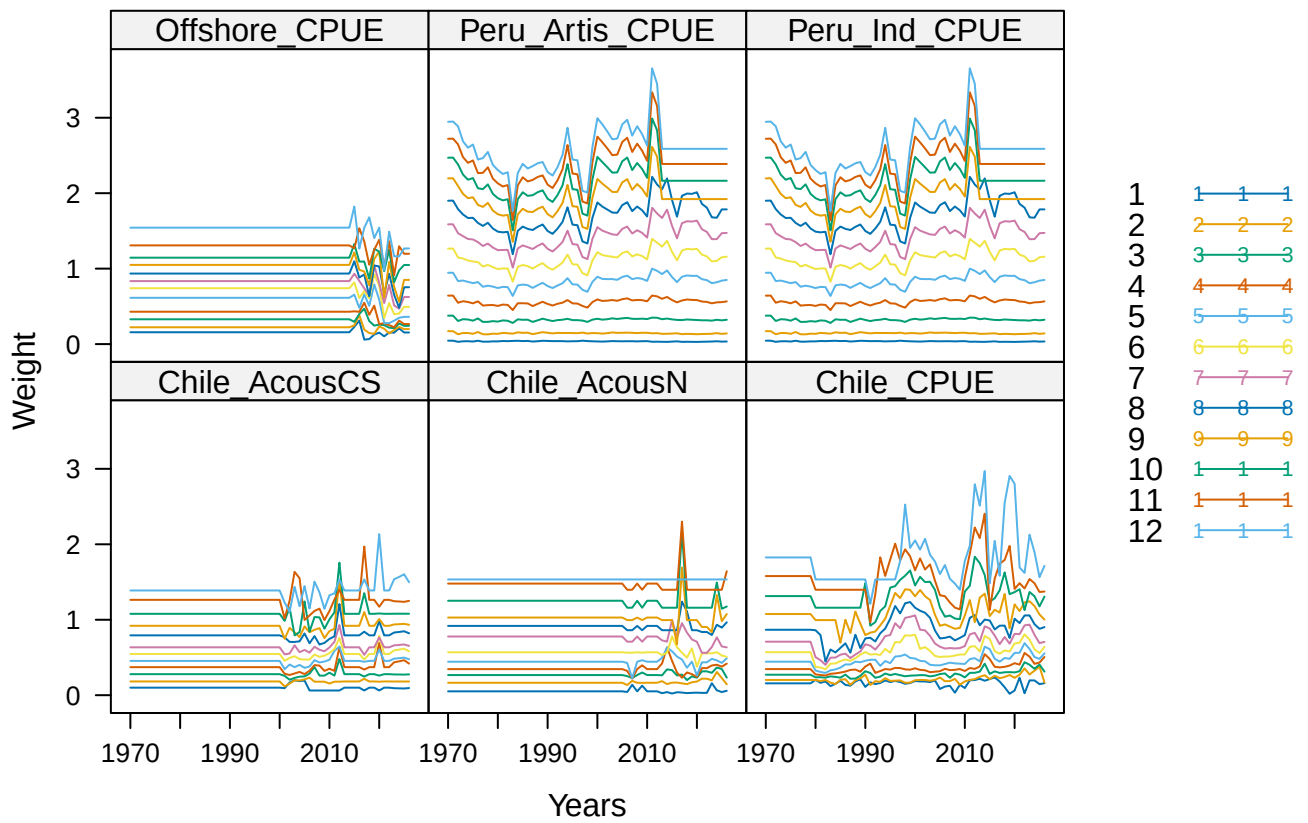


Figure 13: Mean weights-at-age (kg) over time used for the surveys in the JJM models. Each line represents an age from 1 to 12.

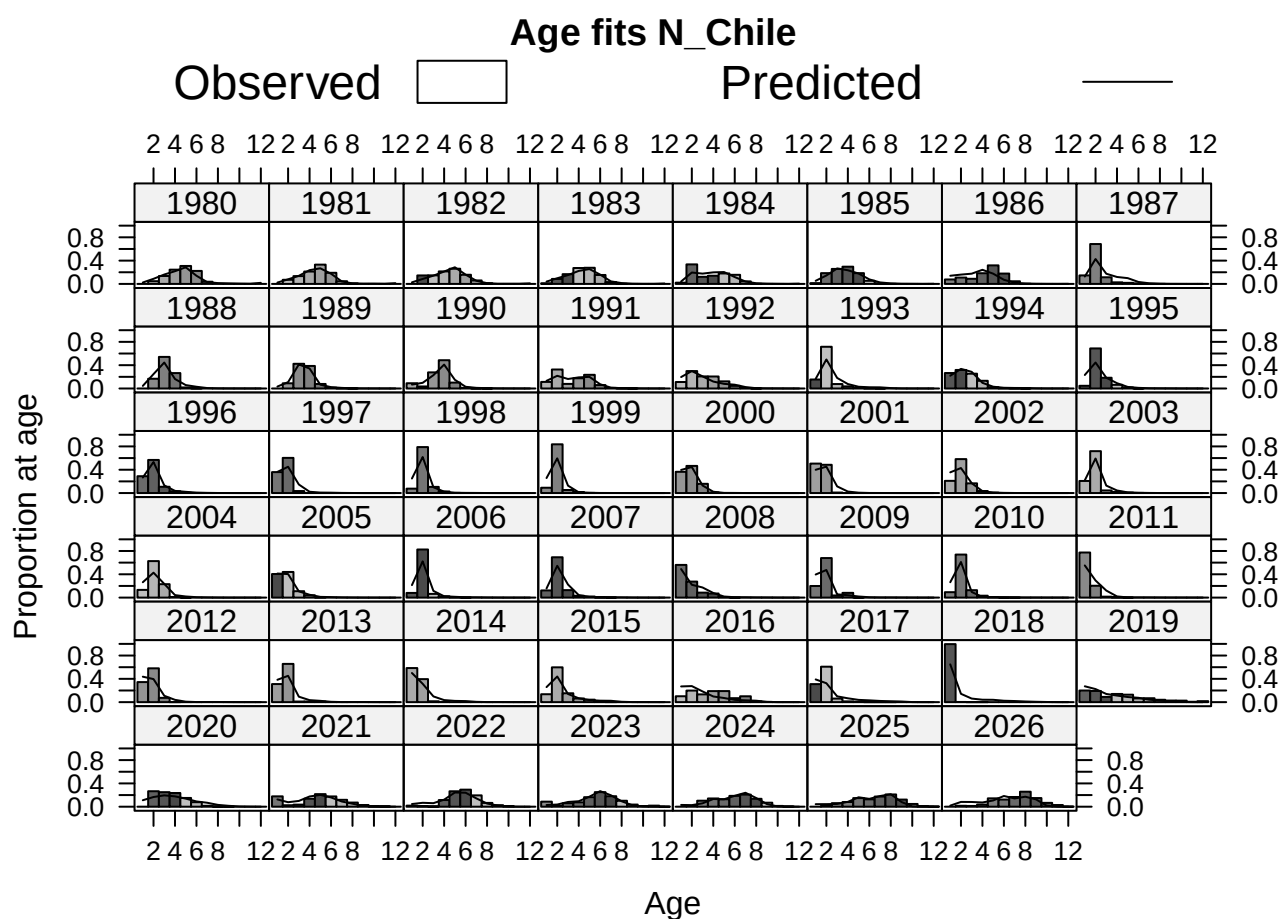


Figure 14: Model h1_1.06 (single-stock hypothesis) fit to the age compositions for the Chilean northern zone fishery (Fleet 1). Bars represent the observed data and lines represent the model predictions.

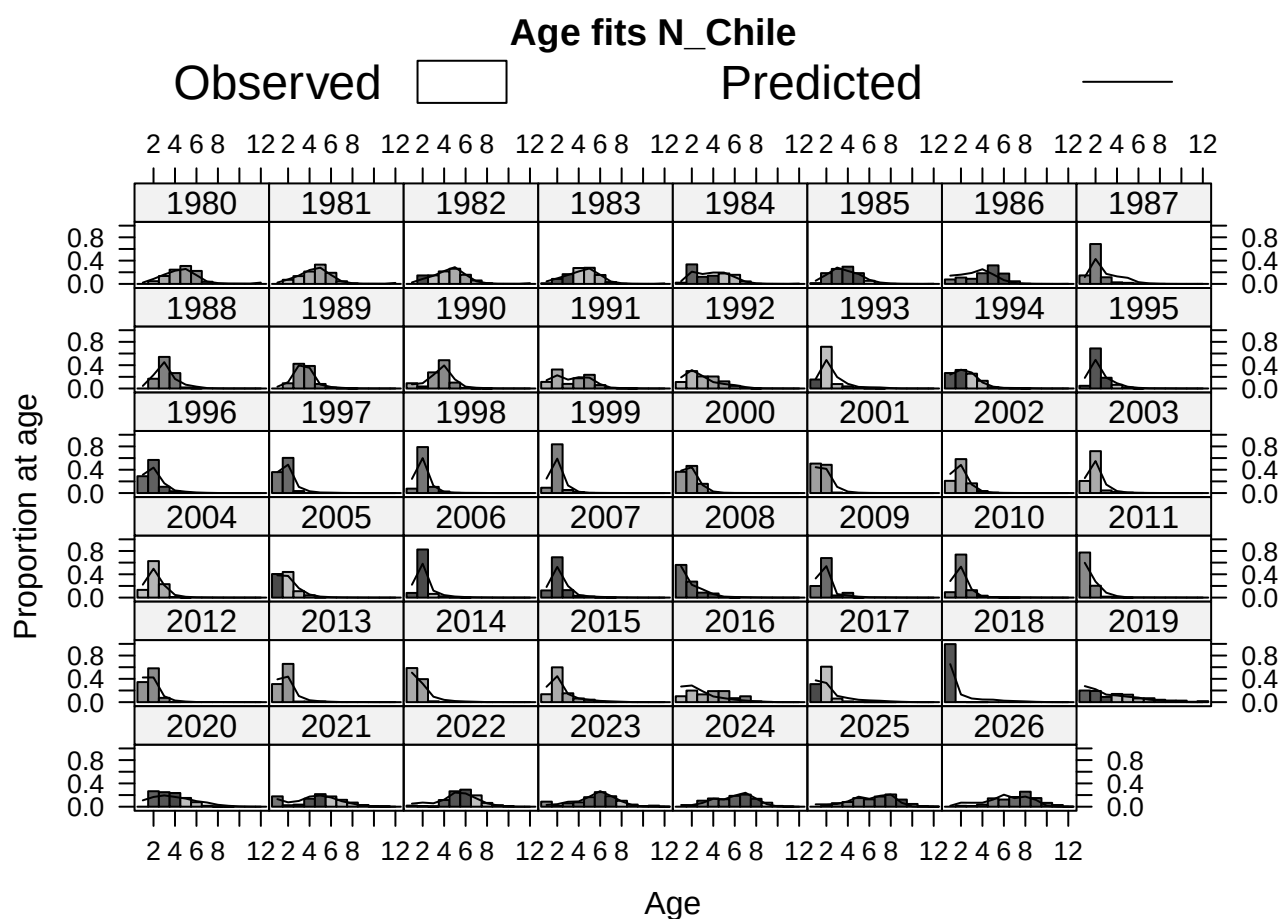


Figure 15: Model h2_1.06 (two-stock hypothesis) fit to the age compositions for the Chilean northern zone fishery (Fleet 1). Bars represent the observed data and lines represent the model predictions.

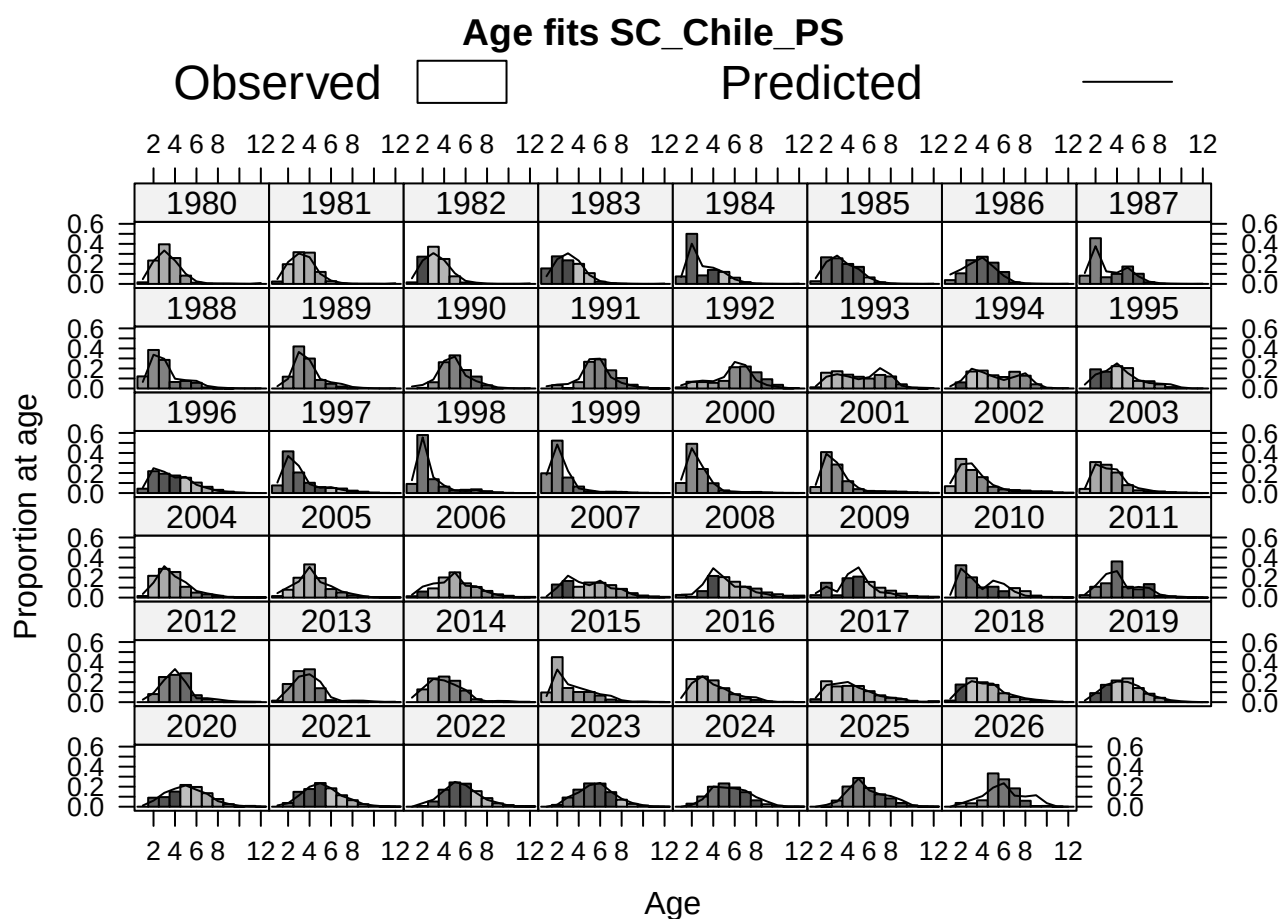


Figure 16: Model h1_1.06 (single-stock hypothesis) fit to the age compositions for the South-Central Chilean purse seine fishery (Fleet 2). Bars represent the observed data and lines represent the model predictions.

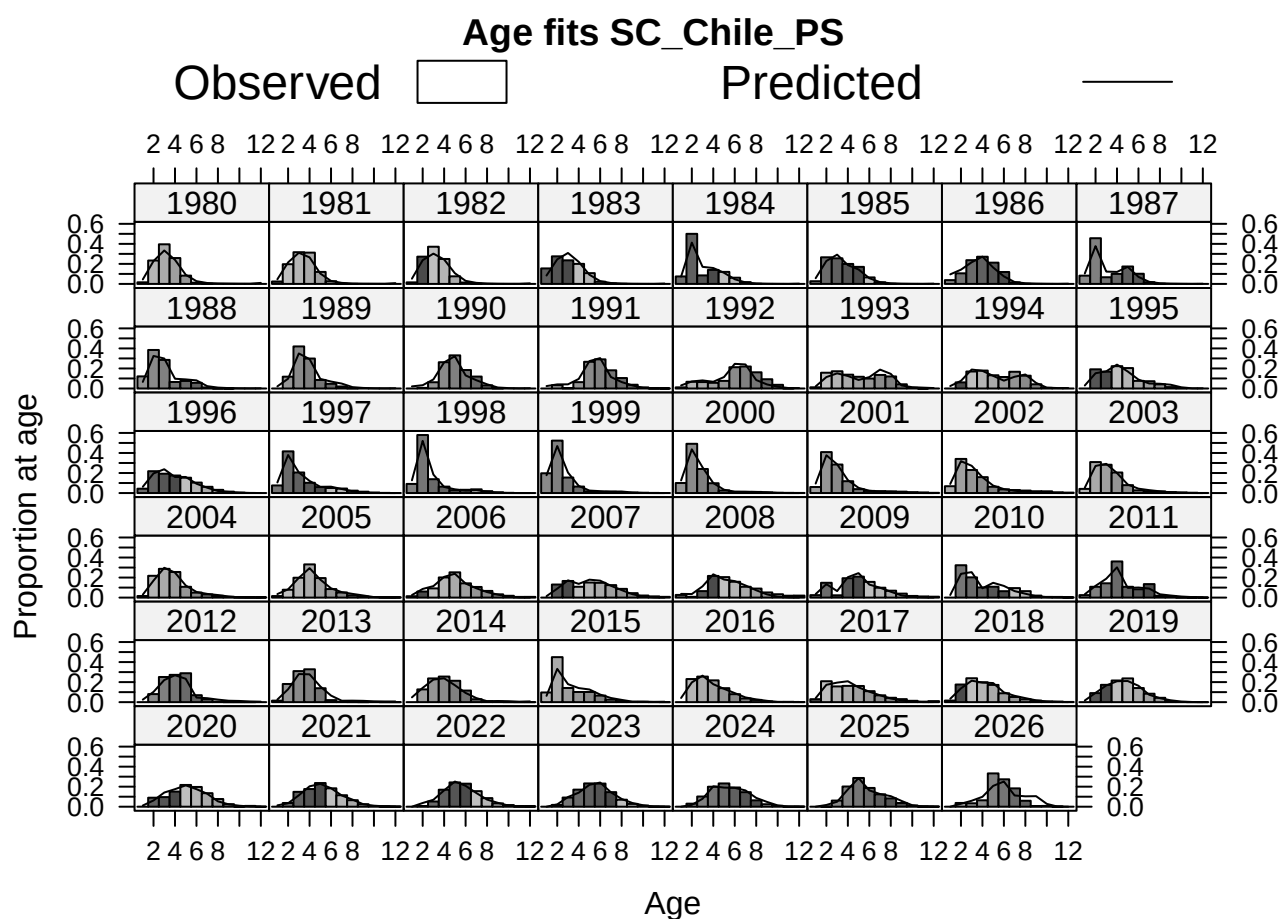


Figure 17: Model h2_1.06 (two-stock hypothesis) fit to the age compositions for the South-Central Chilean purse seine fishery (Fleet 2). Bars represent the observed data and lines represent the model predictions.

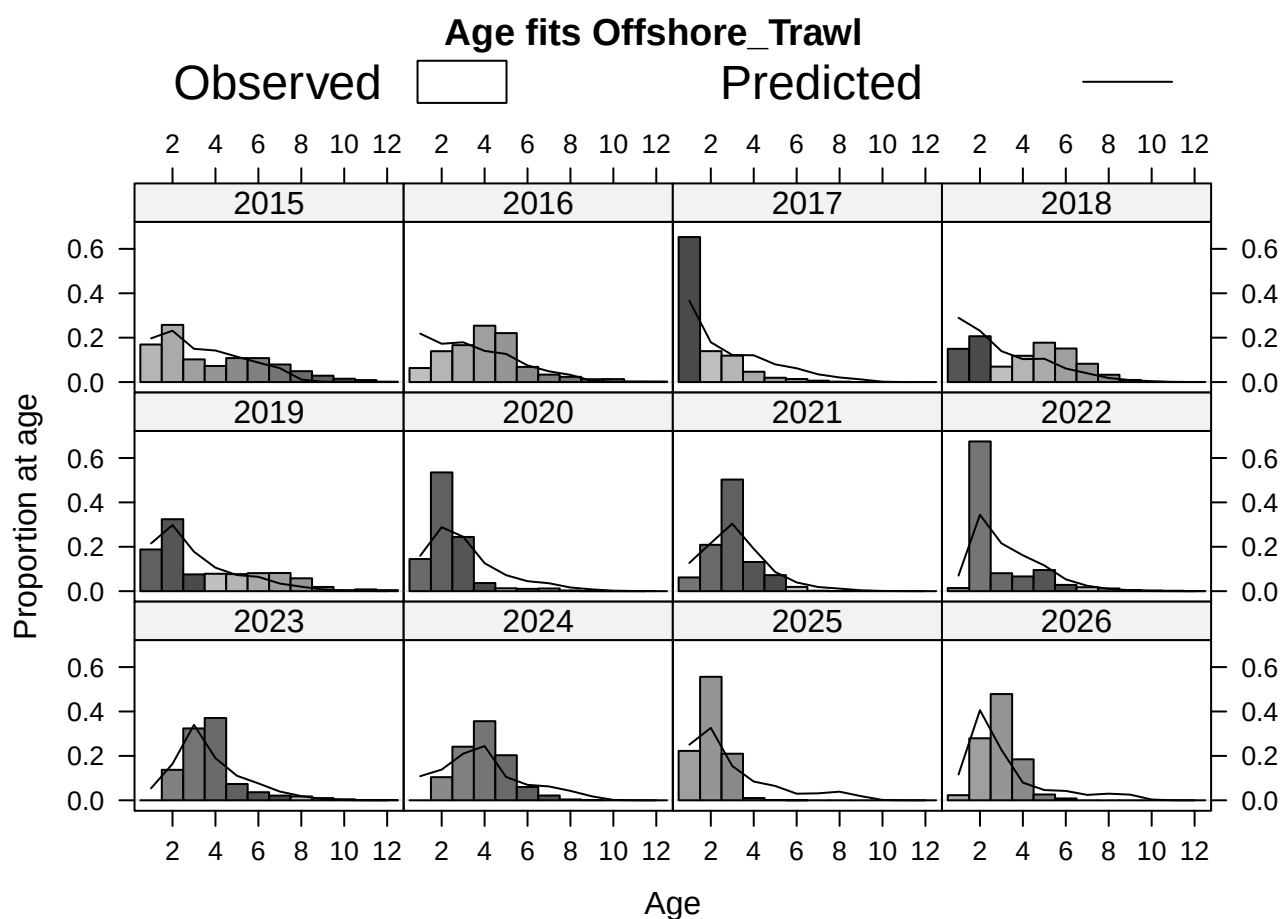


Figure 18: Model h1_1.06 (single-stock hypothesis) fit to the age compositions for the offshore trawl fishery (Fleet 4). Bars represent the observed data and lines represent the model predictions.

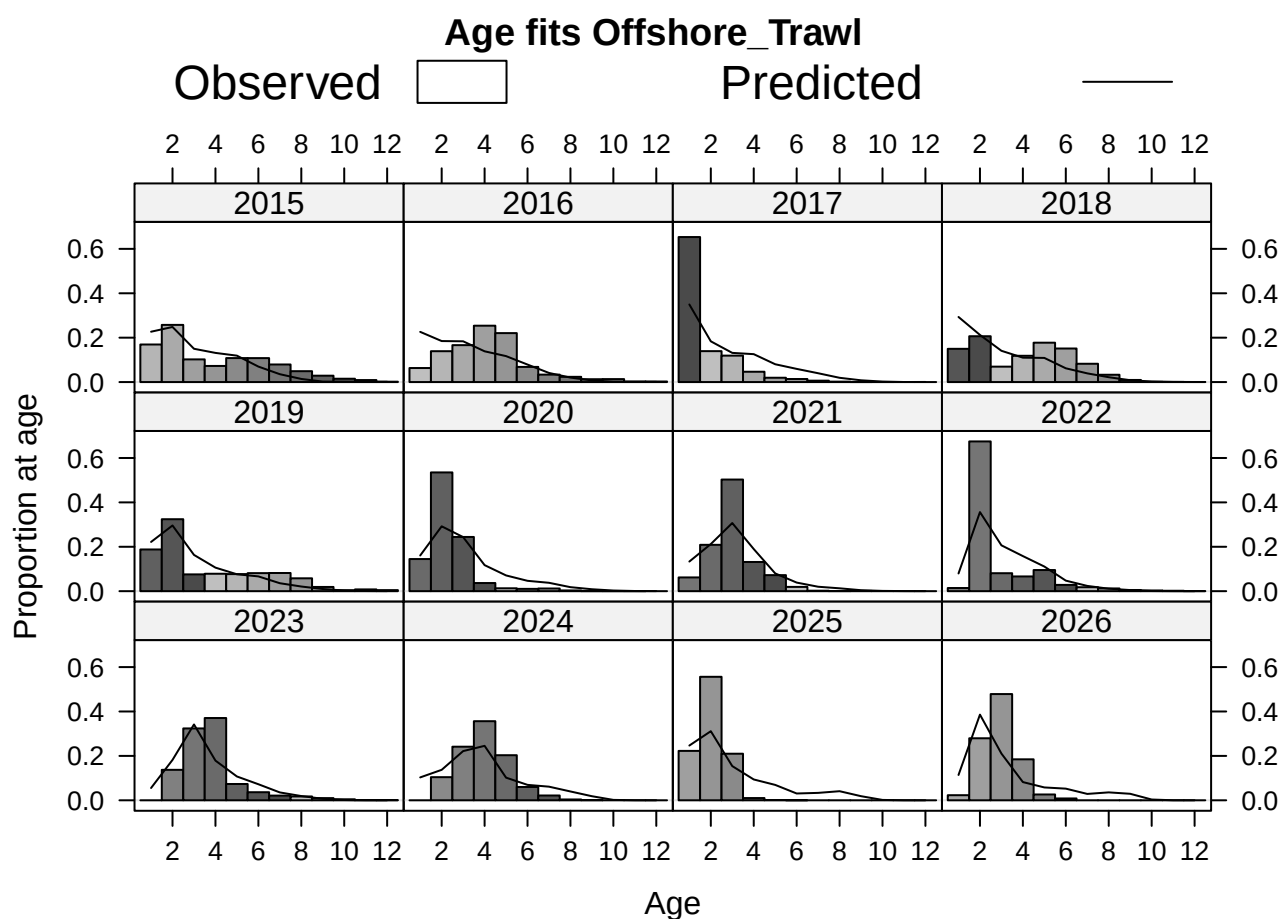


Figure 19: Model h2_1.06 (two-stock hypothesis) fit to the age compositions for the offshore trawl fishery (Fleet 4). Bars represent the observed data and lines represent the model predictions.

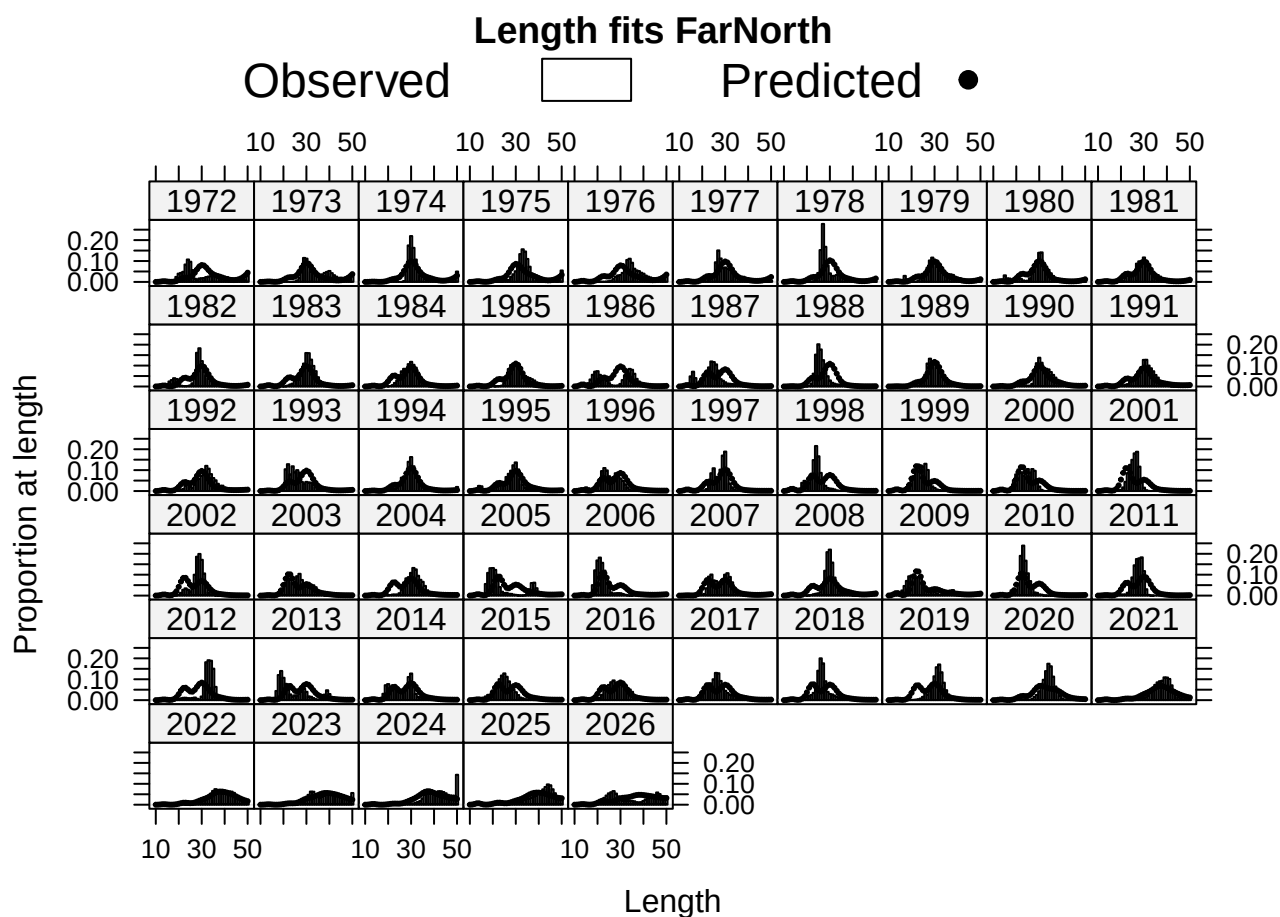


Figure 20: Model h1_1.06 (single-stock hypothesis) fit to the length compositions for the far north fishery (Fleet 3). Bars represent the observed data and points represent the model predictions.

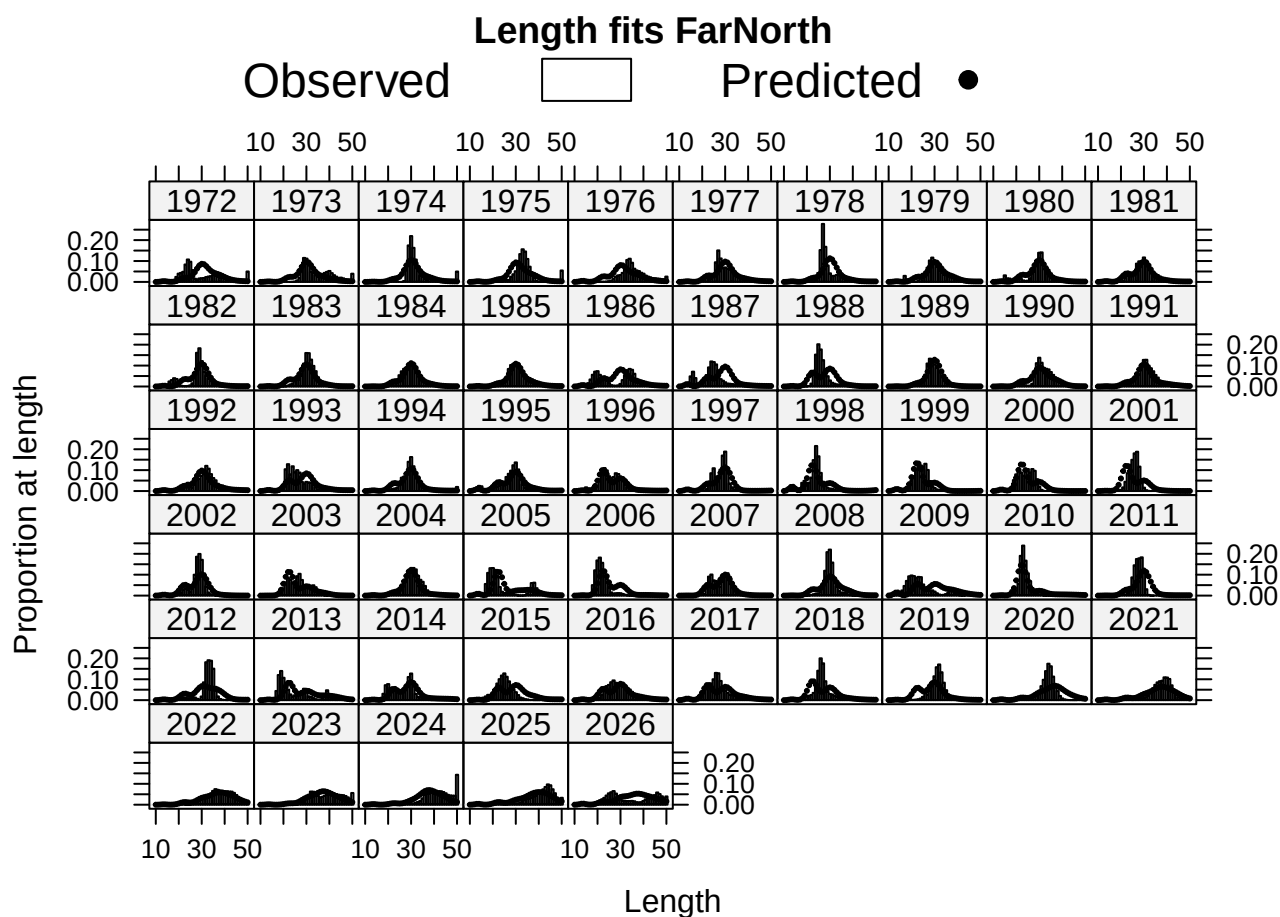


Figure 21: Model h2_1.06 (two-stock hypothesis) fit to the length compositions for the far north fishery (Fleet 3). Bars represent the observed data and points represent the model predictions.

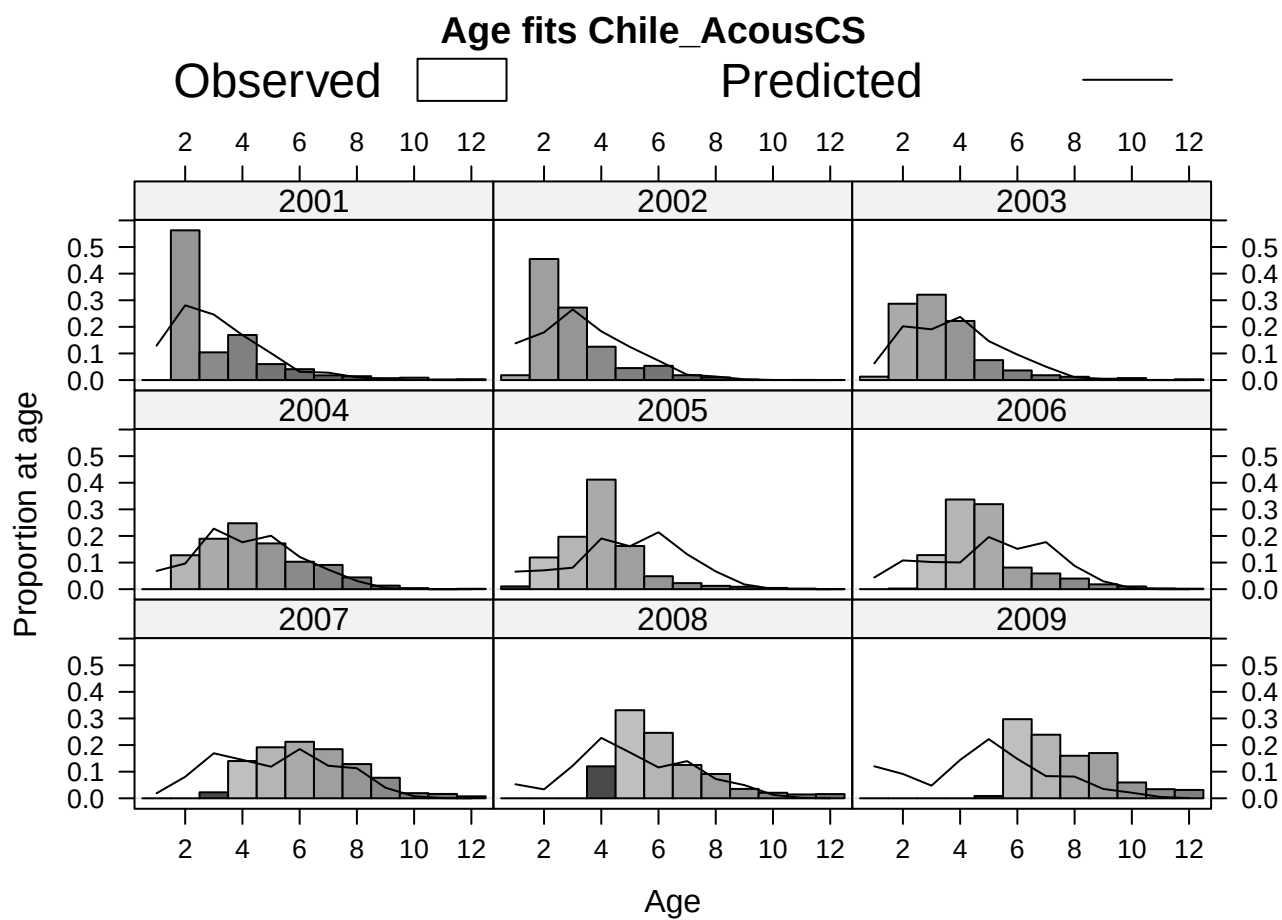


Figure 22: Model h1_1.06 (single-stock hypothesis) fit to the age compositions for the South-Central Acoustic survey. Bars represent the observed data and lines represent the model predictions.

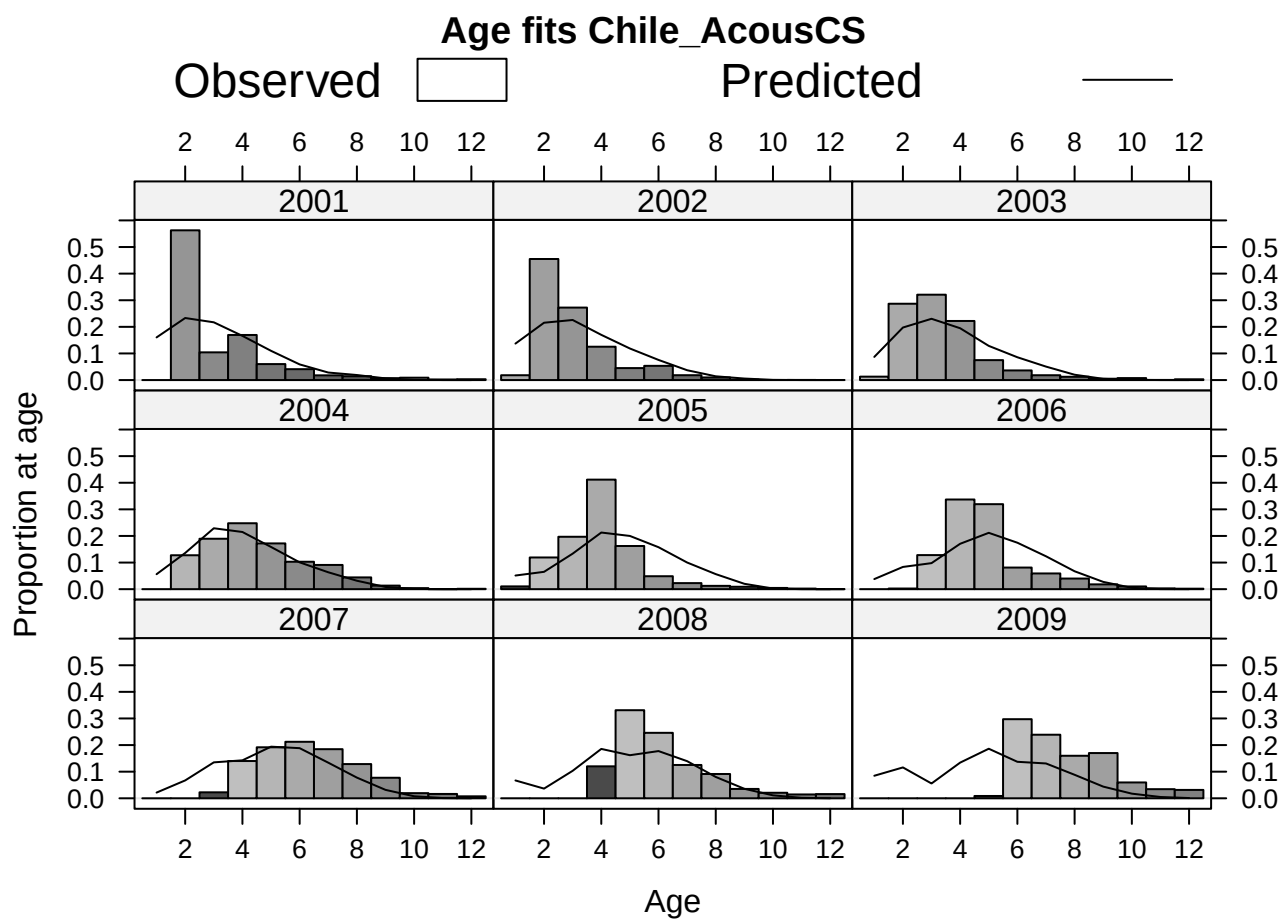


Figure 23: Model h2_1.06 (two-stock hypothesis) fit to the age compositions for the South-Central Acoustic survey. Bars represent the observed data and lines represent the model predictions.

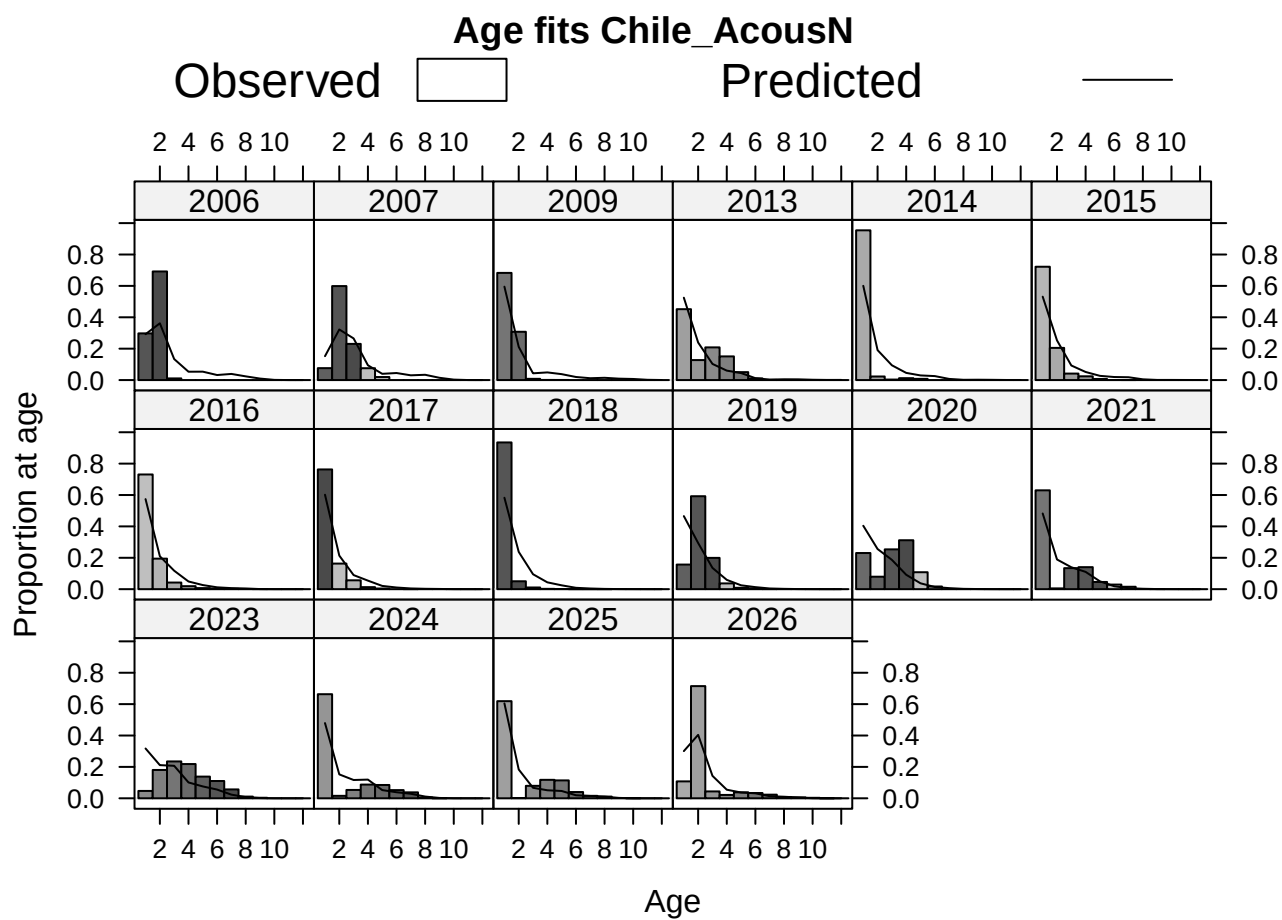


Figure 24: Model h1_1.06 (single-stock hypothesis) fit to the age compositions for the North Chilean acoustic survey. Bars represent the observed data and lines represent the model predictions.

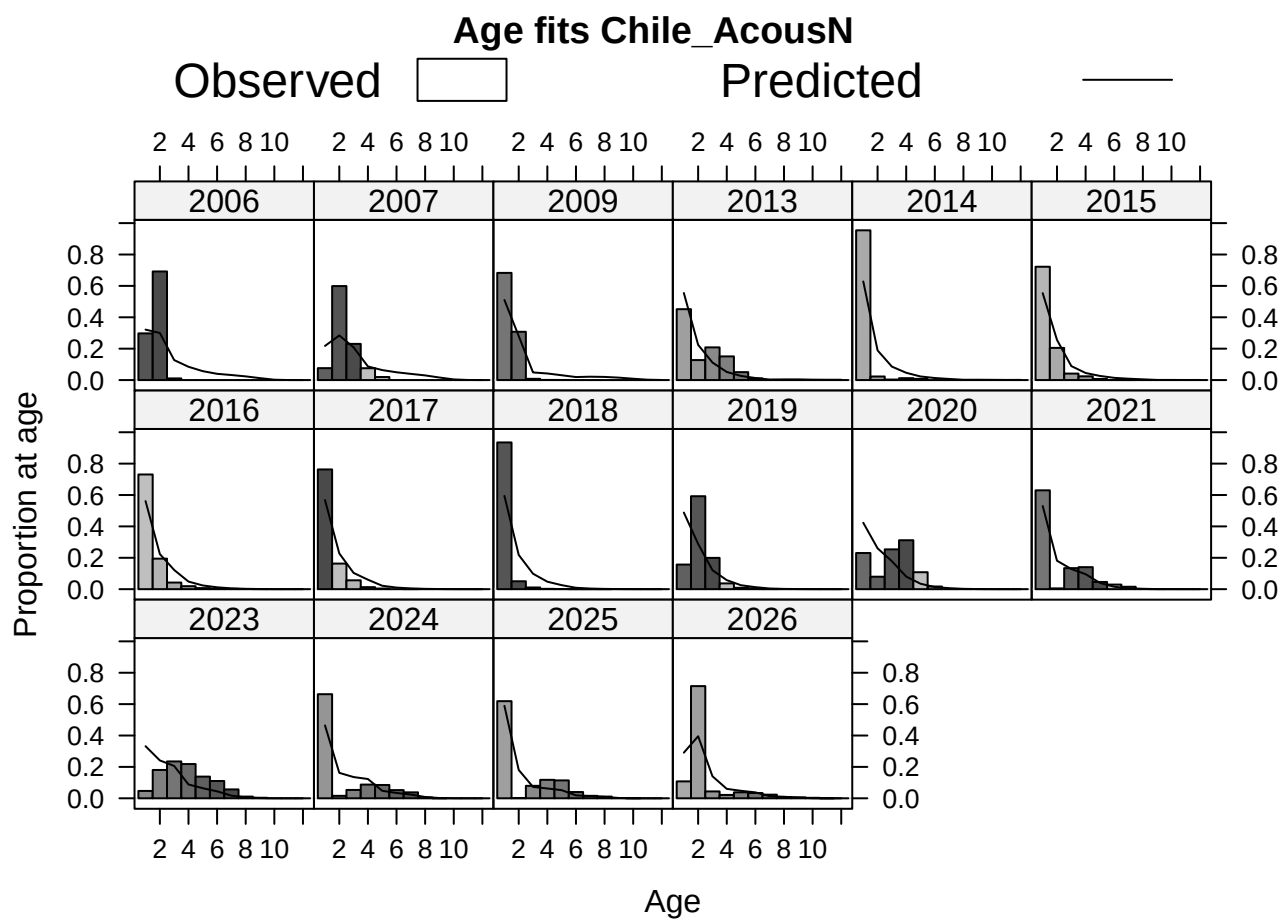


Figure 25: Model h2_1.06 (two-stock hypothesis) fit to the age compositions for the North Chilean acoustic survey. Bars represent the observed data and lines represent the model predictions.

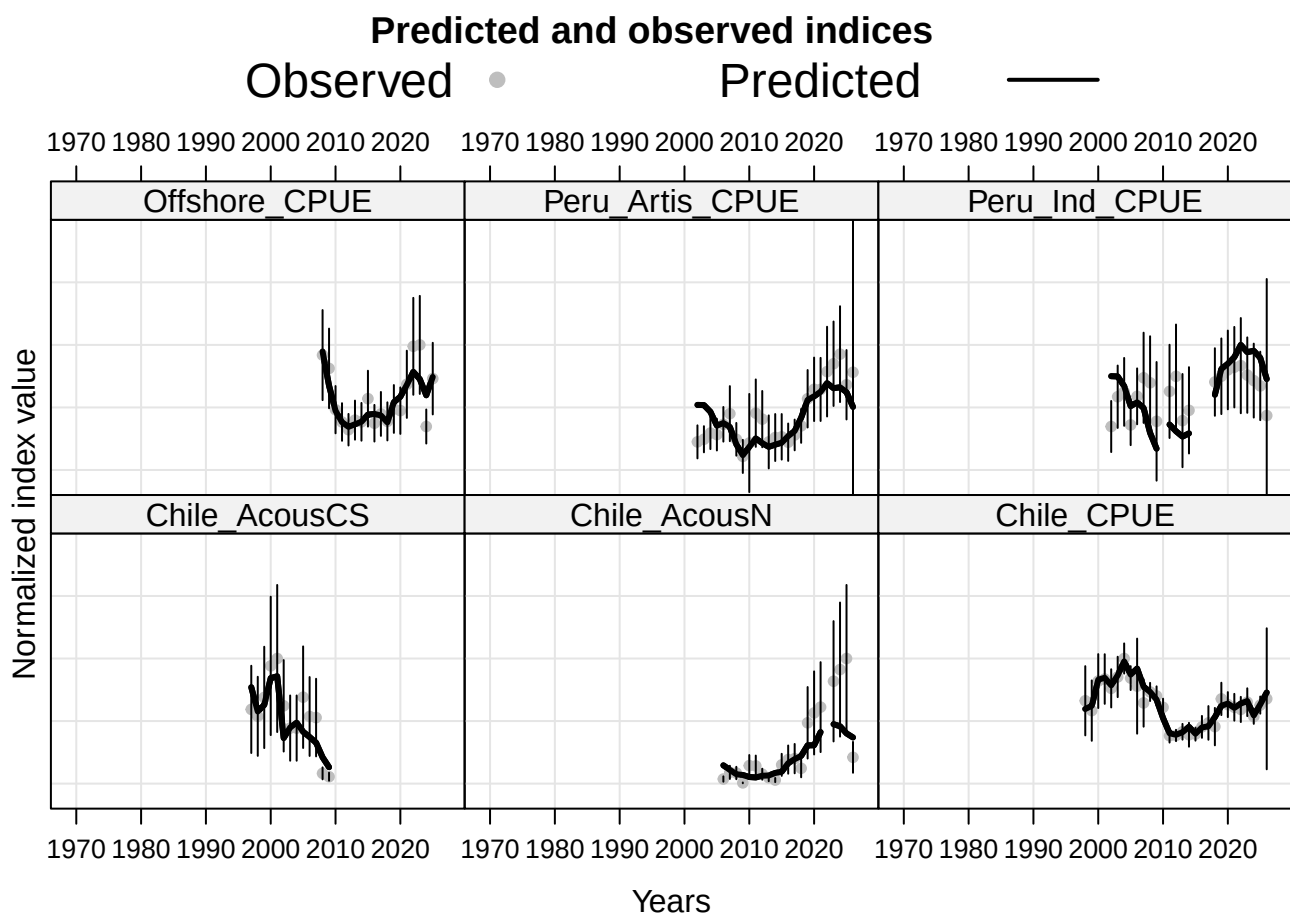


Figure 26: Model h1_1.06 (single-stock hypothesis) fit to different indices. Vertical bars represent 2 standard deviations around the observations.

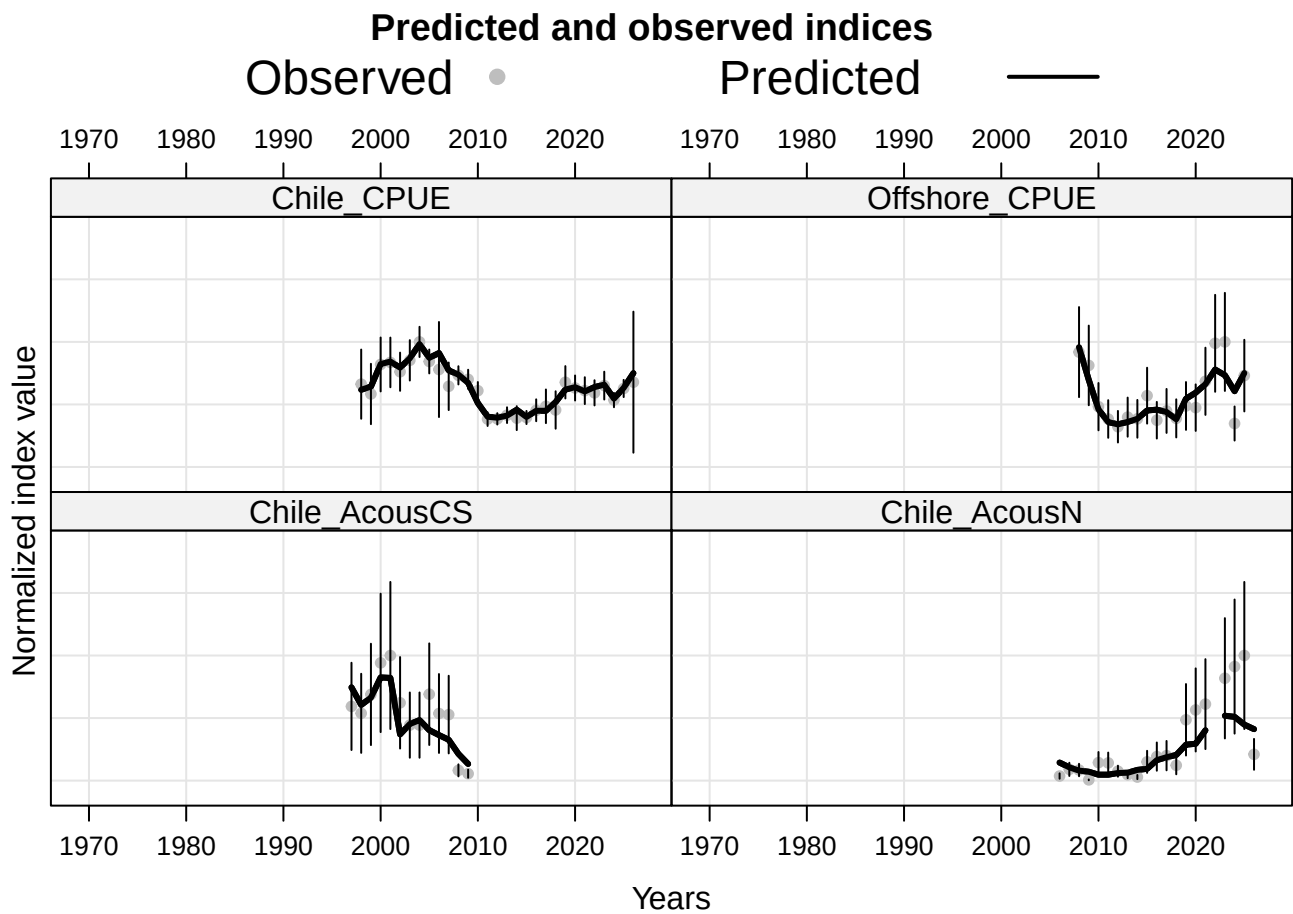


Figure 27: Model h2_1.06 (two-stock hypothesis) fit to indices for the south stock. Vertical bars represent 2 standard deviations around the observations.

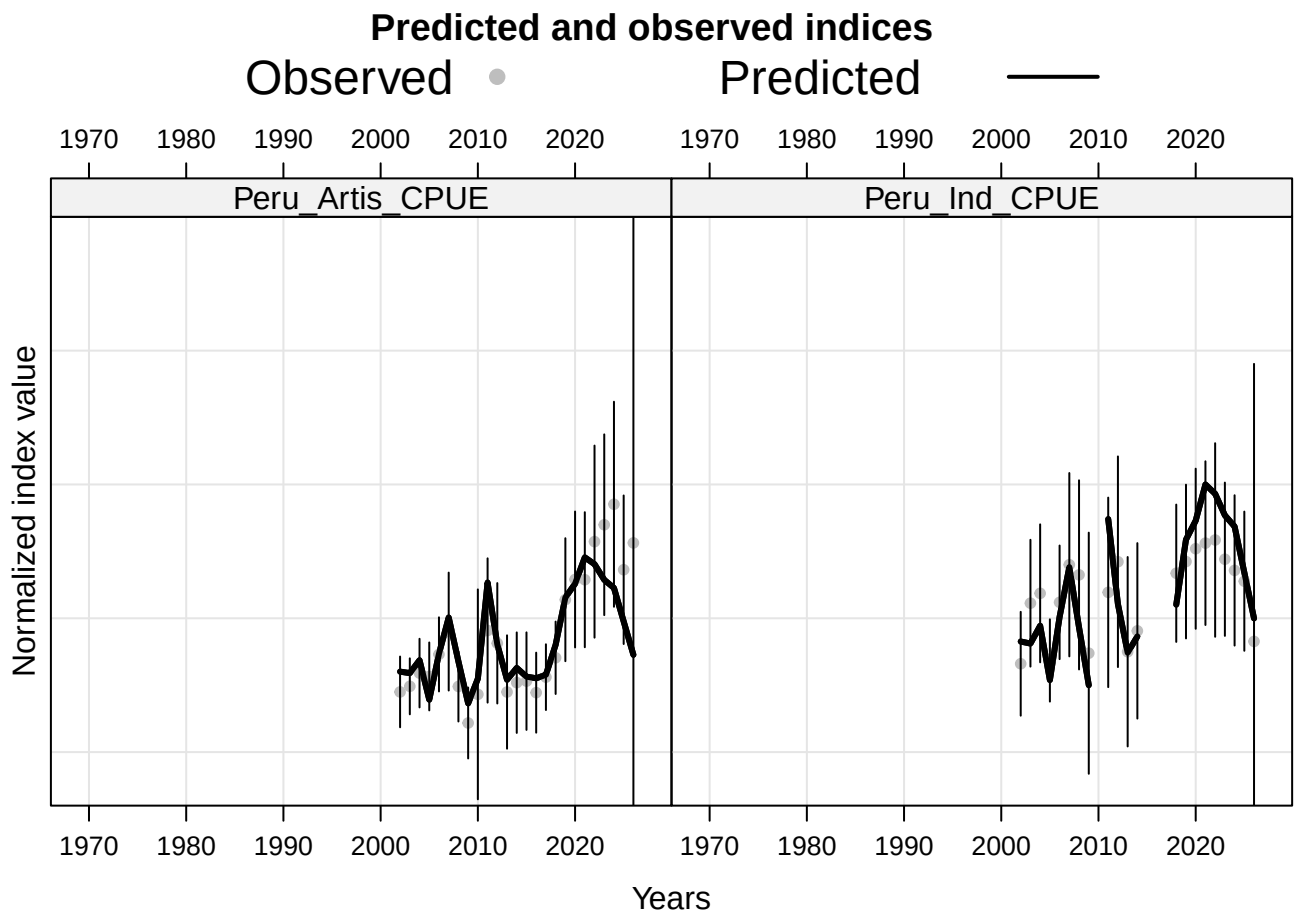


Figure 28: Model h2_1.06 (two-stock hypothesis) fit to indices for the north stock. Vertical bars represent 2 standard deviations around the observations.

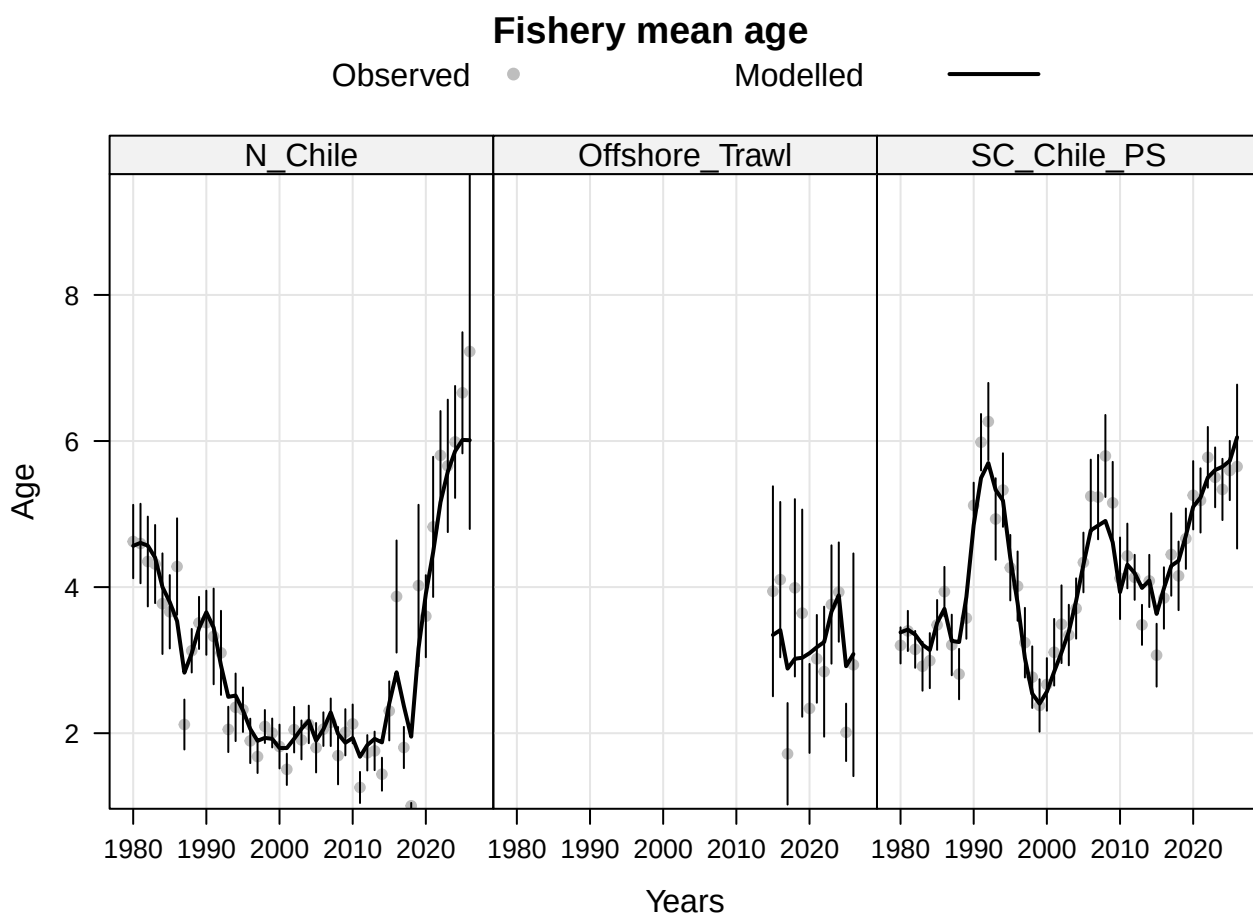


Figure 29: Mean age by year and fishery. Line represents the Model h1_1.06 (single-stock hypothesis) predictions and dots observed values with implied input error bars.

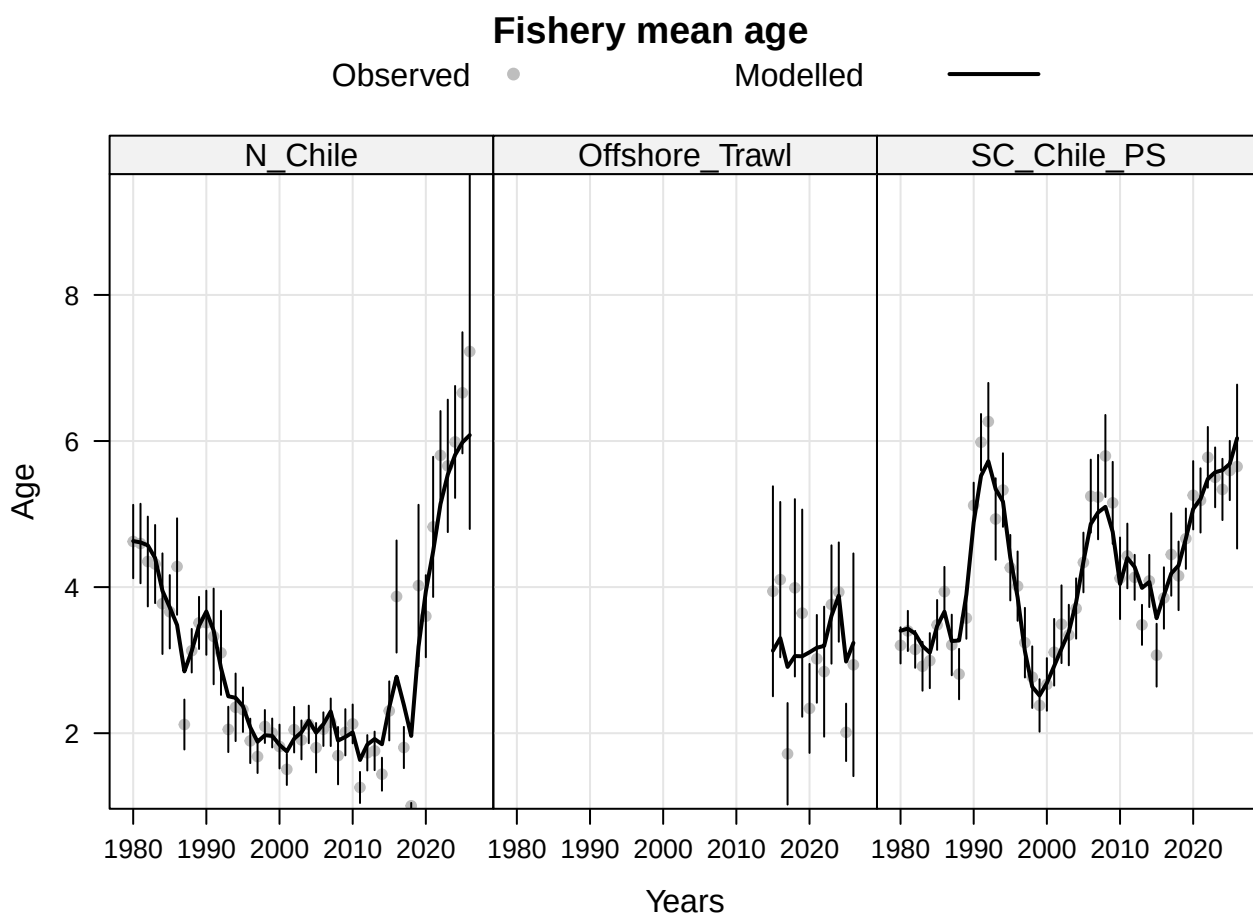


Figure 30: Mean age by year and fishery. Line represents the Model h2_1.06 (two-stock hypothesis) predictions and dots observed values with implied input error bars.

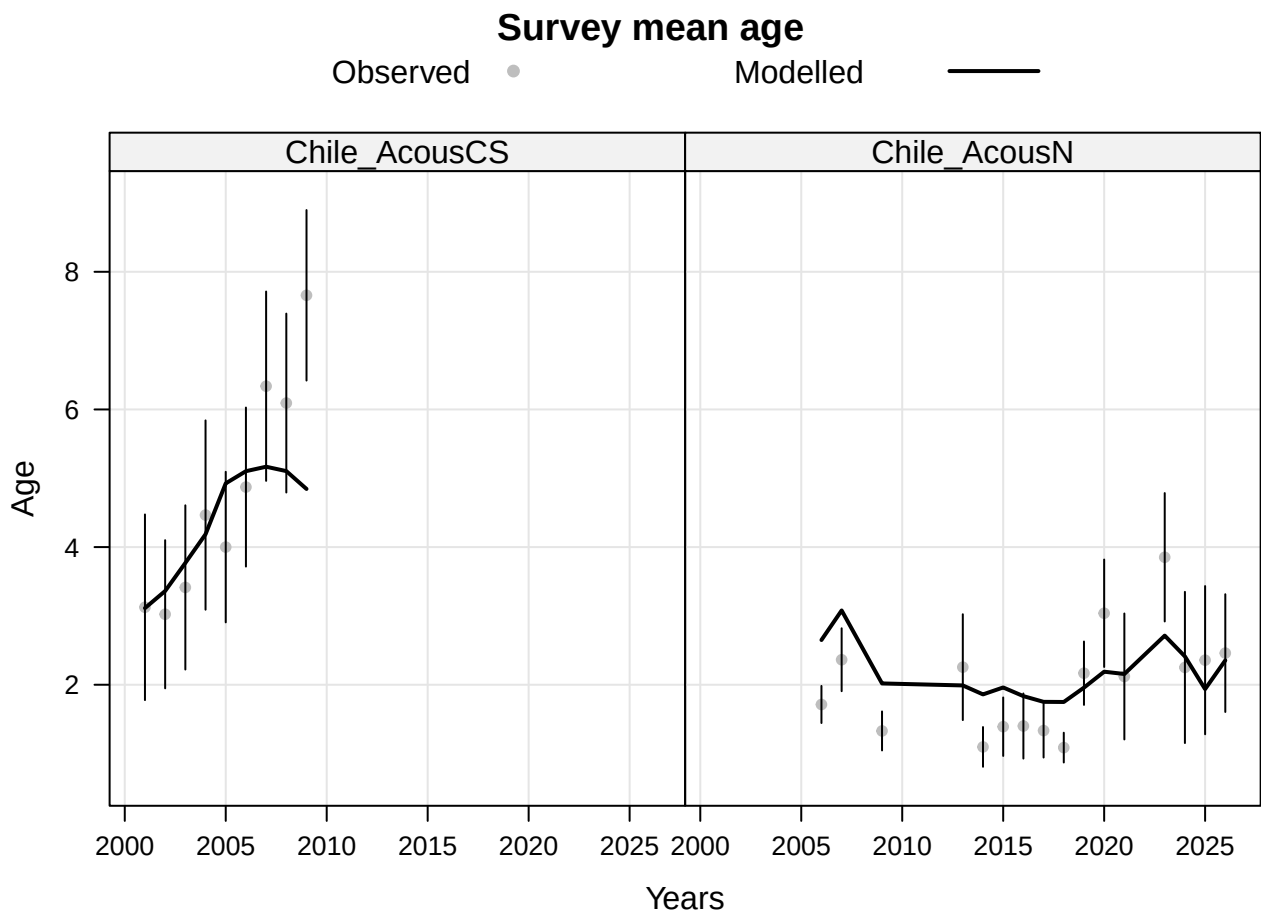


Figure 31: Mean age by year and survey. Line represents the Model h1_1.06 (single-stock hypothesis) predictions and dots observed values with implied input error bars.

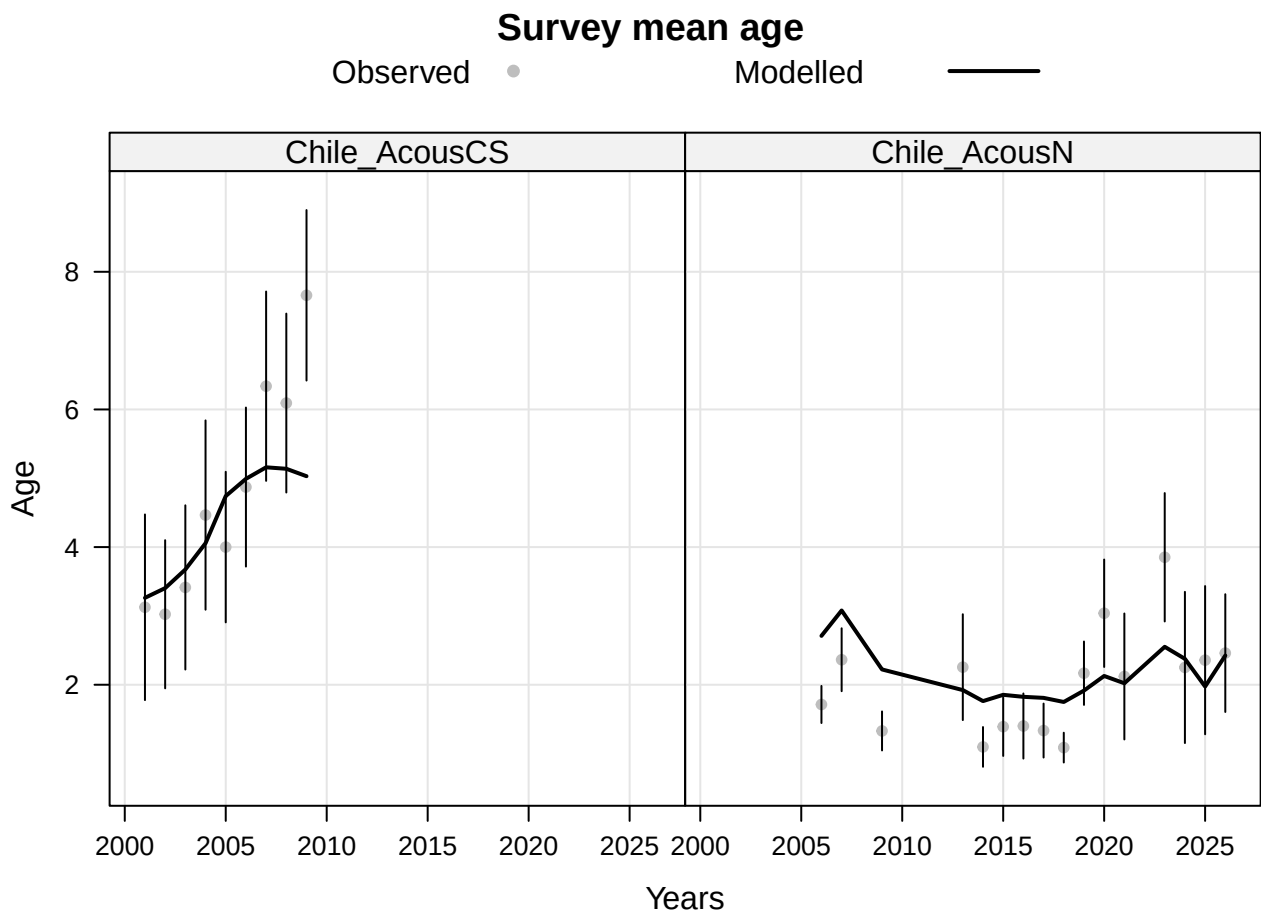


Figure 32: Mean age by year and survey. Line represents the Model h2_1.06 (two-stock hypothesis) predictions and dots observed values with implied input error bars.

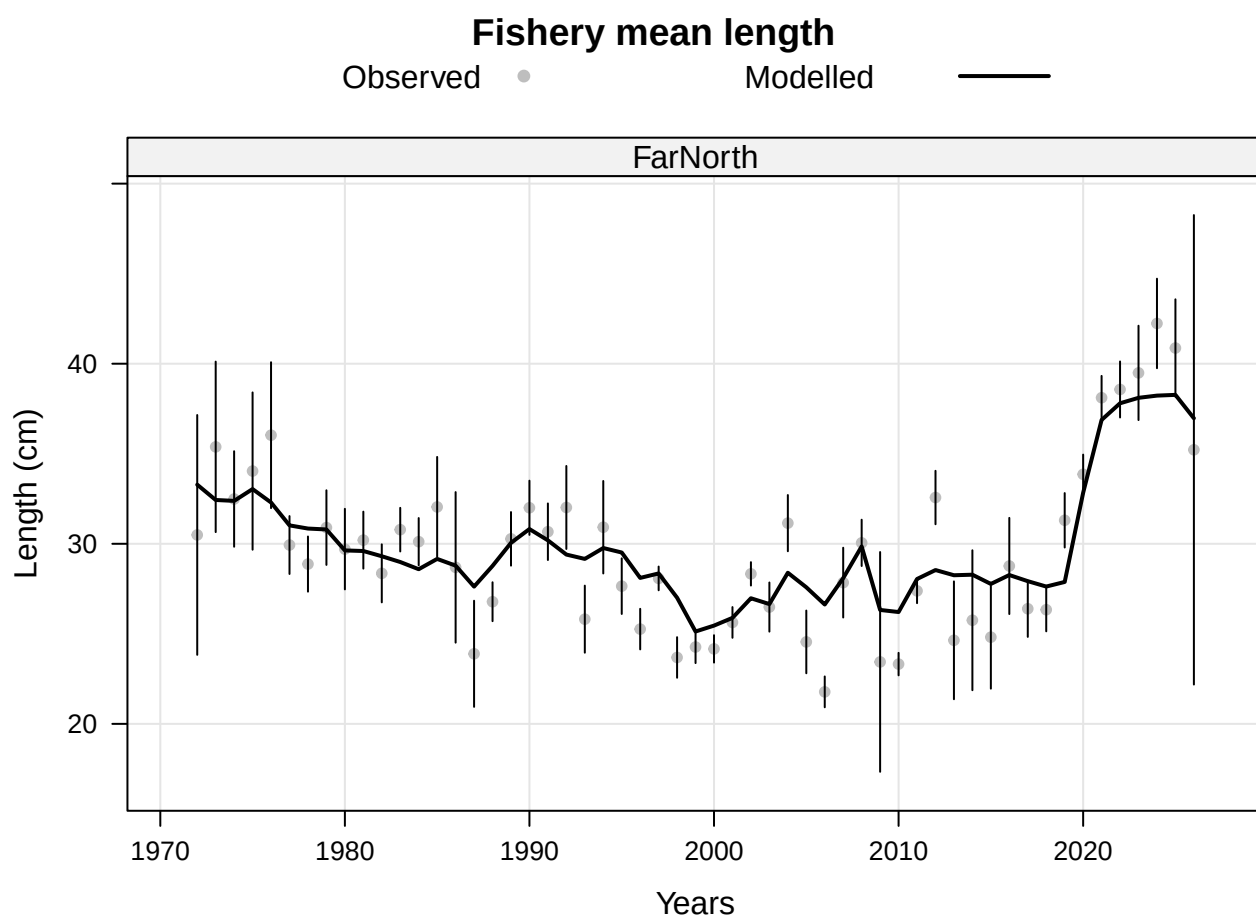


Figure 33: Mean length by year in Fleet 3 (Far North). Line represents the Model h1_1.06 (single-stock hypothesis) predictions and dots observed values with implied input error bars.

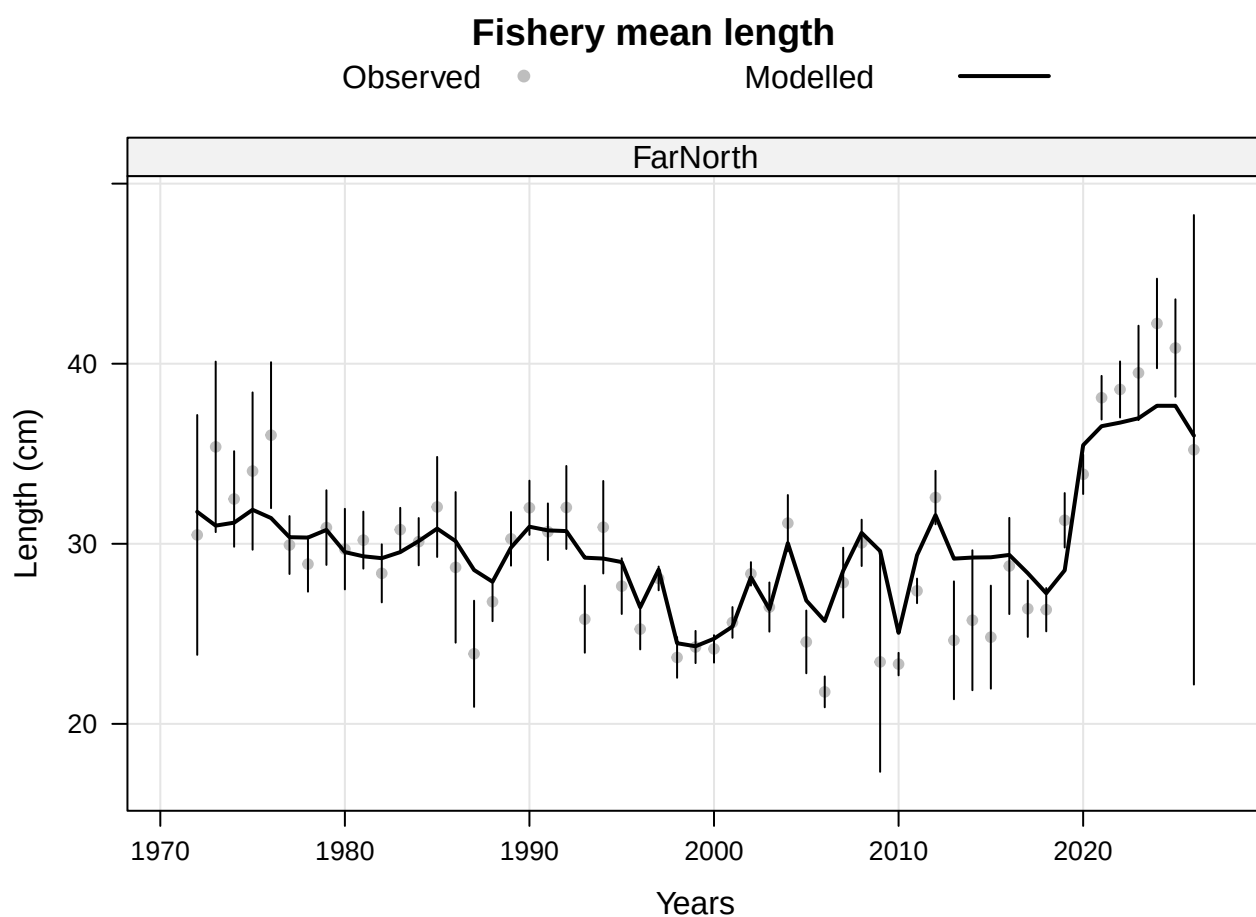


Figure 34: Mean length by year in Fleet 3 (Far North). Line represents the Model h2_1.06 (two-stock hypothesis) predictions and dots observed values with implied input error bars.

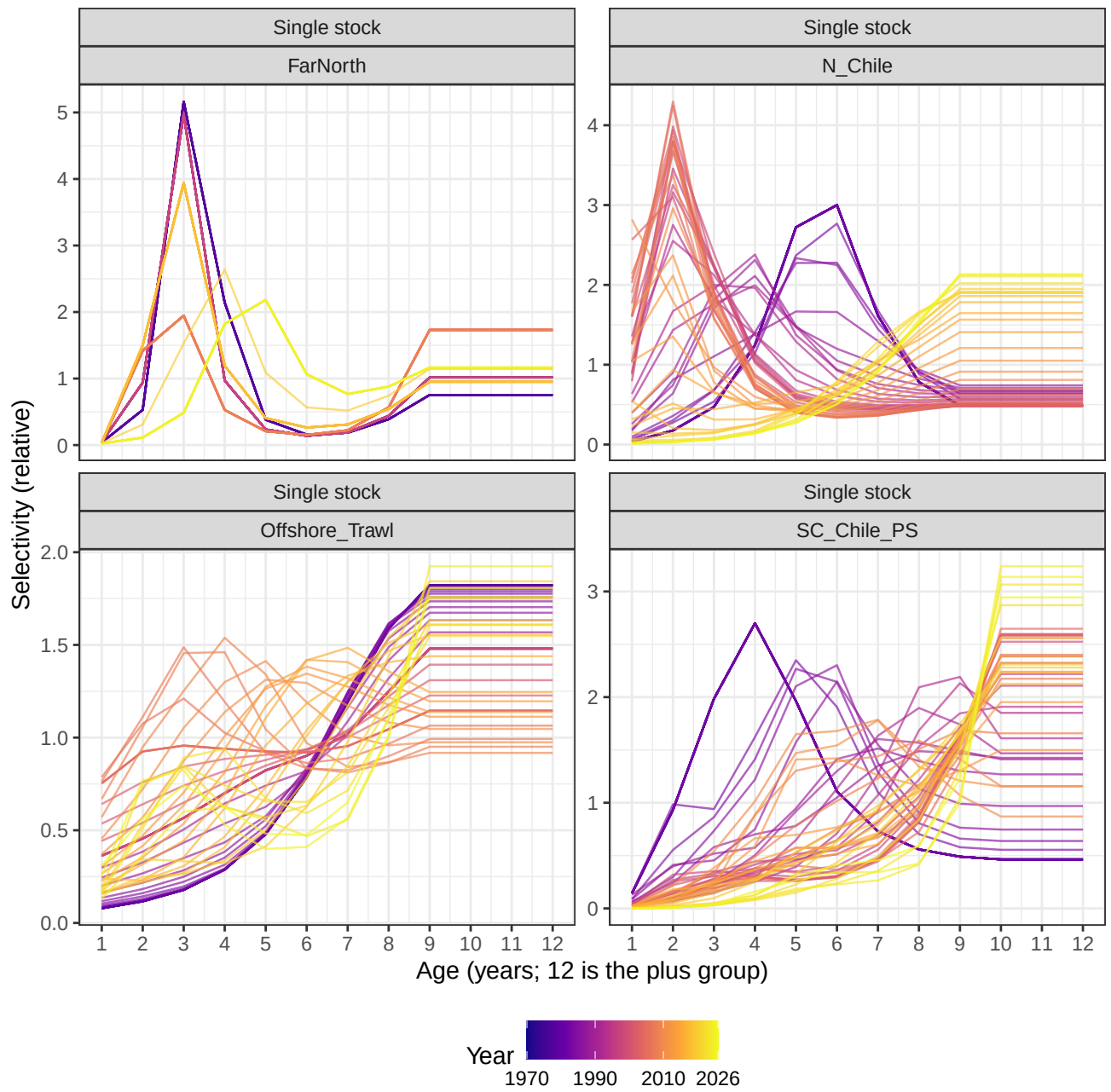


Figure 35: Estimates of selectivity by fishery over time for Model h1_1.06 (single-stock hypothesis).

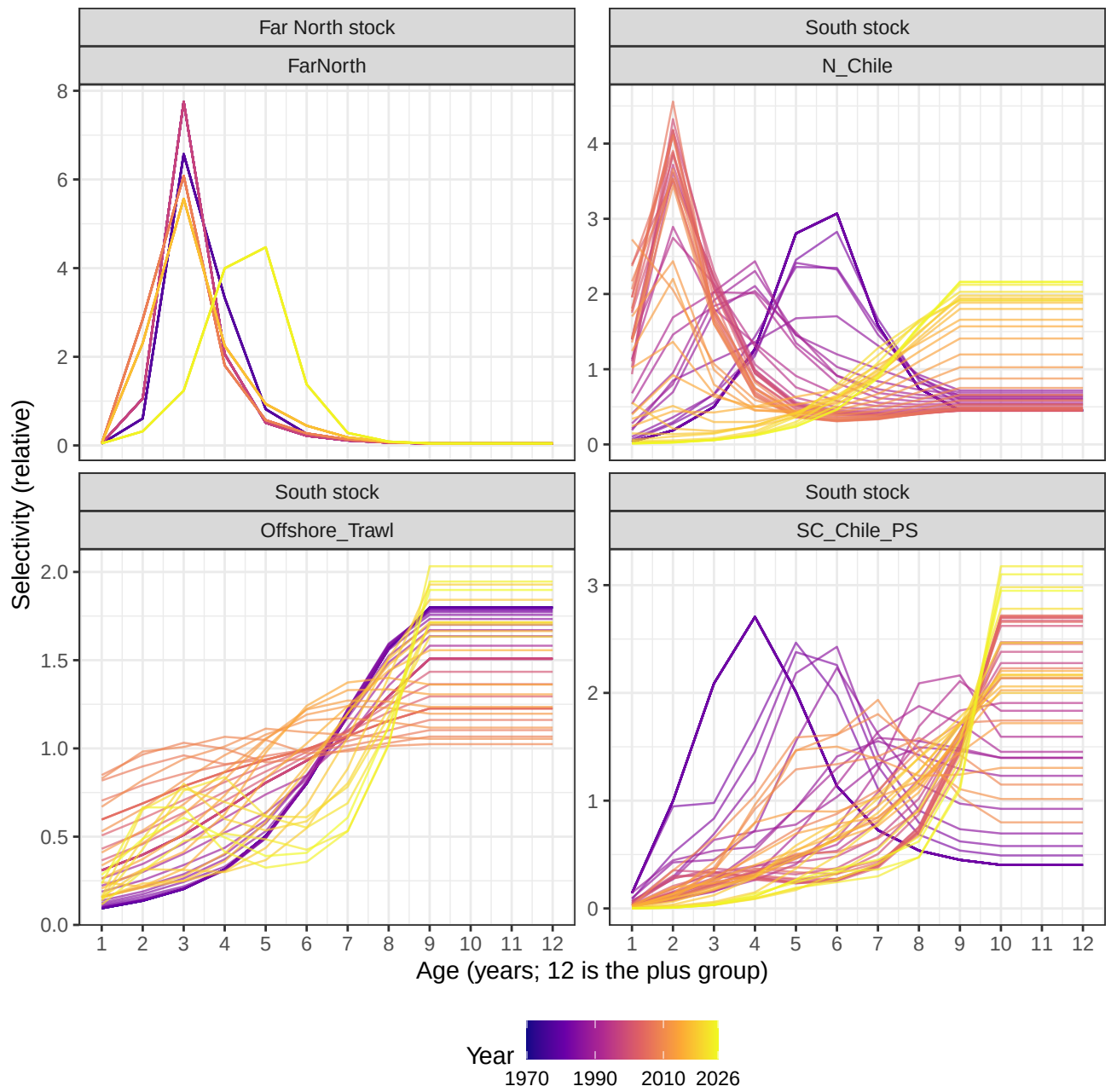


Figure 36: Estimates of selectivity by fishery over time for Model h2_1.06 (two-stock hypothesis).

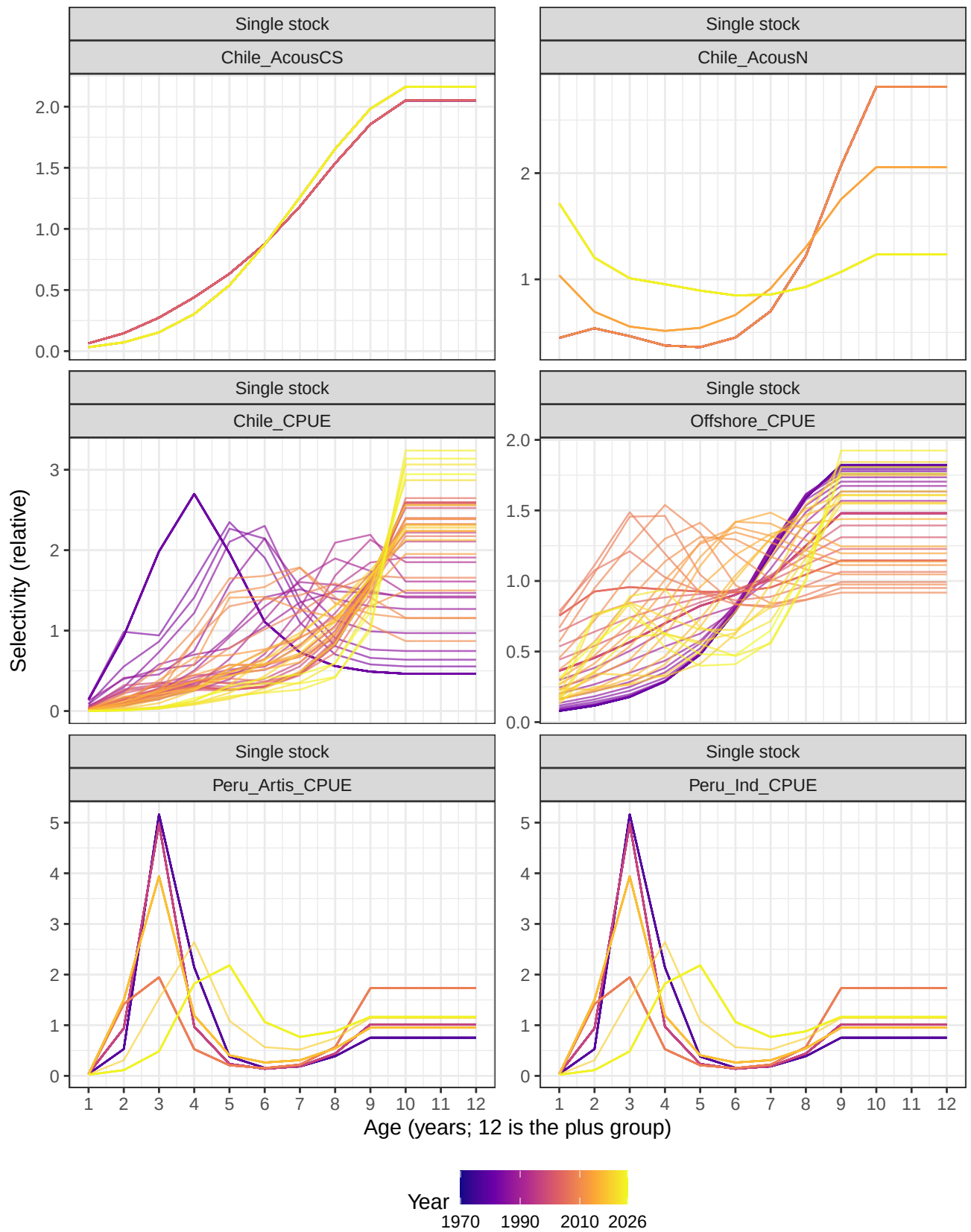


Figure 37: Estimates of selectivity by survey over time for Model h1_1.06 (single-stock hypothesis).

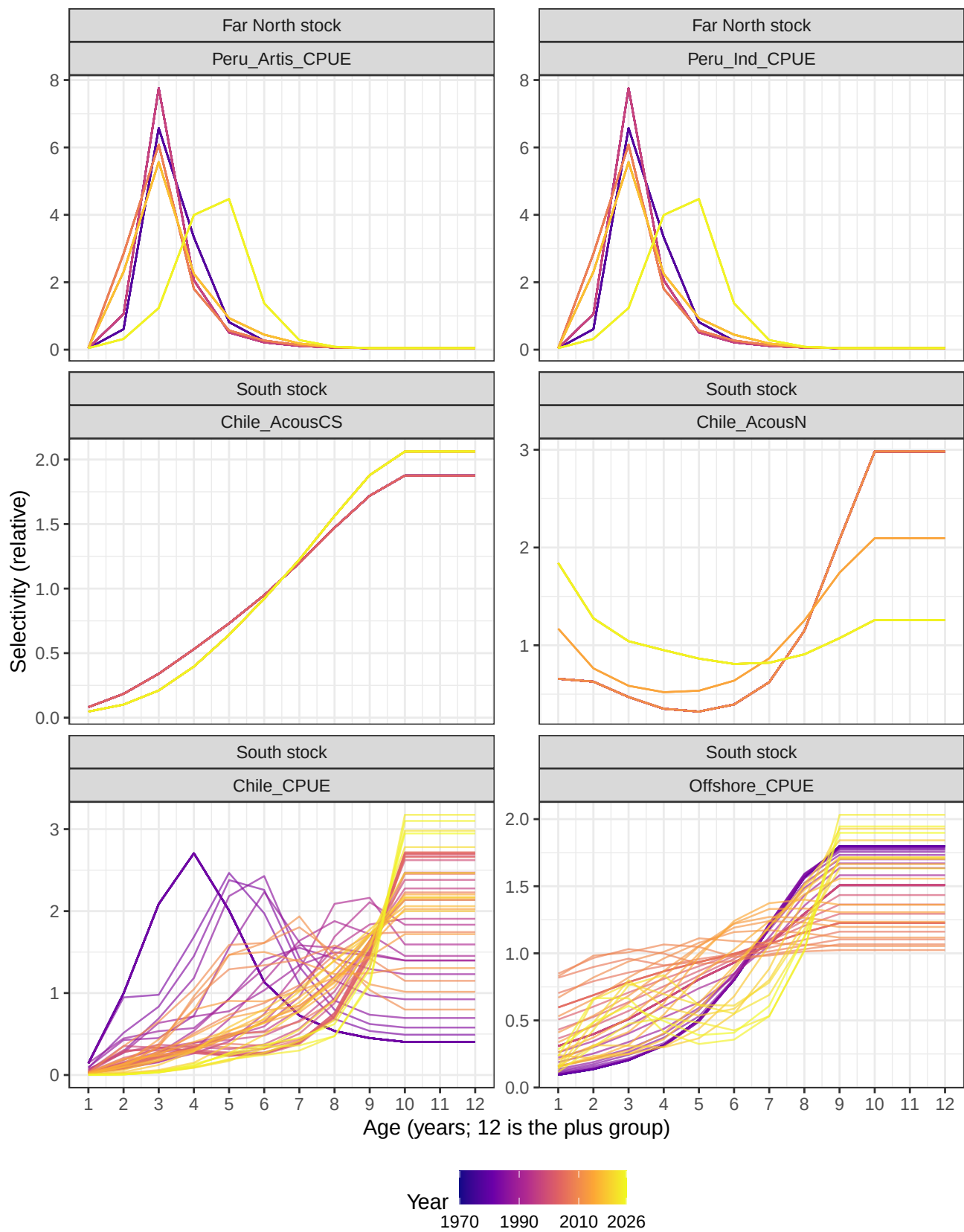


Figure 38: Estimates of selectivity by survey over time for Model h2_1.06 (two-stock hypothesis).

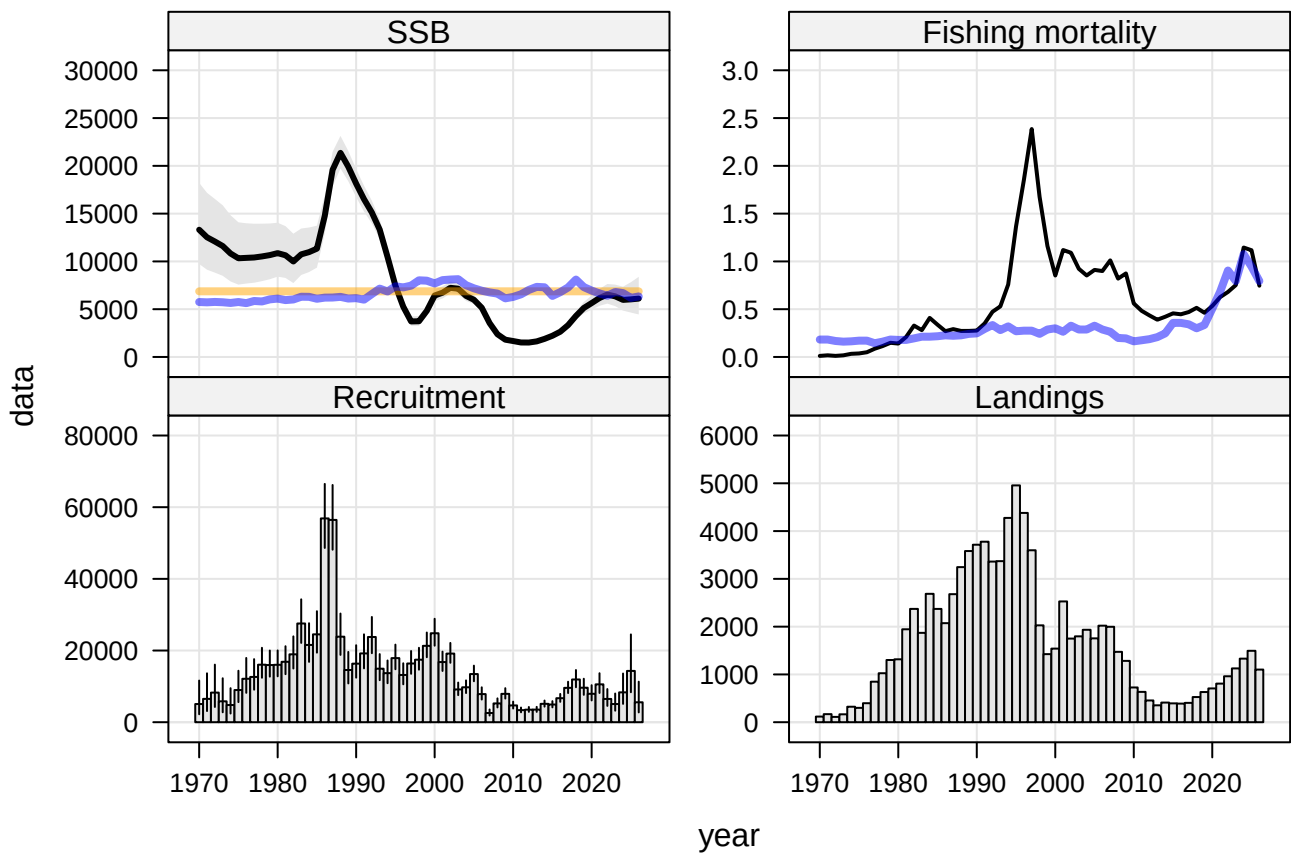


Figure 39: Model h1_1.06 (single-stock hypothesis) summary estimates over time showing spawning biomass (kt; top left), recruitment at age 1 (millions; lower left), total fishing mortality (top right), and total catch (kt; bottom right). Reference overlays follow the existing summary routine, including the recent-average biomass reference; these summaries use the fitted assessment outputs.

Comparing fished with unfished biomass

Fished — Unfished

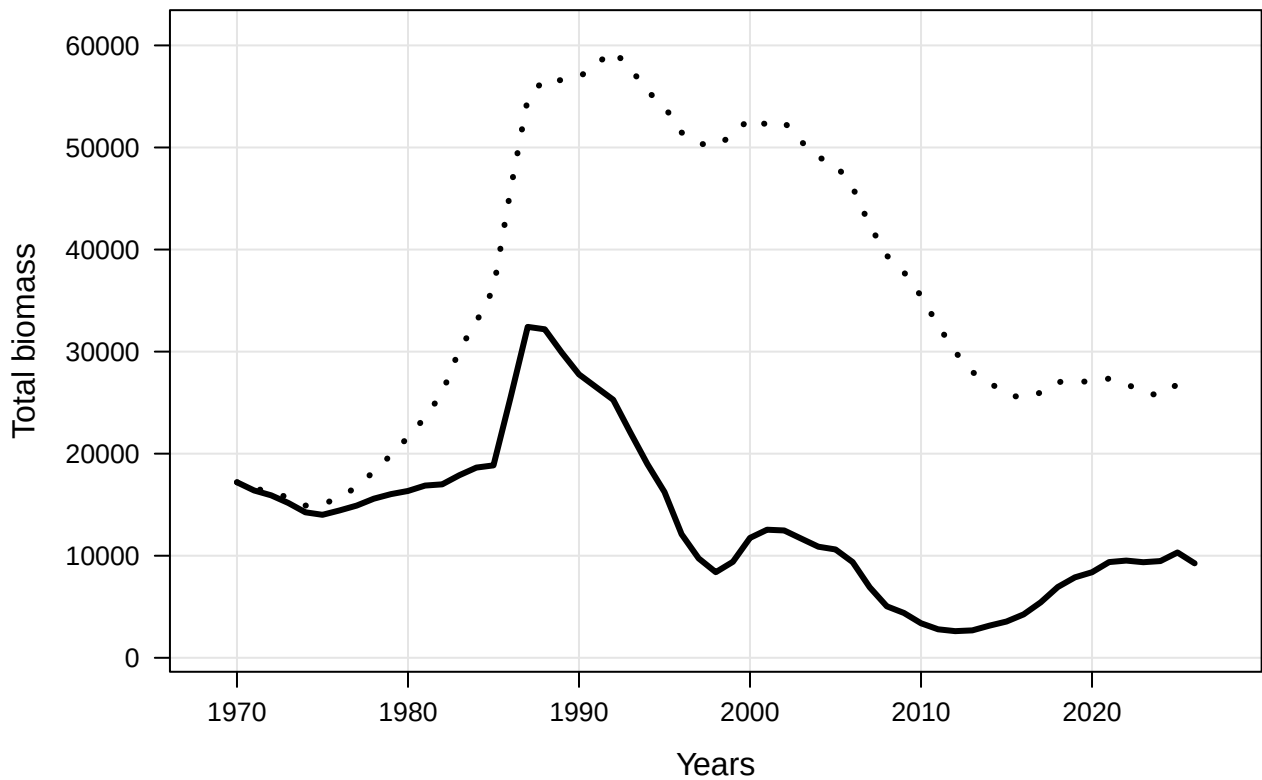


Figure 40: Model h1_1.06 (single-stock hypothesis) results for the estimated total biomass (solid line) and the estimated total biomass that would have occurred if no fishing had taken place (dotted line), beginning in 1970. This year, the ratio of total biomass to total biomass without fishing is 0.36.

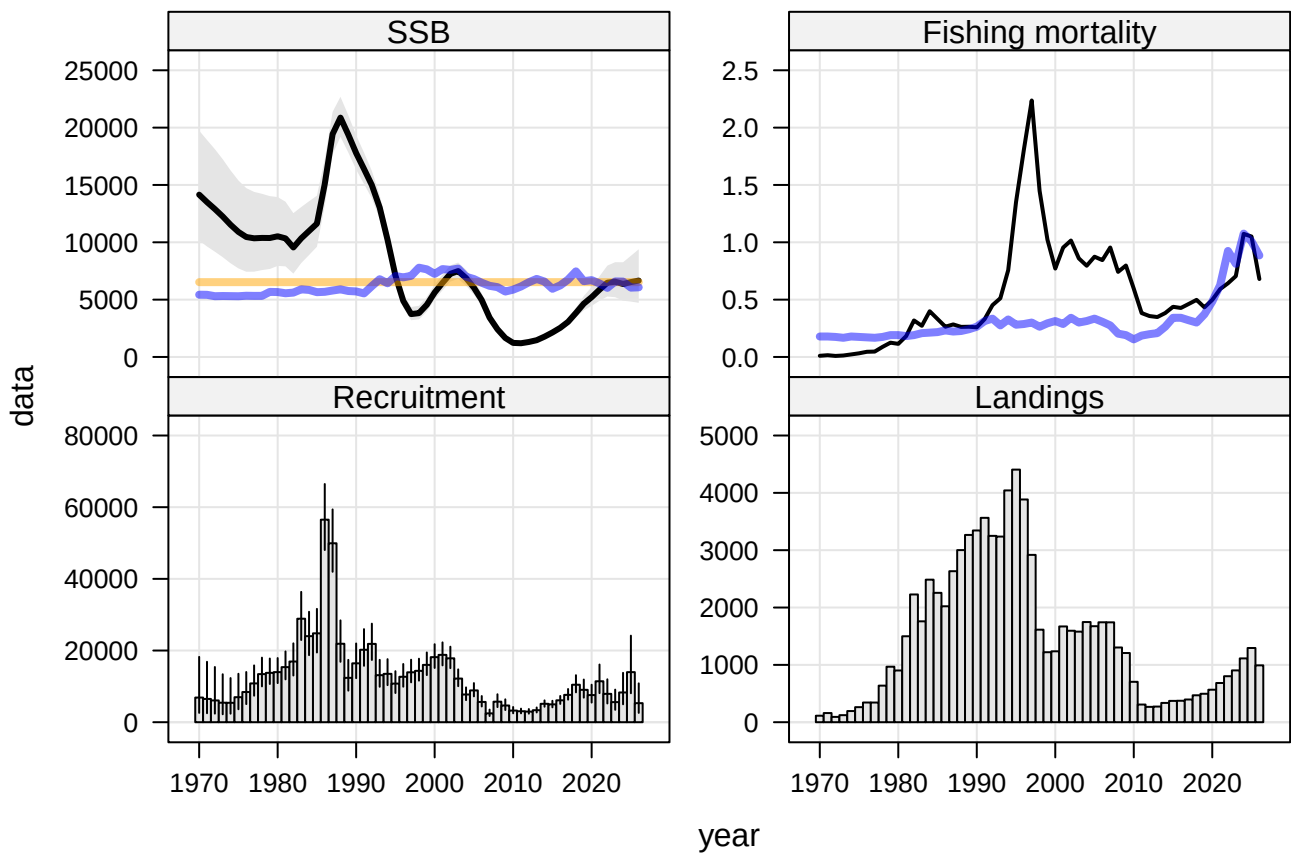


Figure 41: Model h2_1.06 (two-stock hypothesis) summary estimates over time showing spawning biomass (kt; top left), recruitment at age 1 (millions; lower left), total fishing mortality (top right), and total catch (kt; bottom right) for the south stock. Reference overlays follow the existing summary routine, including the recent-average biomass reference; these summaries use the fitted assessment outputs.

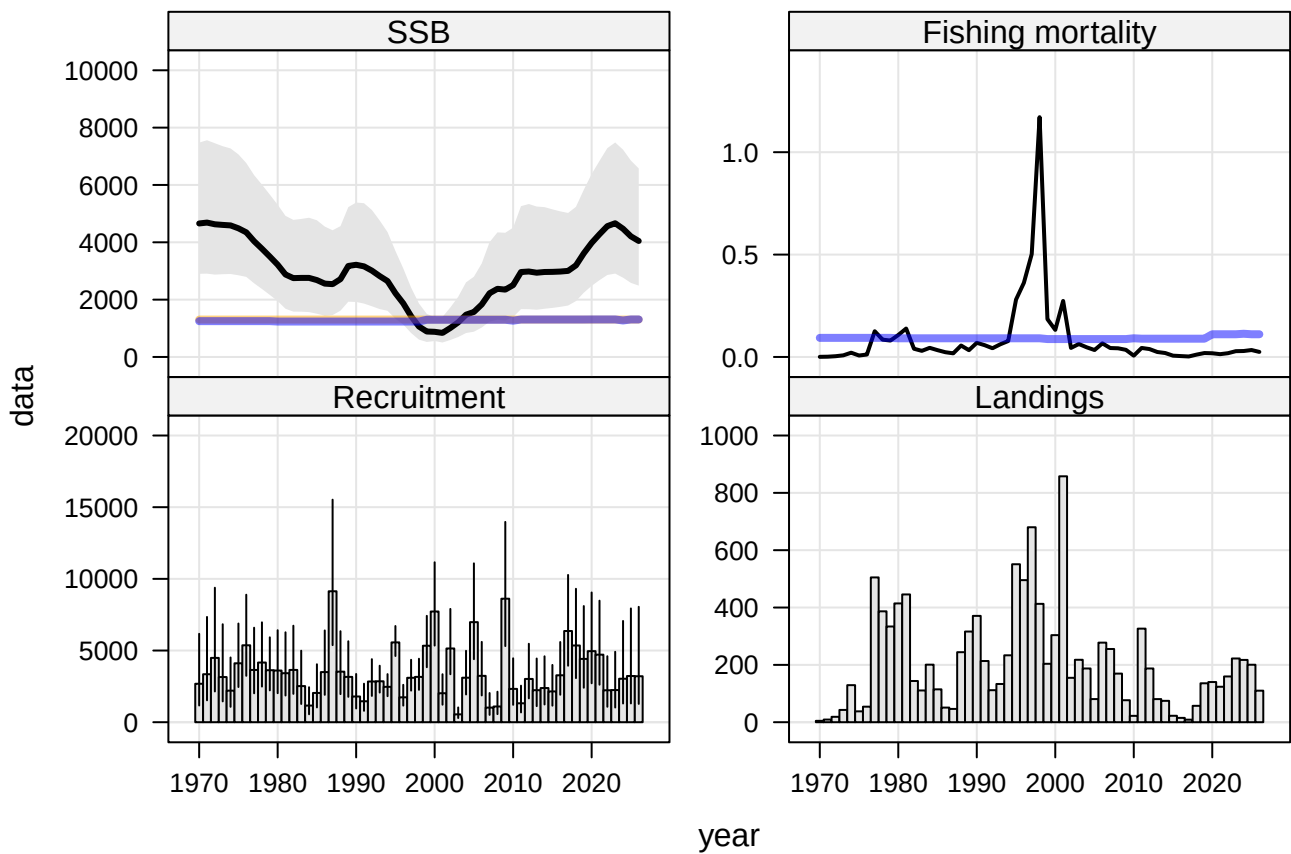


Figure 42: Model h2_1.06 (two-stock hypothesis) summary estimates over time showing spawning biomass (kt; top left), recruitment at age 1 (millions; lower left), total fishing mortality (top right), and total catch (kt; bottom right) for the far north stock. Reference overlays follow the existing summary routine, including the recent-average biomass reference; these summaries use the fitted assessment outputs.

Comparing fished with unfished biomass

Fished — Unfished

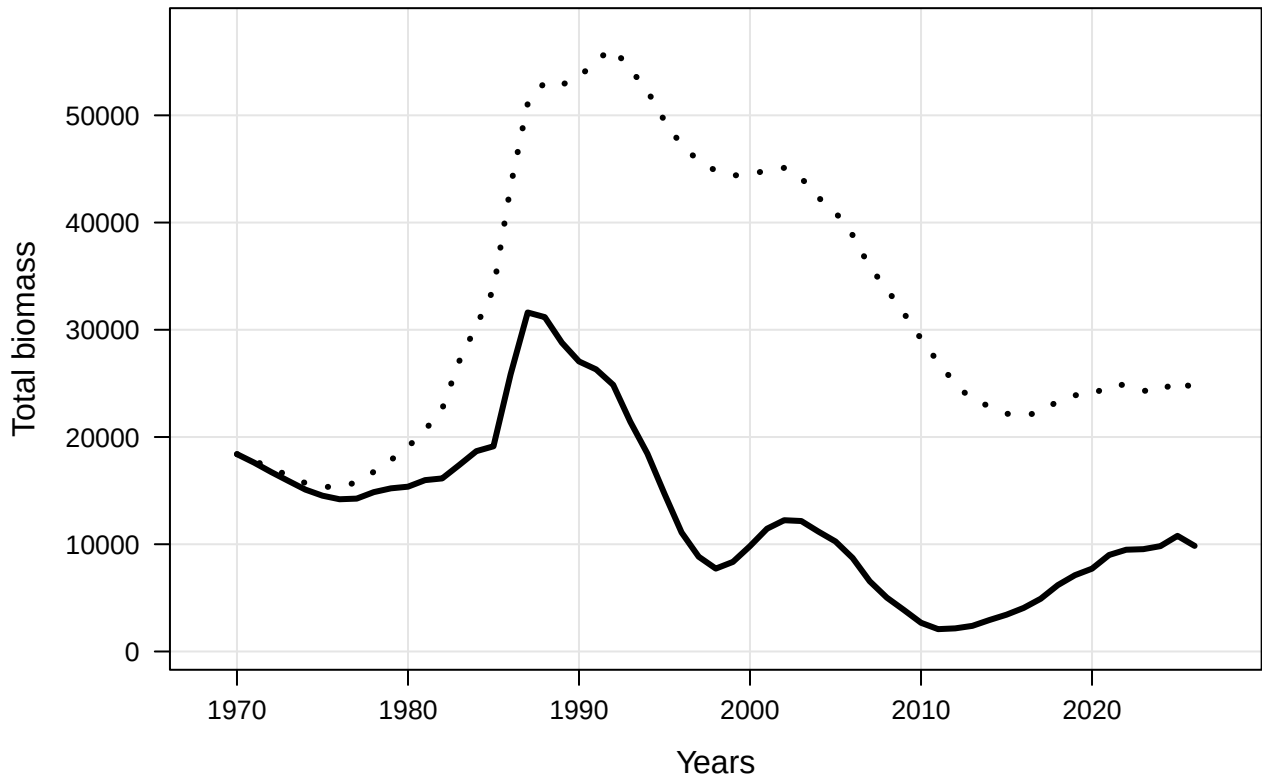


Figure 43: Model h2_1.06 (two-stock hypothesis) results for the estimated total biomass (solid line) and the estimated total biomass that would have occurred if no fishing had taken place (dotted line) for the south stock, beginning in 1970. This year, the ratio of total biomass to total biomass without fishing is 0.4.

Comparing fished with unfished biomass

Fished — Unfished

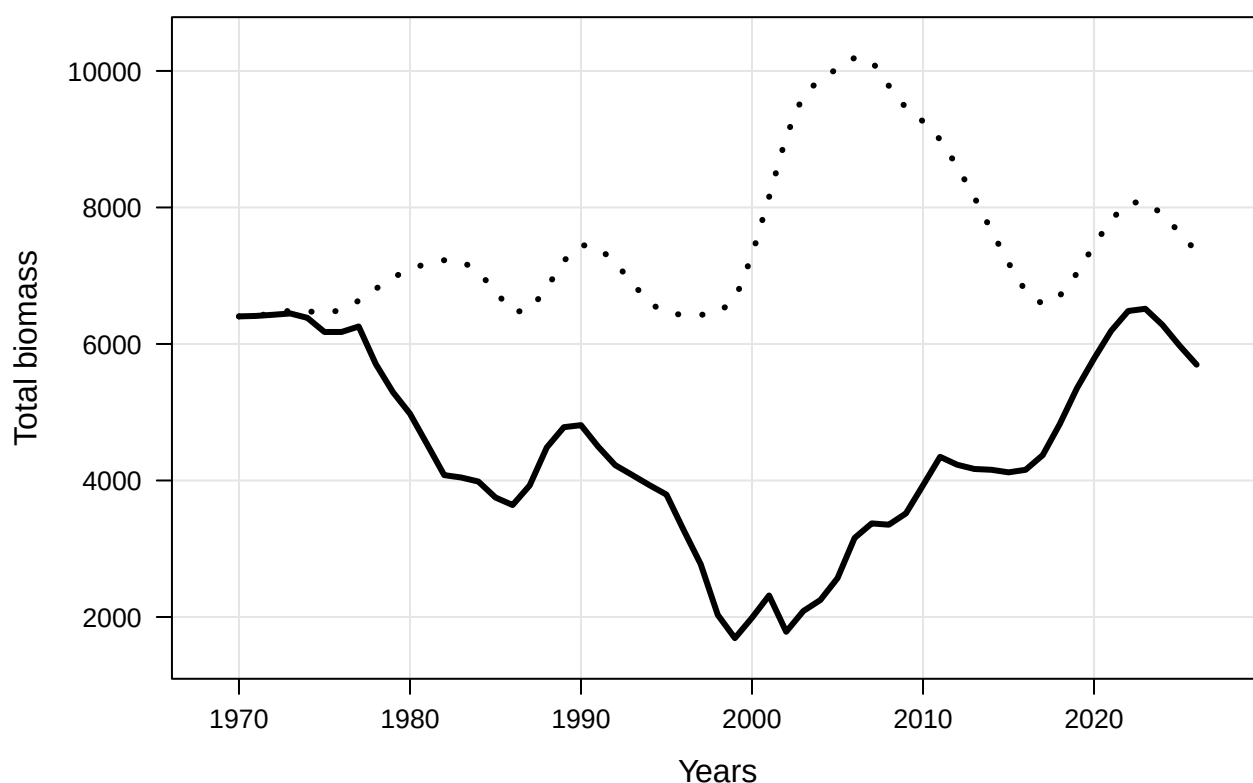


Figure 44: Model h2_1.06 (two-stock hypothesis) results for the estimated total biomass (solid line) and the estimated total biomass that would have occurred if no fishing had taken place (dotted line) for the far north stock, beginning in 1970. This year, the ratio of total biomass to total biomass without fishing is 0.77.

9.1 Projection diagnostics

These figures retain the **earlier five-year model 1.06.1s projection outputs (2027–2031)** for comparison. The updated ten-year projections from the corrected `jjm.tpl`, including the revised scenario-4 calculation, are summarized in Section 4.2. Both sets use steepness 0.65, with recruitment over 2001–2015 for the single stock and southern component and 1970–2016 for Far North.

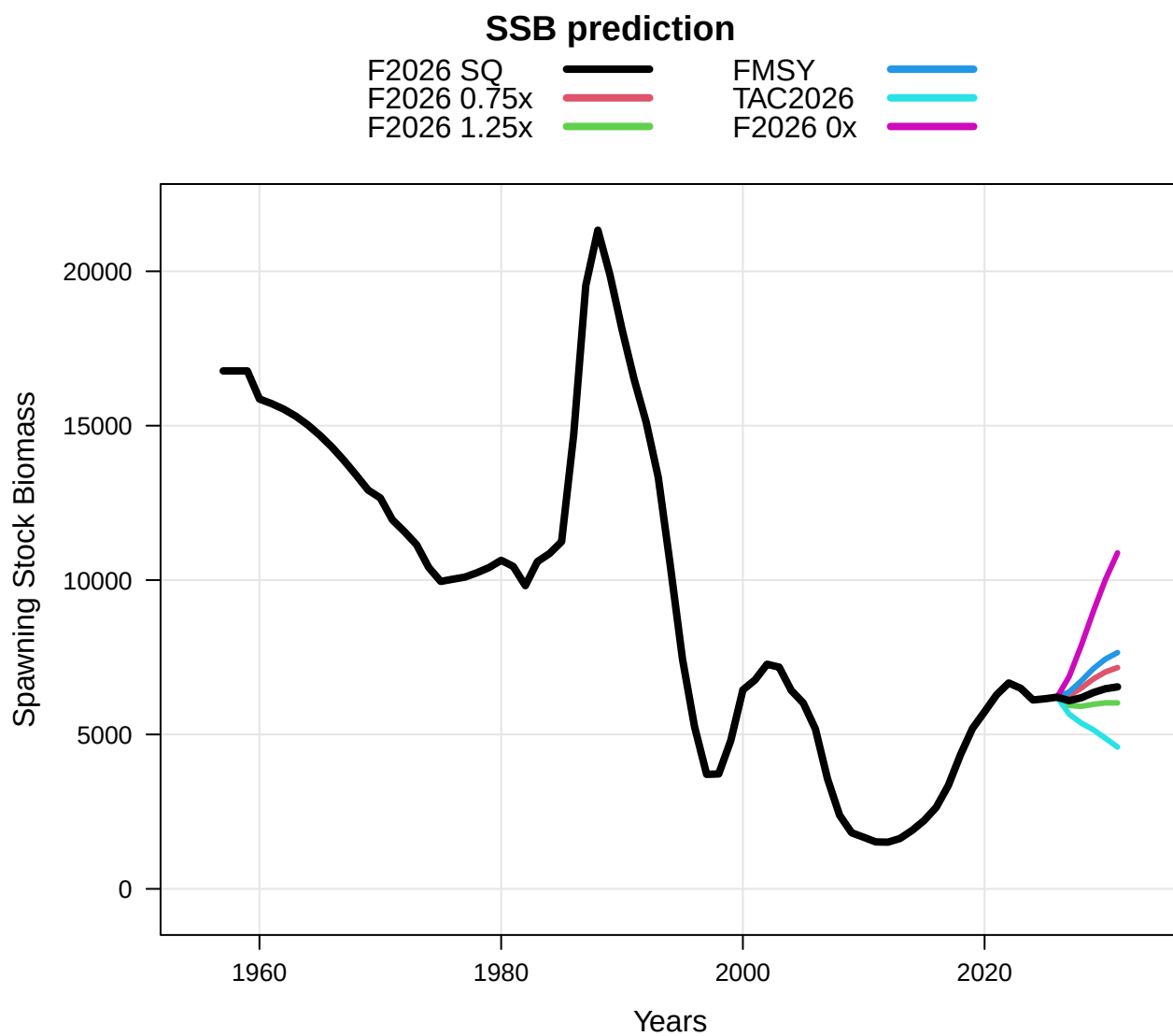


Figure 45: Earlier five-year (2027–2031) spawning biomass projections (thousand tonnes) from h1_1.06.ls under the scenario labels shown in the existing model output.

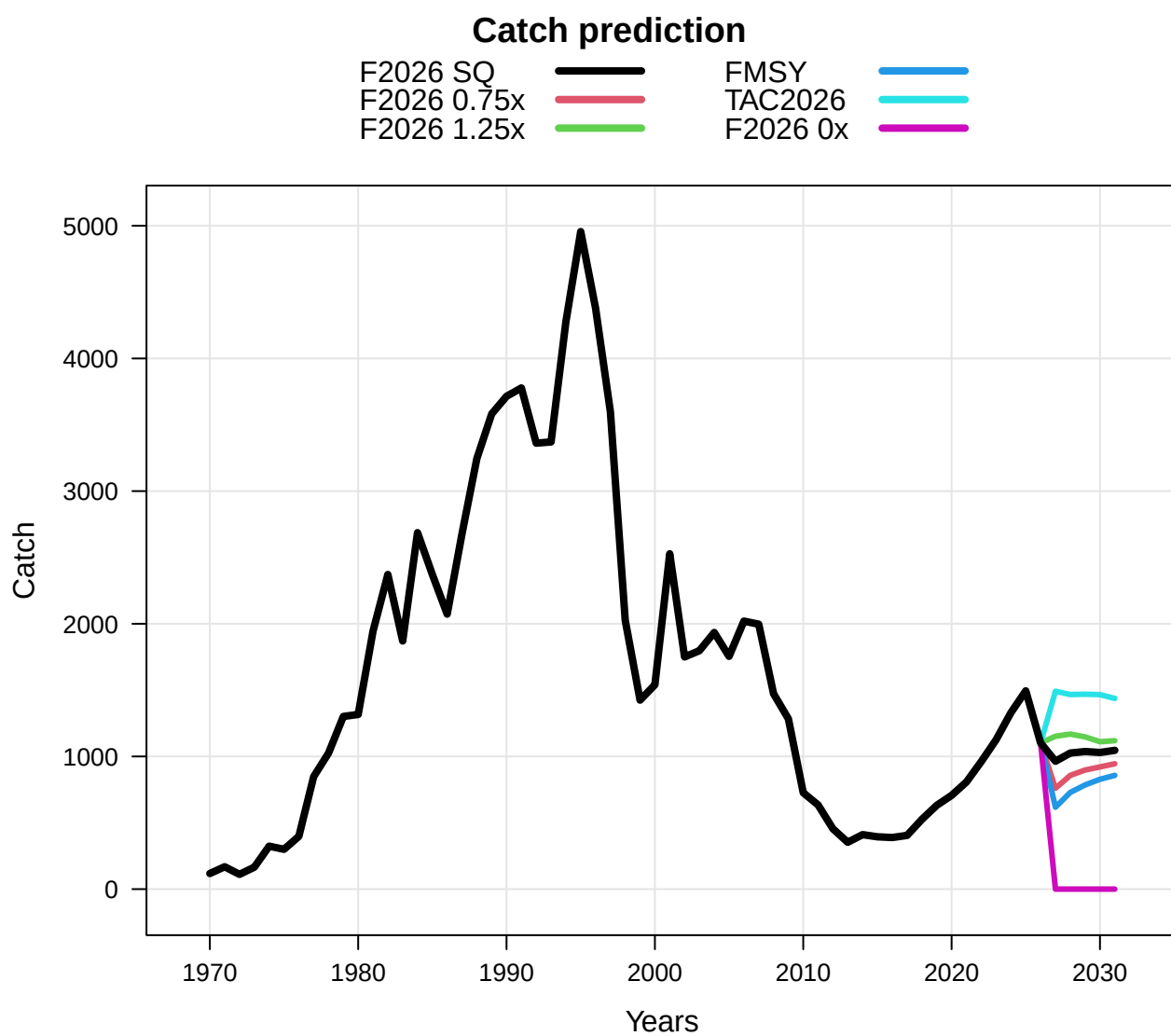
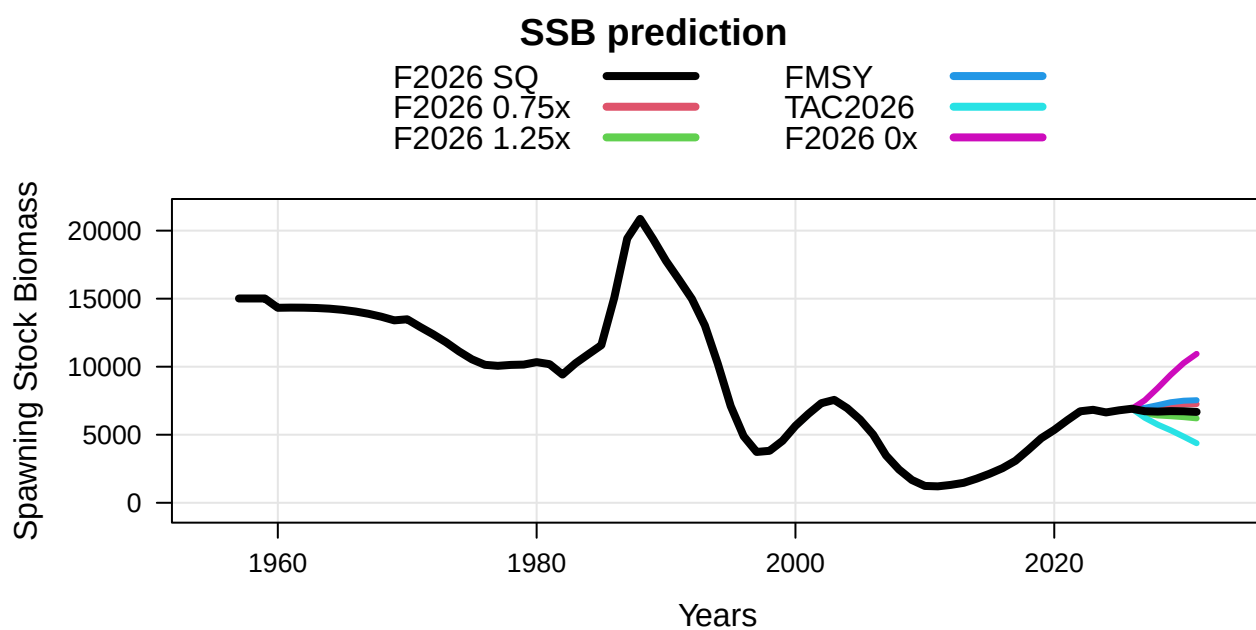
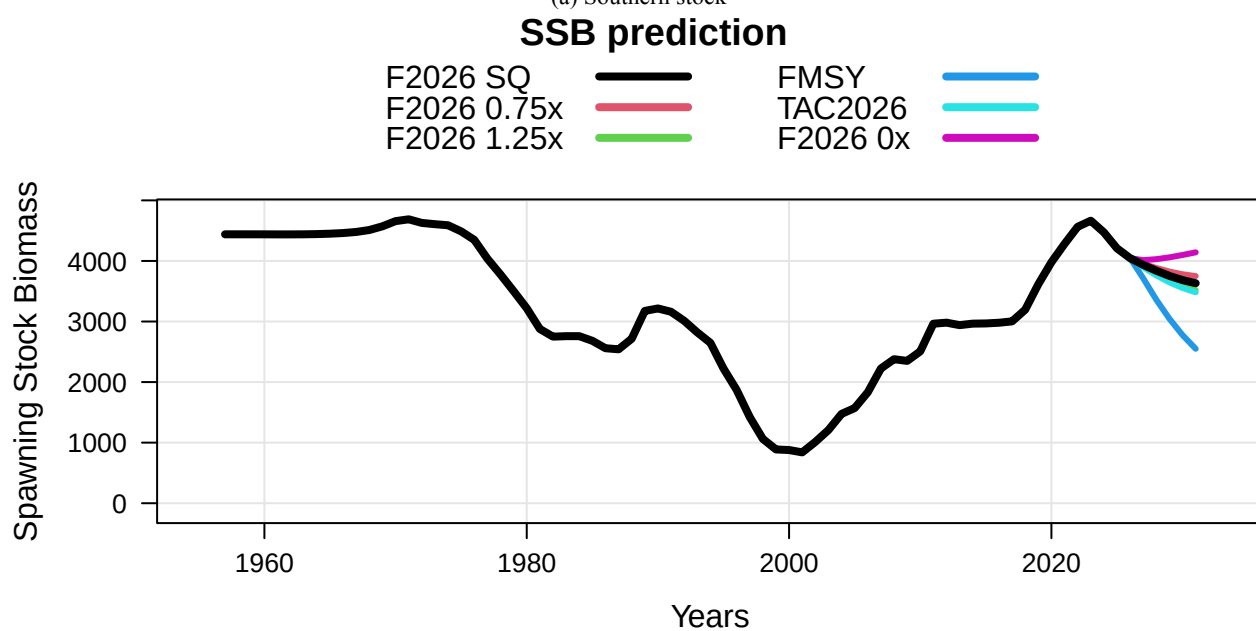


Figure 46: Earlier five-year (2027–2031) catch projections (thousand tonnes) from h1_1.06.ls under the scenario labels shown in the existing model output.

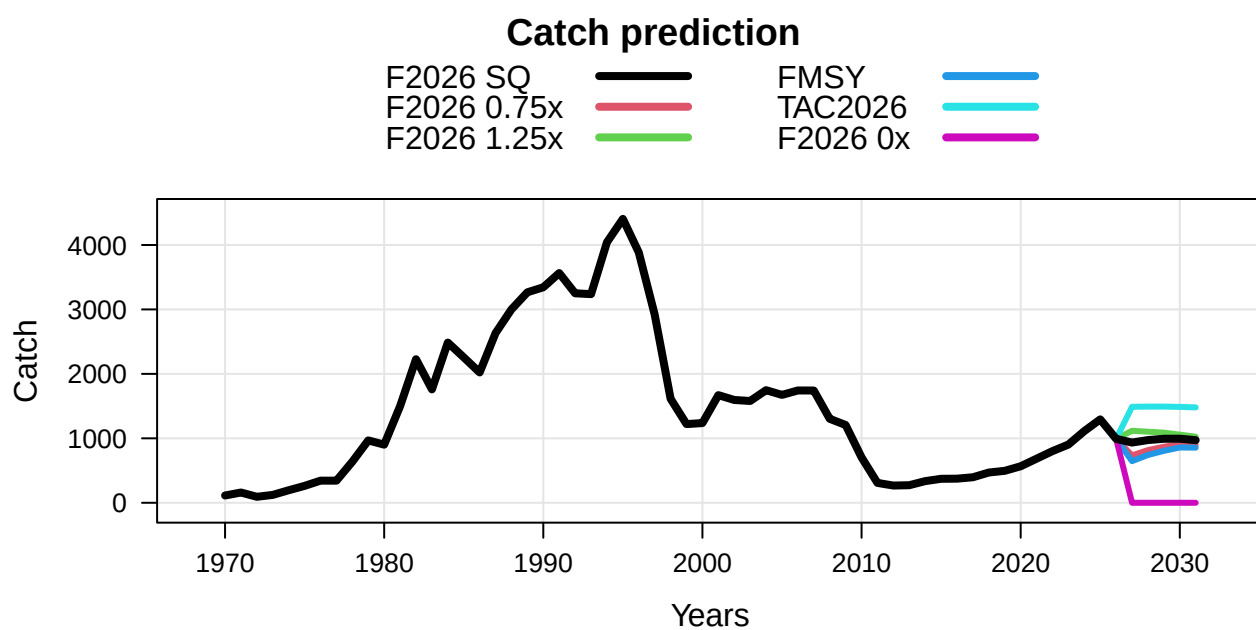


(a) Southern stock

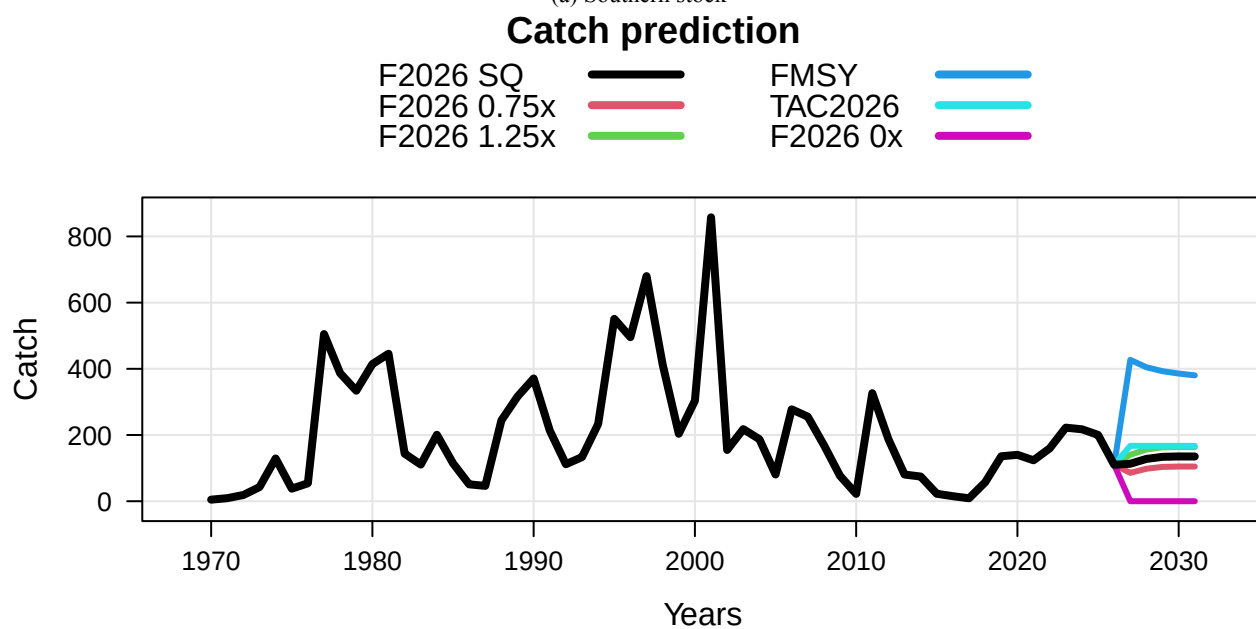


(b) Far North stock

Figure 47: Earlier five-year (2027–2031) spawning biomass projections (thousand tonnes) from h2_1.06.ls: southern stock followed by Far North stock. Scenario labels follow the existing model output.



(a) Southern stock



(b) Far North stock

Figure 48: Earlier five-year (2027–2031) catch projections (thousand tonnes) from h2_1.06.ls: southern stock followed by Far North stock. Scenario labels follow the existing model output.